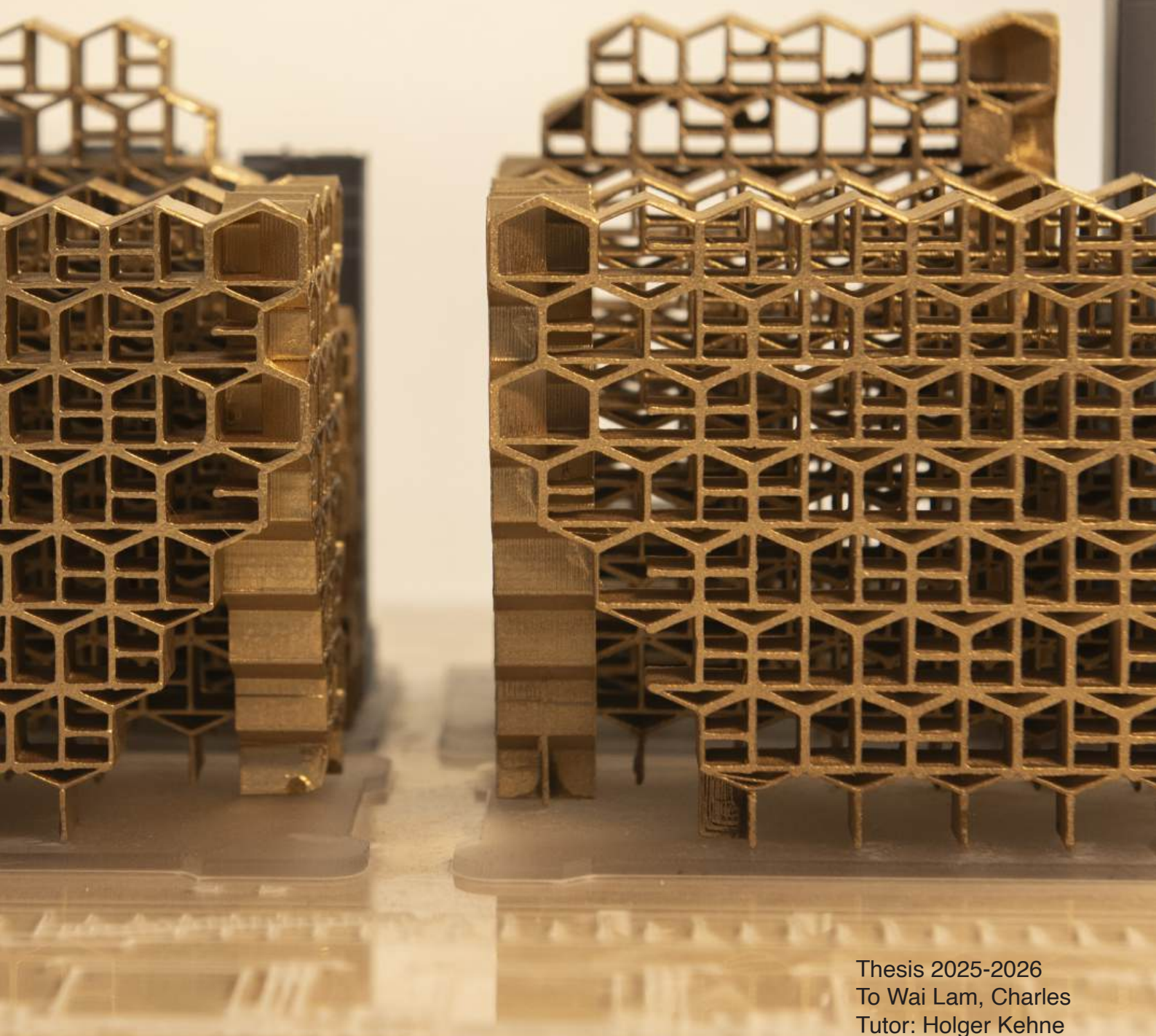


Super Liquid Housing:

Redefining Spatial Politics & Fractional Ownership
through Financial Model



Thesis 2025-2026
To Wai Lam, Charles
Tutor: Holger Kehne

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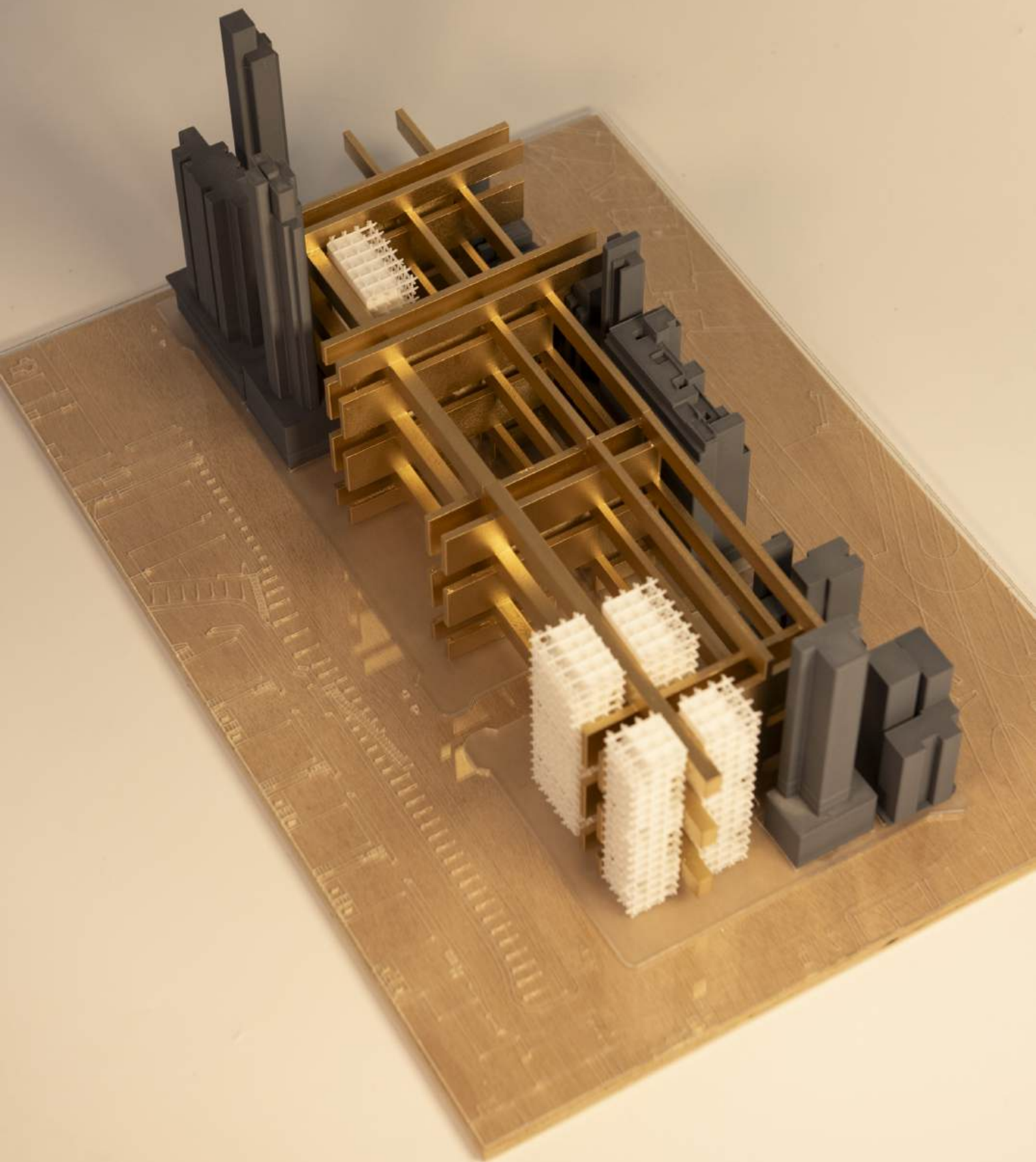
Abstract

This thesis introduces a radical architectural and financial model designed to dismantle the housing crisis as asset trap facing Hong Kong's marginalized "sandwich class". By leveraging the concept of fractional ownership and tokenization in housing, the project aims to decouple the housing of property's assetization with spatial ownership and autonomy.

A system of hyper-durable infrastructural frames was explored with two prototypes that rules interconnectivity, degree of expansion, light and air compliance, and formal expression. The permanent infrastructure "Vault" not only acts as value storage, hedged against a gold-backed reserve to secure long-term liquid value, but also serves as framework for expandable modular living units deployed within system.

Moving beyond rigid property titles, the ecosystem introduces a 100-year spatial bond tied strictly to spatial volume rather than individual occupants, compressing monthly costs and unlocking unprecedented user autonomy. A foundational set up guarantees that half of the volumetric voids remain communal, allowing private spaces to dynamically expand or contract across human lifecycles. With the investment of non-occupant token owners, the community gained more mobile chances to expand in urban scale.

Tested against the paralyzed Civil Servants' Co-operative Building Society (CBS) schemes in To Kwa Wan, Super Liquid Housing establishes a highly scalable, democratic urban paradigm where housing functions flawlessly as a home, not a financial commodity.



The Mortgage is a control tool for settler colonial modernity.

nialism. Its destruction is the end of capitalist

Chapter1:

Agenda

Agenda I **The Post-Commodity Housing**

The housing crisis in Hong Kong is far beyond supply shortages; it is a crisis of debt and a failure in financial model. The contemporary housing market, particularly within hyper-dense and land-scarce urban environments, operates on a fundamental flaw: the conflicts of the physical space as an “asset” (value of which are believed to appreciate) and as spatial utility (which is a human necessity to occupy for safety).

In the current real estate paradigm, a home is not primarily a sanctuary for living; it is a highly leveraged financial commodity. This thesis, *Super Liquid Housing*, sets out to dismantle this conflation by proposing an alternative model of unbundled financial asset value from spatial use.

By relying on credit to achieve shelter to live, middle-class and sandwich class homebuyers rely heavily on mortgages. This causes their quality of living to be inherently vulnerable to volatile capital markets. Mortgages allow ownership to home - a private space home also acts as an asset of an individual or a family. Meanwhile, a relationship between financial institutions and debt-financed homeowners is formed under the system of neoliberalism, thereby subordinating the labor market and household livelihoods to an asset loop financially.

For the sandwich class homeowners burdened with mortgages, they often find themselves trapped in debt peonage and precarity, not to say achieving spatial autonomy. The agenda of this project is to unmask this exploiting financial mechanism under capitalism and neoliberalism, proposing an architectural alternative with a finance model of fractional ownership by rethinking the following design questions:

1. Where does our capital actually flow when purchasing a flat under neoliberalism?
2. Can we reach a paradigm where floor area is no longer strictly equated to financial currency?
3. What truly defines a “home” versus a “place to live”?
4. How can individuals build genuine and long-lasting attachment to a space they inhabit but do not strictly own (rental)?
5. Is it possible to decouple the concept of “home” from “asset” within private housing development?
6. Can quality space be returned to residents as a living environment rather than a speculative commodity?

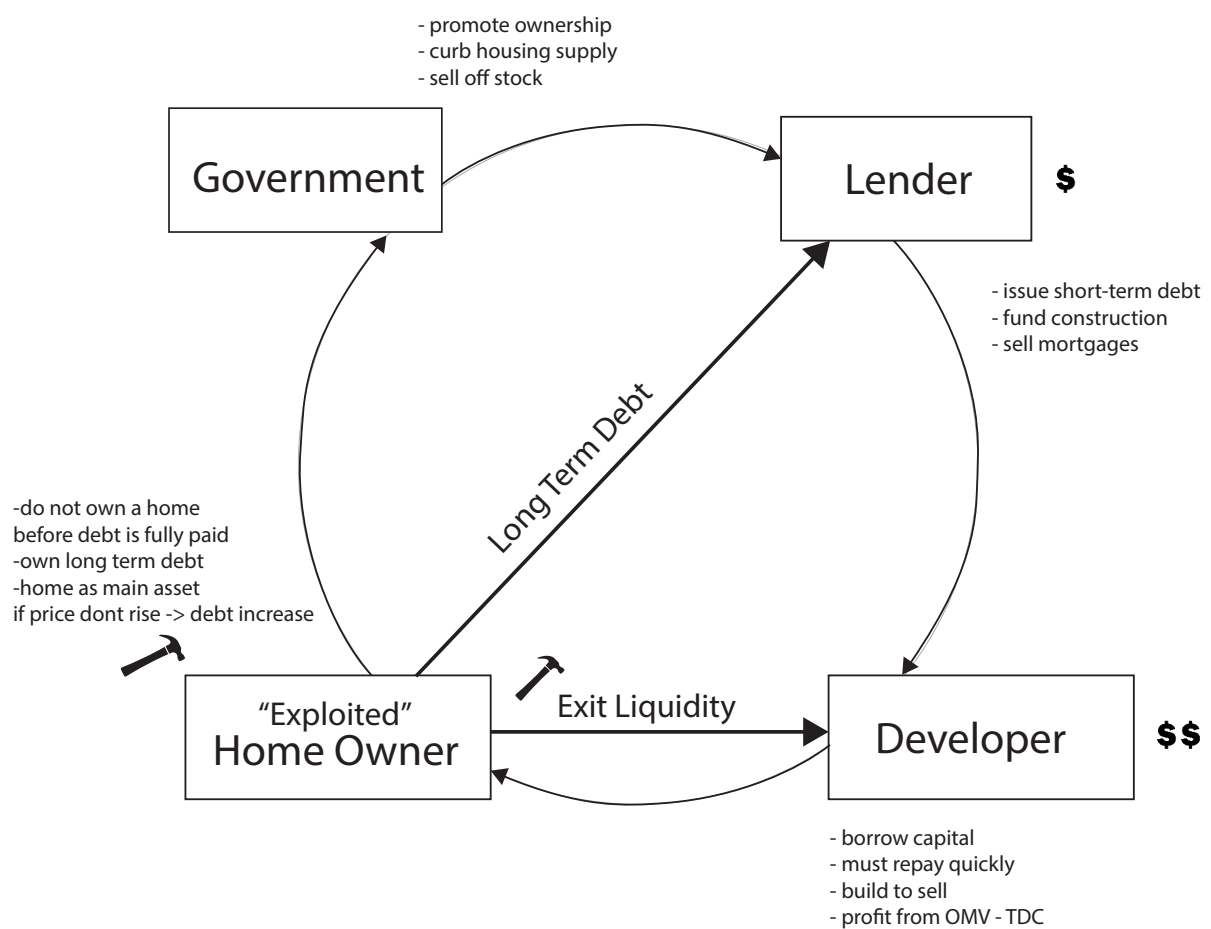
Agenda | **Housing as The Asset Trap**

In capitalism, housing is treated as an **FINANCIAL ASSET** (for speculation/wealth storage) rather than a place to live (home). This duality creates a conflict: for the “home” to be a good investment, its price must rise faster than **WAGES**, making it unaffordable for the next generation.

Neoliberalism primarily operates as a wealth distribution imperative that uses diverse mechanisms to channel **SURPLUS CAPITAL** from the middle and lower classes and concentrate it in a dwindling elite. **MORTGAGE** combined with a restriction on the supply of public housing has caused replacement of real ownership. Over time, this redistribution produced **SOCIAL INEQUALITY**.

To understand the predicament of the **DEBT** of home buyers, one must examine the phenomenon of housing assetisation, which has redefined homeownership, shifting it from merely fulfilling the need for residence and quality of life to increasingly being viewed as a store of value or lucrative investment. In this paradigm, property becomes a physical **WEALTH STORAGE** and a hedge against economic uncertainty. However, this reliance on property as a financial vehicle creates an inescapable Asset Trap.

“When house prices rise faster than wage, housing is not affordable for the next generation.”



Exploitation of the home buyer

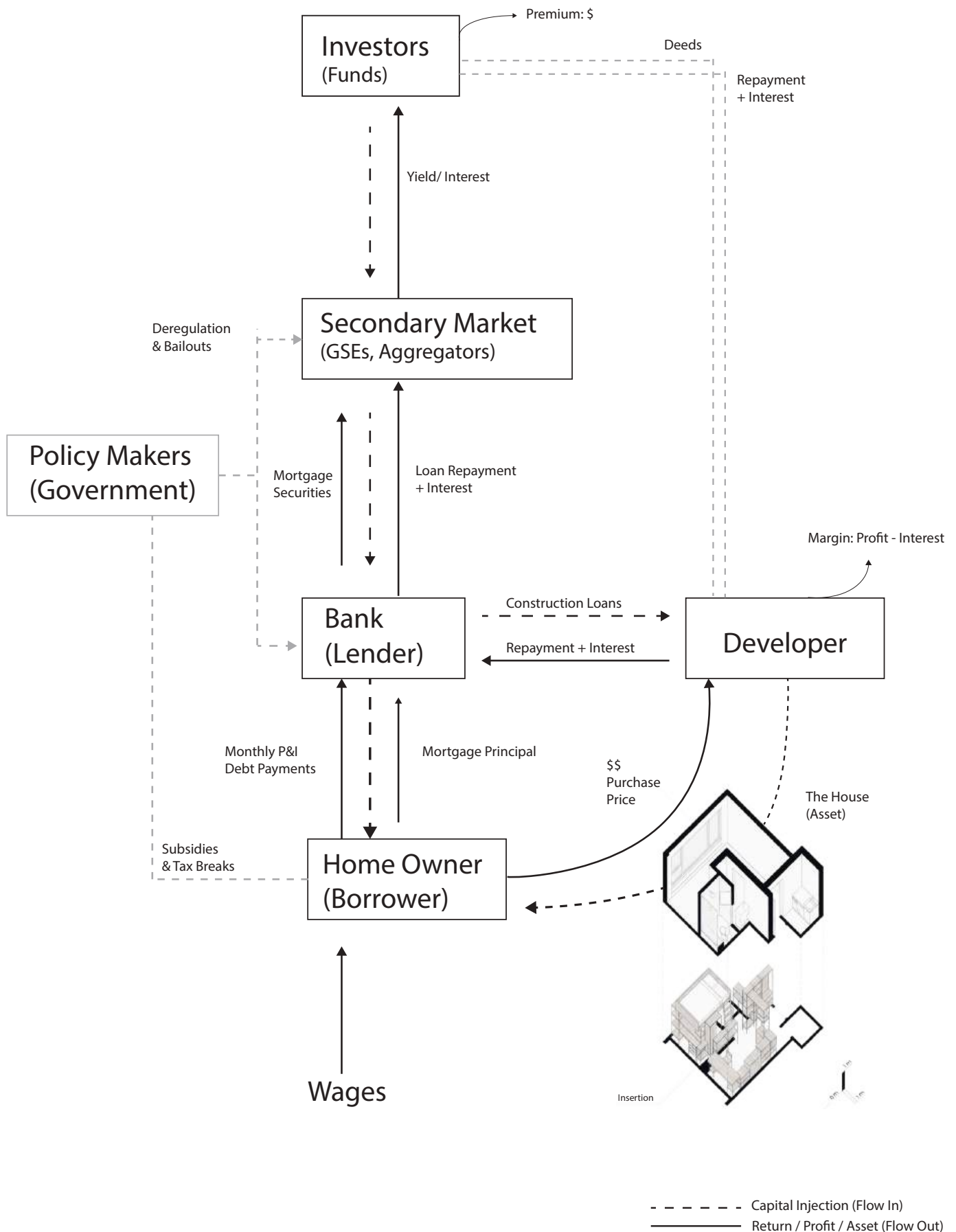
Agenda | **The Mortgage Loop**

Mortgage is a tool of social control, being especially criticized in Neo-liberalism. It forces individuals into decades of debt servitude, limiting their freedom to take risks (like starting a business or protesting). The bank owns the house; you own the debt.

(“Post-Capitalist Housing”, Jack Self)

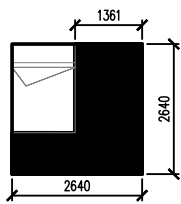
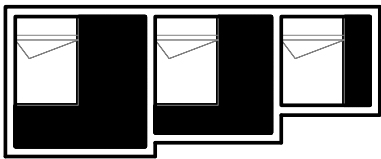
Since homeownership is required to store wealth and access capitalized public infrastructure, families are compelled into The Mortgage Loop. Substantial family wealth has been anchored in the housing market through achieving homeownership, keeping the land-infrastructure-leverage wheel spinning. This loop forces the middle class to continuously funnel their labor wages into mortgage repayments, effectively transferring wealth from the working demographic to the quaternary circuit of capital and rentier classes.

The architecture of the home is no longer a site of living, but a node of financial exploitation.

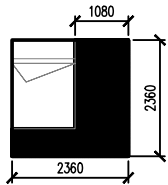


Capital flow chart of purchasing a unit

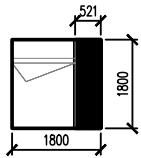
Mininum size of a bed room



90th



80th



70th

Agenda | **A Room Against Ownership**

In Pier Vittorio Aureli's (Dogma) critique of real estate, he contrasts the "architecture of property" with the "architecture of use". Keywords are:

Ownership as domination:

Aureli argues that possessing property goes beyond merely claiming an object; historically, it has served as a mechanism to extract the labor of others and exert power over those who are dispossessed. By transforming a household into property, an owner inherently invests it with economic value designed to generate profit and maintain control.

The concept of "use":

In direct opposition to ownership, Aureli advocates for "use"—which he defines as the temporary appropriation of an object by an individual. Once an object has served its purpose, it is released and shared with others, functioning as the supreme form of living together in common.

The generic room:

To illustrate this architecture of use, Aureli highlights Swiss architect Hannes Meyer's 1924 Co-op Zimmer. Reduced to a single room containing only the bare essentials (a bed, a cupboard, and a foldable chair), the space is intentionally generic and anonymous. By refusing to act as a reflection of an owner's personal identity, the room promises its inhabitant a life completely liberated from the financial and ethical "burden of household property".

Owning property is not just about having an object; it is a mechanism to own the labor of others and exercise power over the dispossessed.

"Use" represents the temporary appropriation and sharing of objects, whereas "ownership" invests objects with economic value to generate profit and control.

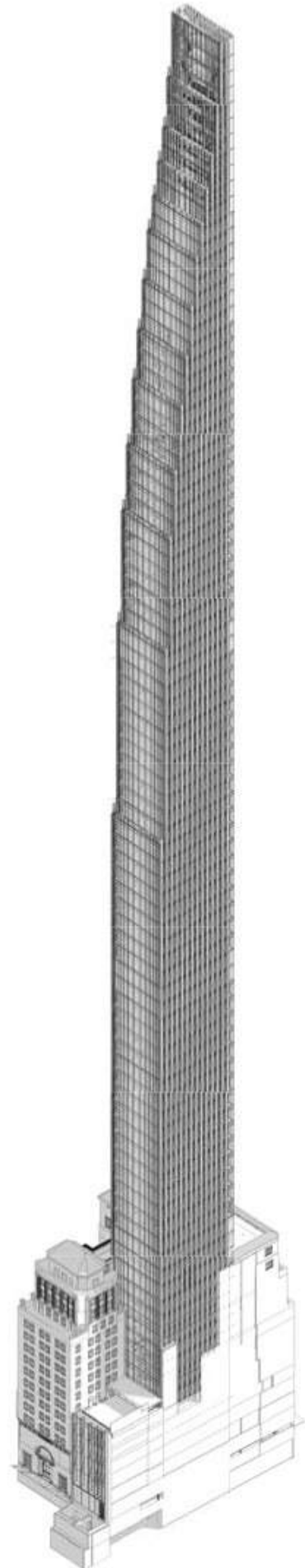
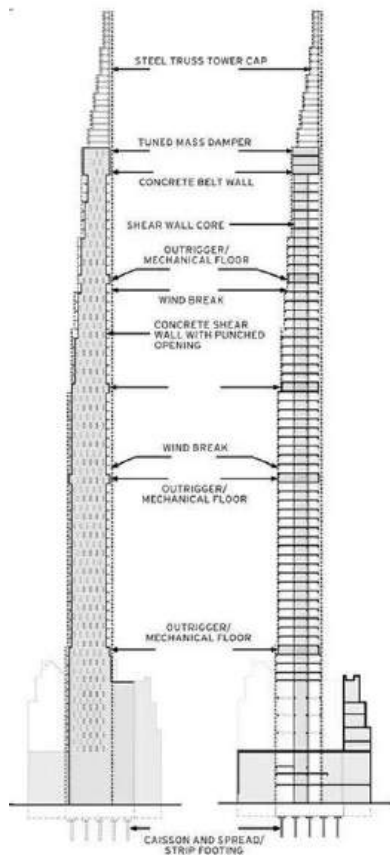
Agenda | **Form Follow Finance**

The main critique of “form follows finance” is that a building’s form is no longer an artistic or design choice, but a strict calculation to **maximize rentable area**. High land costs force developers to build vertically, while the need for natural light limits floorplate depth, and zoning laws further mold buildings into stepped “wedding cake” shapes to protect neighboring property values. This financial focus culminates in ultra-thin “pencil towers” — luxury apartments functioning not as habitats, but as highly leveraged “safe boxes in the sky” designed purely as speculative wealth storage to maximize price-per-square-foot.

In stark contrast stands “**social equity building**,” like Alejandro Aravena’s Villa Verde project, which uses housing as a tool for social mobility by providing low-income residents a basic structure and an empty spatial void. Instead of taking on massive debt, residents use their own “sweat equity” to incrementally build into the void over time, potentially tripling the property’s market value and retaining that profit to enter the middle class.

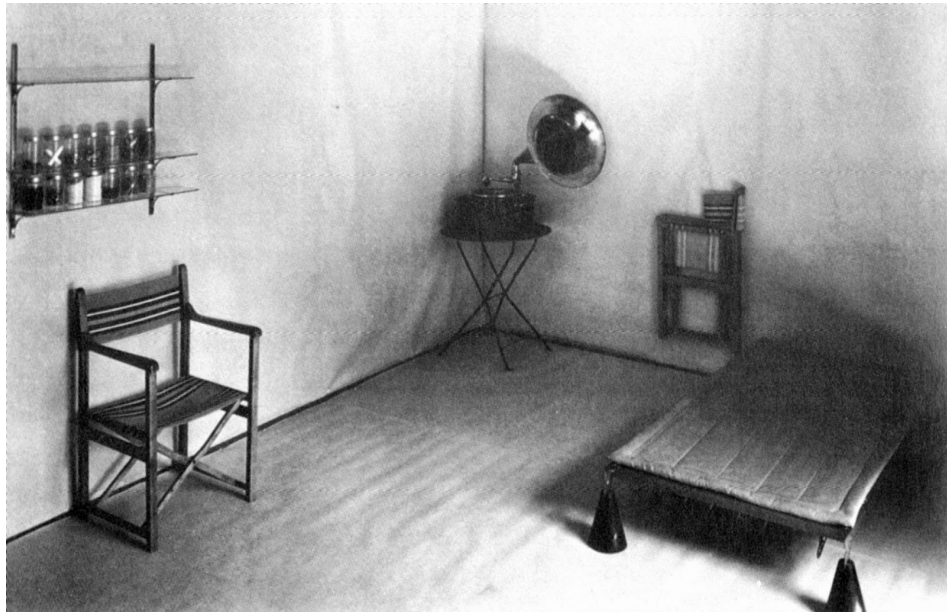


111 West 57 Street Residential Tower, NY



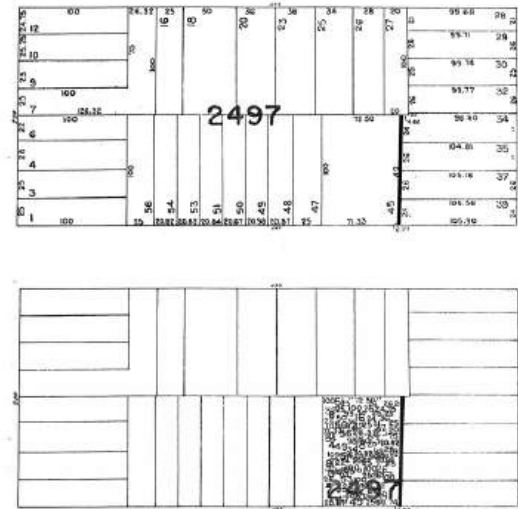


The Ingot is a theoretical project designed for the City of London, situated over the ruins of an ancient Roman forum near London Bridge. Proposed by Jack Self as an example of “derivative architecture,” it envisions a massive 356-meter tower funded entirely by a 50-year bond rather than traditional speculative real estate development’s financing model. The goal of the project is to provide high-quality housing for living-wage workers by separating the building’s financial form from its social function, keeping the property off the speculative market so residents only pay for use rather than ownership.

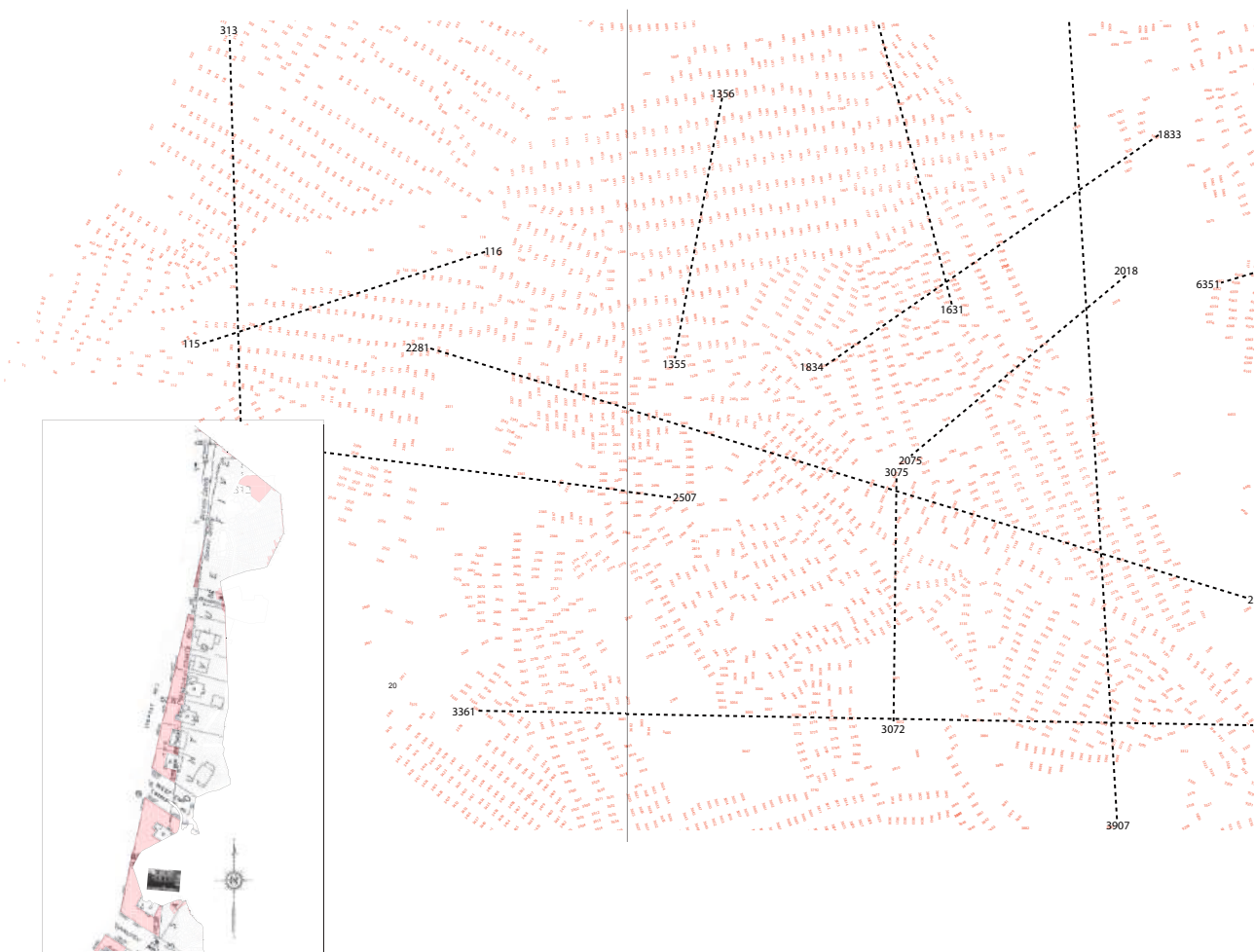
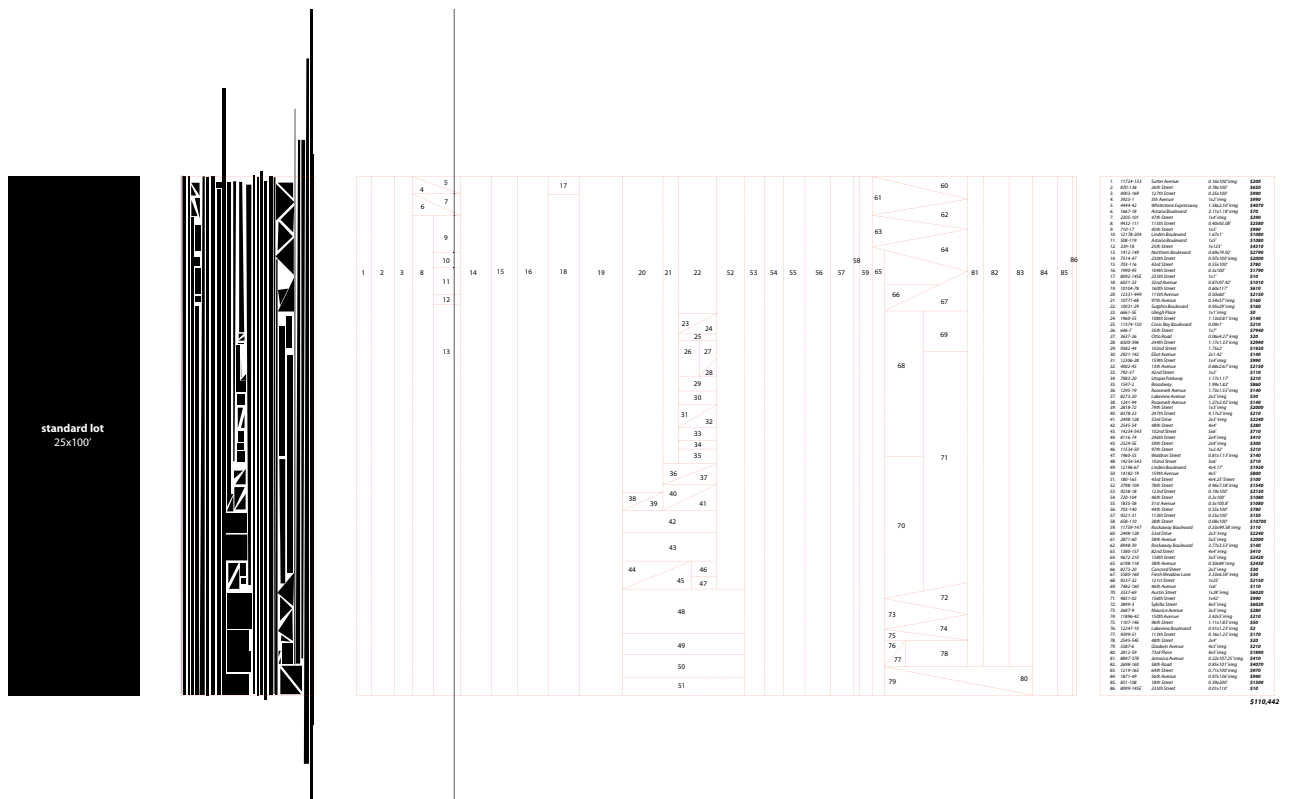


Co-op Zimmer (1924) Designed by the Swiss architect Hannes Meyer, this project is a radical challenge to the “house-as-property”⁴. It is a single room designed for a nomadic worker containing only the bare essentials: a bed, a cupboard, and a foldable chair (with a mock-up also featuring a gramophone)⁴. By minimizing domestic space, the design implies that the rest of the building and the city are communal, offering a life liberated from the burden of household property⁴.

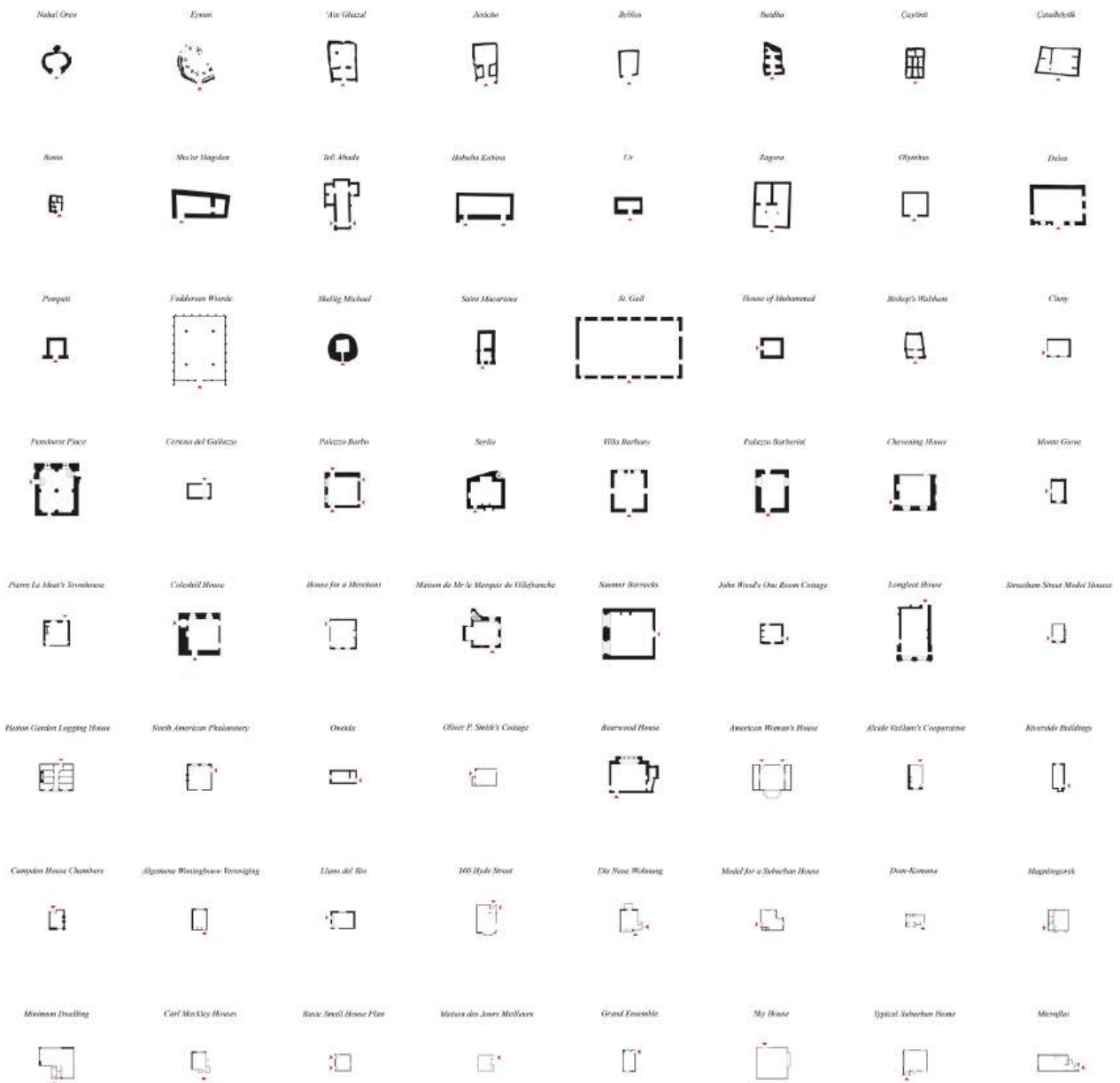
18	4-3937-18	49.42x119.42' 11914 6th avenue	\$490,000
20	4-3937-20	37.58x113' 11918 6th avenue	\$502,000
22	4-3937-22	23.58x113.42' 11920 6th avenue	\$379,700
23	4-3937-23	19.67x113.42' 11922 6th avenue	\$392,900
24	4-3937-24	19.67x113.42' 11924 6th avenue	\$356,900
25	4-3937-25	19.67x113.42' 11926 6th avenue	\$348,900
26	4-3937-26	19.67x113.42' 11928 6th avenue	\$353,900
27	4-3937-27	20.50x113.42' 11930 6th avenue	\$356,600
28	4-3937-28	20.50x113.42' 11932 6th avenue	\$356,600
29	4-3937-29	20.50x113.42' 11934 6th avenue	\$358,600
30	4-3937-30	20.50x113.42' 11934 6th avenue	\$358,600
31	4-3937-31	41.25x113.42' 11940 6th avenue	\$516,000
33	4-3937-33	20.50x113.42' 11942 6th avenue	\$360,600
34	4-3937-34	20.50x113.42' 11946 6th avenue	\$380,600
35	4-3937-35	27.42x113.42' 11948 6th avenue	\$412,500
37	4-3937-37	27.42x113.42' 11950 6th avenue	\$412,500
39	4-3937-39	27.42x113.42' 11952 6th avenue	\$412,500
42	4-3937-42	35.08x103.17' 11964 6th avenue	\$512,000
44	4-3937-44	26.67x103.17' 608 college place	\$440,000
45	4-3937-45	41.25x103.17' 610 college place	\$488,000
47	4-3937-47	30.75x103.17' 612 college place	\$534,000
49	4-3937-49	30.75x103.17' 616 college place	\$534,000



Fake Estates (1973–1974) is a conceptual art and architectural project where the artist Gordon Matta-Clark purchased 15 oddly shaped, unusable parcels of land—or “gutterspaces”—in New York City at property auctions. These micro-parcels were often inaccessible and too small to occupy. The project aimed to satirize bureaucratic banality and expose the absurdity of real estate, property demarcations, and the “American dream” of land ownership.



64 ROOM PLANS, DOGMA



The Project of Autonomy, by Pier Vittorio Aureli and the architectural practice Dogma, is a critical provocation within contemporary architectural theory. It seeks to detach the architecture of the city from the pure forces of urbanization and capital accumulation. Aureli's argument is rooted in the idea that architecture, in its purest form, should resist becoming merely a tool for economic development or spatial commodification.

64 UNIT PLANS, DOGMA



He advocates for “autonomy”—the idea that architecture possesses its own internal logic, history, and formal language, which can be deployed to create spaces that resist the relentless flow of capitalist urbanization. The Project of Autonomy argues that architecture can carve out spaces of political and civic significance, allowing for forms of collective life that are not solely dictated by the market.

Finance Model of Non Public Housing



	Traditioanl Developer	Subdivided Flat	Cooperative
Ownership	INDIVIDUAL DEED to a specific unit.	Landlord owns parent unit; tenant has no title.	Shares in a corp. owns the building.
Usage Right	Exclusive, permanent.	Exclusive but precarious (often informal).	Exclusive, perm. shares are held.
Debt Type	Individual high-leverage mortgage.	None for tenant; high-yield for landlord.	SHARED CORP. GAGE ; low men
Rental Rate	100% Market Rate.	"Poverty Premium" (high per-sq-ft).	Cost-recovery (A Expenses).
Building Height	High (Profit-optimized, approx. 90m+).	Restricted to existing building height.	Mid-rise (Human
Liquidity	Moderate. Subject to market cycles and mortgage approval.	High (for Landlord). Easy to evict/re-lease; Zero for tenant.	Low. Sales often approval and ve members.
Exit Strategy	SALE OF DEED via real estate agent; capital gains realized.	Precarious. Tenant exits via eviction or end of informal lease.	SALE OF SHAR TO CO-OP or ve capped price.
Governance	Property Management Company.	Unregulated / Informal Landlord.	Democratic Men
Maintenance	Owner (Internal) / Management (External).	Landlord (often neglected).	Collective Memb



Cooperative Housing	The Ignot	Timeshare (Hotel)	Fractional Ownership
incorporation that requires member debt.	Usufructuary rights; building is a debt-free asset.	Right to use for a specific time interval.	Deeded percentage of a shared asset.
Member is permanent as long as they pay dues.	Permanent "rights of use" without ownership.	Restricted to specific weeks/dates annually.	Shared / specific period.
COOPERATIVE MORTGAGE (Member debt).	50-YEAR LONG-TERM BONDS (Institutional).	High upfront cost; high-rate financing.	POOLED EQUITY ; often no individual mortgage.
Actual Operating	46% Market Rate (Targeted).	High maintenance fees + upfront cost.	Variable (usually market-adjusted).
Member-centric.	Extreme (350m+) (Density-optimized).	Commercial/Resort scale.	Composite building.
Members must require Board approval for setting of new rules.	Fixed. The 50-year bond locks capital in place to ensure stability.	Very Low. Infamously difficult to resell; often loses value.	High. Shares are smaller and easier to trade than full titles.
EXIT STRATEGIES BACK (Member debt buyer at exit).	Post-Maturity. Asset can be sold for a symbolic sum to residents.	Complex legal exit or "paying off" the management contract.	SECONDARY MARKET SALE OF SHARES or "Exit" at end of LLC term.
Member-Voting.	Algorithm-driven Social Financials.	Management Company.	Managed by a Platform/LLC.
Member-responsibility.	Centralized by the Bond entity.	Resort Service (High fees).	Professional Management.

Agenda | **Atlas of Housing Finance**

Historically, architectural discourse has struggled to resolve the tension between capital and shelter. Precedents such as Alejandro Aravena's work with Elemental introduced the concept of incremental housing, returning spatial agency to the residents. However, Elemental's models largely rely on traditional land ownership and low-density sprawl. Is it possible to adapt a similar finance model of contingency into an urban context?

On the other hand, Jack Self's theoretical project, *The Ingot*, proposed a high-density, hyper-durable structure meant to outlast traditional speculative cycles. This thesis builds upon the two disciplinary foundations with a contemporary intervention of new financial concepts and tools: fractional ownership, and decentralized finance, tokenization. If the whole development is tokenized.

In the Atlas, private and cooperative housing development models around the world are examined and compared, including financial driver, profit driven logic and debt reliance.

111 West Steinway Tower

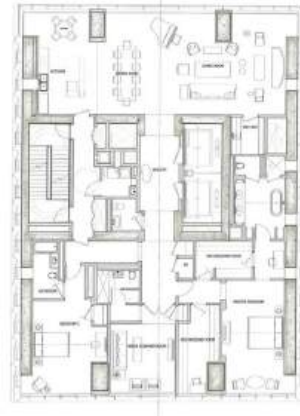
Urban Area, New York

Private
development

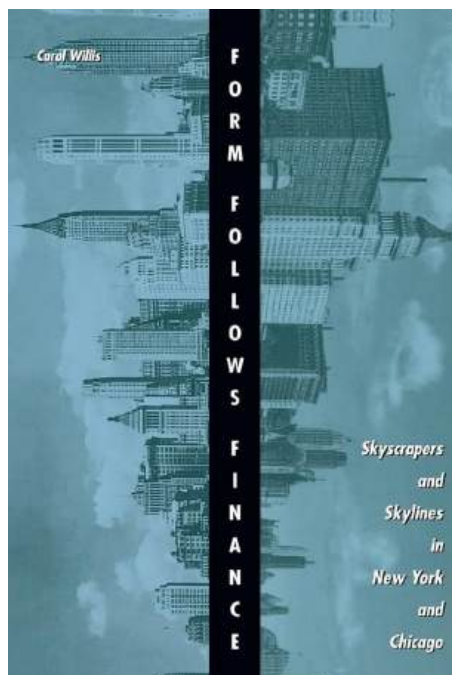
podium
tower



form follow finance
/capitalism



Location : New York City (Manhattan), USA.
Land Type: urban
Site Area: 59 x 75 feet (18 x 23 meters).
GFA: 53,140 m²



The Economic Logic of "The Envelope"

Willis explains that the shape of a building isn't just an aesthetic choice; it's a calculation of **Maximum Rentable Area**.

- **Vernacular of Capitalism:** In New York and Chicago, the "form" was dictated by the need to extract every possible cent from a plot of land.
- **The Floorplate:** Developers discovered that natural light only penetrates about 25–30 feet into a room. This created a "standard depth" for offices and living spaces, dictating the width and shape of the building's footprint.

2. Zoning as a Financial Tool

Legislation like the **1916 Zoning Resolution** in New York wasn't just about urban planning; it was about protecting real estate values.

- **The Setback:** By requiring buildings to step back as they got taller to allow light to reach the street, the law created the iconic "wedding cake" silhouette.
- **Efficiency vs. Height:** Developers had to balance the cost of building higher (expensive elevators and structure) against the premium rent of "prestige" top floors.

3. Land Value and Verticality

The price of the dirt determines the height of the sky.

- In areas with high land costs, the only way to make a profit is to stack "living units" vertically to spread the land cost across more square footage.
- **The "Spec" Building:** Willis emphasizes that many icons were built "on spec"—meaning they weren't designed for a specific tenant but as a generic commodity to be sold or leased. This led to standardized, modular living and working spaces.
- **The "Skinny" Supertalls (Billionaire's Row):** How advances in concrete technology combined with the "wealth storage" financial model (using apartments as assets rather than homes) created ultra-thin towers.
- **Air Rights:** The financialization of "empty space" above existing buildings and how it allows for "zombie" towers that wouldn't be possible under standard logic.
- **Floor Area Ratio (FAR):** The ultimate mathematical formula where $\text{\$Total Floor Area} = \text{Site Area} \times \text{FAR}$. This is the "DNA" of modern architectural form.

property as wealth storage



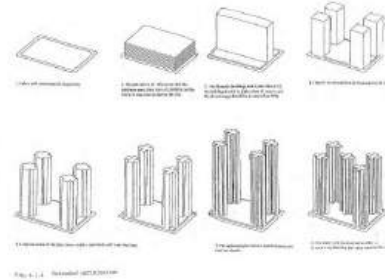
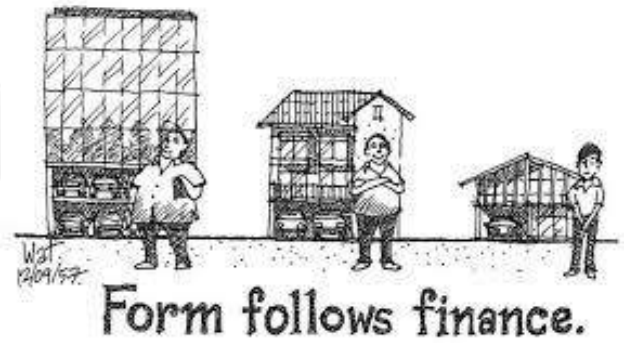
Location : Kwun Tong, Kowloon, **Hong Kong.**
Land Type: urban
Site Area: 21,754 m²
Residential gfa: 139,000 m² (1,999 units).
Retail gfa: 17,200 m²

Private

Hong Kong

form follow finance

Grand Central (凱匯)
Wong Tung &
Partners Limited



Hong Kong
Podium Tower

plot ratio

set back

Steinway Tower

Financial Driver: Speculative Wealth Storage.

The primary driver is the global demand from High-Net-Worth Individuals (HNWIs) for "safe-haven" assets. These apartments function like "safeboxes in the sky"—they are often purchased as investments rather than primary residences.

Profit-Driven Logic: Maximizing Price-Per-Square-Foot (PPSF).

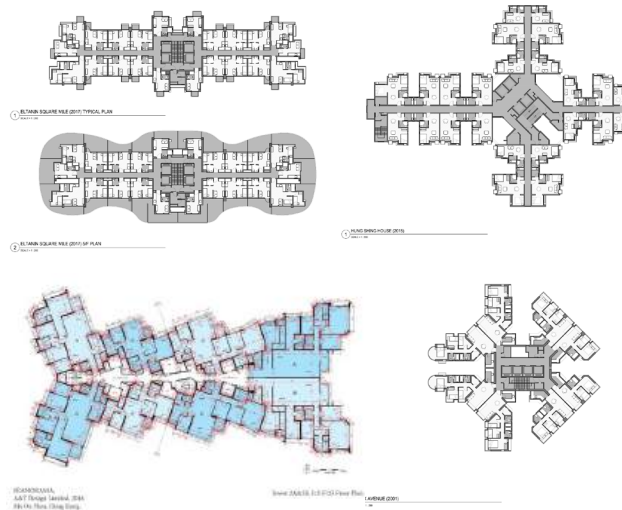
The logic is to produce as many "full-floor" units as possible with unobstructed views of Central Park. By reaching a height of 1,428 feet on such a tiny site, the developer maximizes the value of the "air rights" purchased from neighboring buildings.

Debt Reliance: Extreme (Layered Financing).

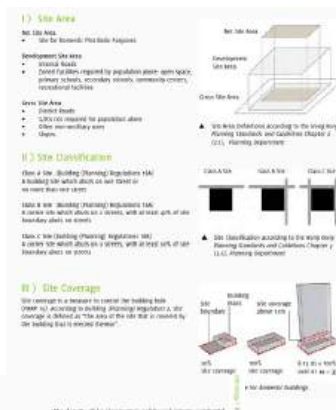
The project is famous for its complex and volatile financial history. It relied on a combination of senior debt and high-interest **mezzanine loans**. In 2017, the project faced a "capital call" and a foreclosure crisis during a stall in construction, highlighting the high-risk nature of the private tower model.

Architectural Form: The "Super-Slender" Pencil Tower.

With a height-to-width ratio of 1:24, it is the world's thinnest skyscraper. It features a "feathered" profile with setbacks on the south side and massive **concrete shear walls** on the east and west to resist wind loads.



long corridor



V J Plot Ratio

- Plot ratio is the ratio of GFA to building.
- Building volume: the higher the plot ratio, the more GFA a developer is able to build up to and maintain profit margin.
- Plot ratio is the ratio of GFA to building.

Building	Plot Ratio	Building Volume	Plot Ratio	Building Volume	Plot Ratio	Building Volume
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2
Steinway Tower	1.2	1.2	1.2	1.2	1.2	1.2

Maximum permitted site coverage and plot ratio in relationship to building height for domestic buildings.

high leverage debt

height-to-width ratio

Traditional Developer Model (Hong Kong)

The Hong Kong model is defined by hyper-density and the **podium-tower typology**, where a high-rise residential tower sits atop a commercial podium.

- Profit-Driven Logic:** Developers typically seek a **20% profit margin**, making sites unviable if costs cannot be squeezed or densities increased.
- Vertical Integration:** This model optimizes space-saving and efficiency but often results in a "blurred zone" of anonymity and lack of privacy for residents.
- Debt Reliance:** For the past 40 years, this system has relied on rising property values to justify high household borrowing, essentially replacing wage growth with debt.

Subdivided Flats (The "Poverty Premium")

Common in Hong Kong, these represent the most extreme exploitation of the "property as domination" concept.

- Usage vs. Property:** Tenants pay for the **use** of a tiny space (often just a bed-space), but the rent per-square-foot often exceeds that of luxury apartments.
- Informal Debt:** Tenants are often excluded from formal credit, living in a cycle of "predatory" rental where they have no rights to the underlying asset.

form follow finance

social/ autonomy
low cost



co-
building

refuges



Location : Constitución, Chile.
Land Type: refuges
Site Area: 120,000 m²
GFA: 27,500 m²



Villa Verde Project
Alejandro Aravena (Elemental)
Low Entry Barrier



Liquidity of Scale: Most social housing is a "**dead end**" asset (it stays the same size and value). Villa Verde is "liquid" because it can **expand** or contract its footprint based on the economic capacity of the inhabitant.

Incremental Housing

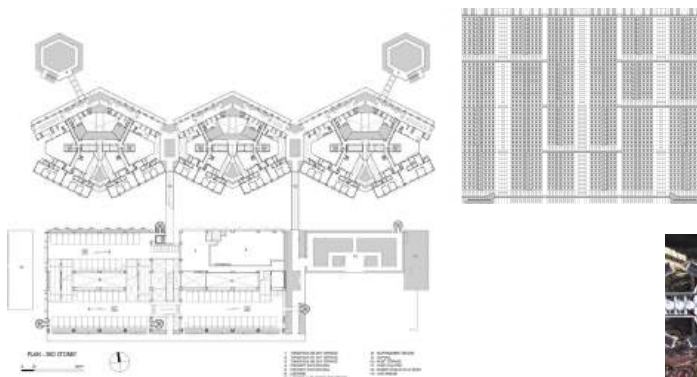
Financial Driver: Public-Private Partnership (PPP). The project was funded by a combination of government housing subsidies and private investment from the **Arauco Forest Company** for its employees.

Profit-Driven Logic: Social Mobility / Equity Building. The "profit" is not for the developer but for the **resident**. A social house that starts at \$20,000 can reach a market value of \$60,000 once the resident finishes the other half, effectively moving the family into the middle class.

Debt Reliance: Low for Residents. Most families used state subsidies that require very little to no mortgage debt. The "debt" is replaced by "**Sweat Equity**"—the resident's own labor and savings used to complete the house over time. The Chilean government provided only \$7,500 per family for land, infrastructure, and the house. This was not enough to build a "good" house.

Living Space Impact: Programmable Growth. Provides a functional kitchen, bathroom, and one living area. The architecture is a literal **financial diagram**. The concrete part is the state's investment; the "empty" space is the resident's future equity. As the family's finance grows, they build into the void.

- **The Void:** Acts as a "bank account" in the form of space. As the family grows or saves money, they build into the void. This prevents the "slumification" of social housing because the expansion is pre-planned and structurally supported.



SkyVille provides the **physical infrastructure** for internal hard drive by removing structural beams from the living space, allowing the "user" to redefine the architecture without compromising the "asset."

Location : Queenstown, Singapore.
Land Type: high density
Site Area: 29,392 m²
GFA: 113,960 m²



Public



Standardized Liquidity

SkyVille @ Dawson
WOHA Architects

Profit-Driven Logic: Sovereign Land Value / Non-Speculative. A high-density public housing project commissioned by the Housing & Development Board (HDB). Units are sold to citizens at subsidized rates (often 30–50% below market value). The Singapore government owns the land and uses a "99-year leasehold" model. The project budget is fixed by the state but subsidized to ensure affordability for the "sandwich class."

Debt Reliance: High (State-Backed).

Construction financed via HDB's budget and government bonds. Most residents use their **Central Provident Fund (CPF)**—a mandatory state-managed savings account—to pay for their mortgages, making the "debt" part of a controlled, national financial ecosystem.

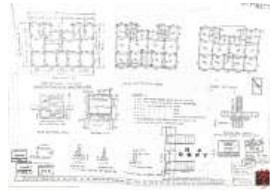
Living Space Impact: The finance dictated extreme density (building high to save land), but the "social finance" mandate forced the creation of **un-rentable social space** (the gardens). This creates a "vertical kampung" (village) feel that private speculative finance would never allow because it doesn't generate "sellable" square footage.

- **No Internal Corridors:** Every unit is "one unit thick," ensuring cross-ventilation and natural light, reducing the reliance on air-conditioning.



Civil Servants' Cooperative Building Societies

土瓜灣：公務員合作社的前世今生



公務員建屋合作社計劃	
年份	合作社區
1942	二戰期間，中區城市發展局，為服務與戰時居民
1946	香港人以公屋形式，向政府申請，以解決戰後「公屋」問題
1950	香港政府成立「香港房屋委員會」，以解決戰後「公屋」問題
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1970	香港政府成立「香港房屋委員會」，以解決戰後「公屋」問題

Location : To Kwa Wan (Kowloon City), Hong Kong.
Land Type: high density urban area of the 1950s/60s.
Site Area: 16,473 m²
GFA: 27,500 m²

"One-Large-Deed"

Hong Kong: Kau Pui Lung Road Redevelopment
The Project: A current Urban Renewal Authority (URA) project to redevelop 1950s Civil Servants' Cooperative Building Societies (CBS). **Tai Chung Co-operative Society** employed their own architects; historically, firms like **W. Szeto & Partners** were active in similar scales

Financial Driver: Government-Aided Mutualism. The colonial government granted land at **1/3 of the market value** and provided low-interest loans (typically **3.5%**) for construction. The original 1950s co-ops were 5-story walk-ups on high-value land. Today, the "finance" of the city demands a higher **Plot Ratio**.

Profit-Driven Logic: Non-Profit / Internalized Equity. The original logic was strictly for **occupation**, not speculation. The "profit" for members was the ability to own a home for life at a fraction of market cost. However, the logic has flipped into a **"liquidity trap"**: to sell or redevelop, members must pay back the "land premium" (the 2/3 discount) to the government at **today's** market rates, which often exceeds the value of the old flat itself.

Debt Reliance: High (Initial Government Debt). The societies were built using government loans that were repaid over 20 years. Today, the "debt" is effectively an **unpaid land liability** to the government that prevents the asset from being liquidated.

Liquidity Paradox: The CBS proves that a "discount" on land isn't a gift if it comes with an alienation restriction. It creates "dead capital" that cannot be utilized for urban renewal without massive state intervention.

co-building

Architectural Form: Low-Rise Tenement Modernism. Typically 5-story walk-up blocks (no elevators) with high ceilings, spacious balconies, and a "shophouse" footprint but purely residential. They are characterized by wide stairwells and "cross-ventilation" layouts typical of pre-AC Hong Kong.

Living Space Impact: The original residents had large balconies and "extra" room (because land was cheap in 1950). The new form follows **Modern Land Premium Logic**: units are smaller, "efficient" (tiny), and maximized for the market to pay back the government's "land premium" tax. It is a case study in how **rising land value shrinks the private living room**.

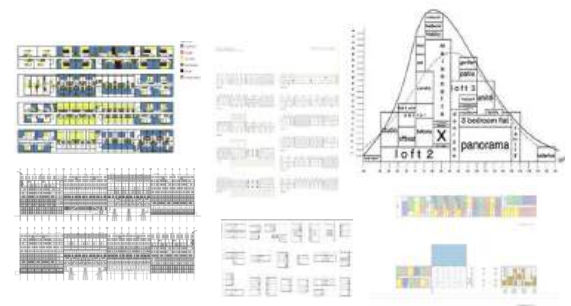
Silodam
MVRDV



Financial Driver: Diversified Market Targeting. The project was a collaboration between the city and private developers to revitalize the docks. It was designed to satisfy a complex set of "points" required by Dutch housing regulations, balancing social housing with high-end private sales.

Profit-Driven Logic: "Risk Hedging through Variety." By packing 157 different apartment types into one building, the developer effectively "de-risked" the project. They could capture the entire market—from students and low-income renters to wealthy professionals and home-office workers—all within a single construction cycle. If one segment of the real estate market cooled, the others remained liquid.

Debt Reliance: High (Commercial Development). As a speculative mixed-use project, it relied on standard construction financing, but the "pre-sale" of the unique, high-end units was used to secure the debt and subsidize the inclusion of social units.



Programmatic Volatility

Location : Amsterdam, Netherlands
Land Type: 10-meter-wide dam pier, industrial harbor.
Site Area: 2,400 m²
GFA: 19,500 m²

Social Collision: By stacking different economic classes and demographics to share the same elevators and corridors, the architecture attempts to create a "micro-city" where social friction leads to a more vibrant community.

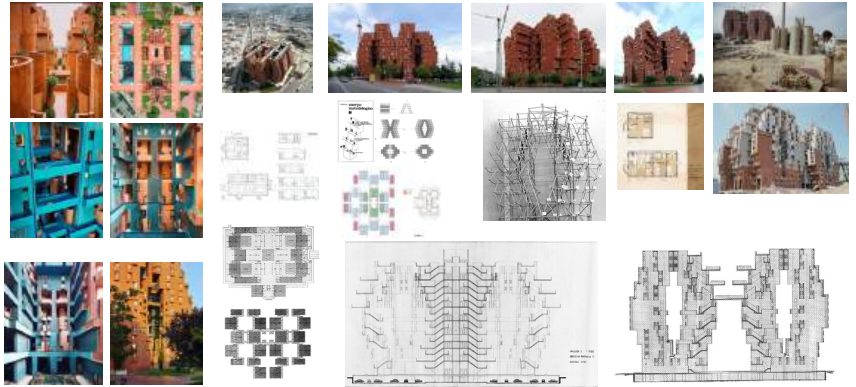
neighborhood in a box

structure / spatil configuration



Walden 7
Ricardo Bofill

Utopian Housing



Location : Sant Just Desvern, Spain.
Land Type: suburban
Site Area: 1.1 hectares
GFA: 31,140 m²

Financial Driver: Experimental Social Subsidization. The project was developed by **CEVSA** (a private developer) but utilized **VPO (Vivienda de Protección Oficial)**—Spanish state-subsidized housing status. This allowed the project to access low-interest state loans in exchange for providing housing to lower-income and middle-class residents.

Profit-Driven Logic: Hybrid / Crisis Management. The original intent was to prove that "utopia" could be profitable through high-density modularity. However, due to the complexity of the geometry and the economic instability of the mid-70s, the developer nearly went bankrupt. The "profit" was ultimately realized in the long-term cultural capital and the high resale value of the unique units today.

Debt Reliance: High & Volatile. The project was plagued by debt during construction. To save costs, the original exterior was finished with ceramic tiles that began to fall off within years due to structural movement and poor adhesive, leading to a massive multi-decade debt cycle for the residents' association to fund the iconic terracotta-colored renovation seen today. those deep inside the block—has access to light and air, mitigating the oppressive density of the 446 apartments.

Unité d'Habitation Le Corbusier's



Location : Marseille, France.
Land Type: Urban
Site Area: 3.5 to 4 hectares
GFA: 50,000 m².

cluster apartment volumetric space

modular plugin

Public

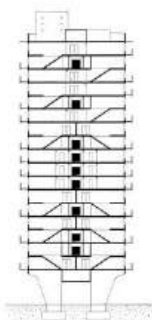


Financial Driver: State-Led Post-War Reconstruction. Funded by the French Ministry of Reconstruction and Urbanism (MRU) to address the severe housing shortage after WWII.

Profit-Driven Logic: Zero-Profit / Social Utility. The building was a "laboratory" for the state. The goal was not to make money but to create a repeatable prototype for "modern man." The "profit" was the health, efficiency, and social harmony of the French citizenry.

Debt Reliance: High (State Bonds). The project was entirely financed by the French government through post-war reconstruction bonds. It was famously expensive to build (over budget and behind schedule), which is why it remained a rare "monument" rather than a mass-produced solution.

Architectural Form: The "Vertical City" on Pilotis. A massive concrete "slab" lifted off the ground by heavy V-shaped pillars (*pilotis*). The structure is essentially a giant "concrete rack" into which prefabricated modular apartments were "slotted."



Living Space Impact: The "Bottle in a Rack" / Interlocking Duplex.

- **The Section:** The most famous innovation is the **interlocking L-shaped duplex**. Two apartments wrap around a central "internal street" (corridor). One apartment goes over the corridor, the other under it. This means every resident has a "double-height" living room and windows on both sides of the building for cross-ventilation.
- **Internal Streets:** The building includes a "commercial street" on the 3rd and 4th floors with shops, a hotel, and a bakery, aiming to eliminate the need to leave the building for daily tasks.
- **Rooftop Public Realm:** The roof is a "communal landscape" featuring a running track, a paddling pool, a kindergarten, and an open-air theater.

Collective Infrastructure as Value

Co-living

Coop Housing at River Spreefeld

Carpaneto Architekten + Fatkoehl Architekten +
BARarchitekten

social ecology
programming



Location : Berlin, Germany
Land Type: former industrial zone
Site Area: Approx. 7,400 m²
GFA: 13,000 m²



Spreefeld: The "Option Room"

This project consists of three buildings on the banks of the Spree river, financed through a "solidarity-based" model where wealthier residents pay slightly more to subsidize those with less.

Financial Driver: Equity-Based Mutualism. The project was funded by the residents themselves, who formed a cooperative to purchase the land from the city. The primary driver was to "de-commodify" the riverfront, preventing it from becoming a private luxury enclave.

Debt Reliance: High (but Socially Structured).

The project relied on long-term bank loans (approx. 70-80% of the total cost), but these are serviced by the reliable, long-term monthly payments of the members. It also utilized "Internal Solidarity Loans" where wealthier members lent money to the coop at low interest to help lower-income members cover their entry equity.

- **Profit-Driven Logic: Zero-Profit / Asset-Lock/ "Cost-Rent."** The cooperative wanted to ensure that the riverfront remained public, which is "lost revenue" in a speculative model but a "social asset" in a co-op.
- **Living Space Impact:** The buildings feature "Option Rooms"—unprogrammed spaces on the ground floor that are rented out cheaply for community use (cafes, workshops, daycare).

Mehr als Wohnen

Duplex Architekten, FuturaFrosch
Müller Sigrist Architekten Pool
Architekten Miroslav Šik

Financial Driver: Collaborative Mutualism.

It was founded by over 50 different Zurich housing cooperatives to celebrate the 100th anniversary of municipal housing. The primary driver is "cost-rent" (Kostenmiete), where rents are calculated solely to cover construction, maintenance, and interest, rather than to generate profit.

Profit-Driven Logic: Non-Profit / Circular Equity.

Any "surplus" is reinvested into a "Solidarity Fund" to subsidize rents for lower-income residents within the same complex. The logic is long-term stability and social diversity rather than short-term capital gains.

Debt Reliance: Low-interest

Like most Swiss cooperatives, it relies on "Patient Capital." They utilize high-leverage but very long-term, low-interest loans often guaranteed by the Federal Office of Housing or the Canton of Zurich. Members also contribute "mandatory shares" (equity), but the bulk is financed through these stable, non-speculative debt instruments.

Architectural Form: The "Urban Quarter".

Instead of a single monolithic block, it is a cluster of 13 distinct buildings with varying heights and facades. The forms are deep-plan and "chunky," designed to maximize volume while maintaining a porous ground floor that connects to public squares and paths. By sharing large portions of the GFA, the project provides a much higher standard of living (including guesthouses, rooftop saunas, and workshops) while keeping the private rent-per-square-meter low.



social/ urban form

structure/
volumetric space

Location : Zurich-Leutschenbach, Switzerland
Land Type: former industrial zone
Site Area: Approx. 40,000 m²
GFA: 60,000 m²

Co-living

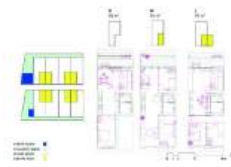
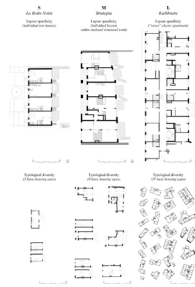


the Social Liquidity



Housing Zollhaus

Enzmann Fischer Partner with Bollhalder Walser Architektur



Location : Zurich (District 5), Switzerland

Land Type: next to central railway tracks

Site Area: 5,000 m²

GFA: 12,500 m²

cluster apartment volumetric space



furniture: physical liquidity

social/ urban programming



Kalkbreite is built directly over a tram depot. Its financial model is "at-cost," meaning rent covers only the loan and maintenance, not a developer's margin.

Financial Logic: Reinvested Cooperative Capital. It was funded largely through the financial success and "solidarity funds" of the original Kalkbreite project. It operates on a **Non-profit Mutualism** model where the goal is long-term urban sustainability rather than financial return. By minimizing individual private space, they keep the "share price" (the amount a resident must pay to join) low.

Profit-Driven Logic: Anti-Speculative / Cost-Rent. The building is a "permanent asset" that cannot be sold on the open market. Rents are strictly calculated based on the actual costs of construction and maintenance (**Kostenmiete**).

Co-ops

Debt Reliance: Moderate (Institutional). Financing came from a mix of resident equity (mandatory shares), cooperative reserves, and low-interest loans from the pension funds of the City of Zurich and the Canton.

- **Living Space Impact: Spatial Liquidity / "Hallenwohnen."** This allows for a much higher **Net-to-Gross** efficiency. The money "saved" on 50 separate stoves and sinks is reinvested into a public roof garden and a cinema.
- **The Hall Apartment:** The most radical feature is the "Hall Dwelling," where 275 m² of double-height space is delivered as an empty shell. Residents use mobile "living towers" on wheels to define their private bedrooms, allowing the floor plan to be "liquid" and change over time.
- **Reduced Footprint:** By providing high-quality shared spaces (roof gardens, professional kitchens, guesthouses), the project achieves a private living area of only **32 m² per person** (significantly lower than the Zurich average), without sacrificing perceived quality of life.

reconfiguring "mobile tower"

Social housing



The Social Palace *Familistère de Guise* Jean-Baptiste André Godin

Location : Guise, France.
Land Type: Industrial Urbanism
Site Area: 4 hectares
GFA: 30,000 m²

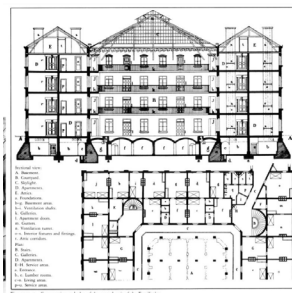
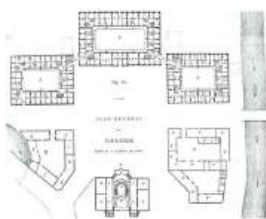
social/ urban form spatial configuration

Financial Driver: Industrial Paternalism & Social Reform. Godin believed that "the equivalents of wealth" (light, air, education, hygiene) should be distributed to the workers who produced the capital.

Profit-Driven Logic: The Association of Capital and Labour. Instead of paying outside shareholders, Godin turned his workers into partners. Profits from the factory were used to buy back the company from Godin himself and distribute shares to the workers. Eventually, the workers collectively owned the factory *and* the Social Palace. If a worker left, they could sell their shares back to the association or to other workers, effectively "liquifying" their housing stake.

Debt Reliance: Zero / Self-Funded. The project was entirely financed by the surplus value generated by the Godin foundry. It was a closed-loop system where industrial success directly funded architectural expansion.

- **Architectural Form: The "Glass-Roofed Phalanstery."** It consists of three large rectangular blocks. Each has a massive central courtyard covered by a glass and iron roof. This created a "temperate" indoor public square where children could play and workers could socialize regardless of the weather.
- **Living Space Impact: Total Social Integration.** Every apartment had access to light and air from two sides (street and courtyard). The "Palace" included a theater, schools, a swimming pool, and a cooperative laundry—amenities that were unthinkable for the working class in the 1850s.



Asset-Backed Security

The Mitchell-Lama program is the "gold standard" of the equity-limiting model. It represents a massive intervention by the state to create a middle-class "buffer" in the urban housing market, decoupling homeownership from speculative profit.

Financial Driver: The Limited-Profit Housing Companies Act (1955). The state and city provided long-term, low-interest mortgages (often as low as 1%) and significant real estate tax exemptions (often 90% abatement) to private developers or non-profit unions in exchange for keeping rents/maintenance low.

Profit-Driven Logic: Limited Profit / Non-Speculative. Profits are capped (traditionally at 6% for rental owners), and for cooperatives, the logic is "**No-Profit**." If a resident sells their unit, they only get back their initial equity plus a small, predetermined percentage. They cannot sell it at market rate. This keeps the unit affordable for the next "middle-income" family on the waiting list.

Debt Reliance: High (State-Backed). The development was financed via massive state bonds. Co-op City famously had a \$400+ million mortgage in the 1960s, which led to a massive rent strike in the 70s when the state tried to raise maintenance costs to cover debt service.

- **Architectural Form: "Tower-in-the-Park" / Brutalist Modernism.** Tall, concrete, and brick towers arranged in clusters with vast lawns, shopping centers, and schools. It utilized "skip-stop" elevators in some buildings to save costs and maximize unit space.
- **Living Space Impact: Stable Middle-Class Sanctuary.** It provides large, standardized apartments with "suburban" levels of light and air within the city limits. However, the sheer scale can lead to a sense of "urban isolation," which the cooperative combats through its own internal security force, newspaper, and community governance.

Co-ops

Equity-Limiting Co-ops *Mitchell-Lama Housing* *(New York)*

Location : Bronx, New York, USA.
Land Type: former site of the Freedomland USA theme park
Site Area: 20% of 130 hectares
GFA: 15,372 residential units



Co-op City (Bronx)



The "Buyout" Risk

Today, many Mitchell-Lama buildings are voting to "buy out" of the program to become market-rate. This turns "illiquid" social housing into "liquid" private assets, which makes the original residents rich but destroys affordable housing for the future.



La Borda: The Timber Democracy residents as architecture cooperative

housing as a social infrastructure

environmental
structure

Location : Barcelona, Spain.
Land Type: former industrial plot
Site Area: 1,000 m²
GFA: 3,000 m²



Democratic



Financial Driver: Ethical Finance & Mutualism. The land was leased from the Barcelona City Council for **75 years** at a low annual fee. The construction was funded primarily by **Coop57** (an ethical finance cooperative) and the residents' own equity. The budget was extremely tight, and the group used "participatory loans" (crowdfunding from the community). To save money, they decided to avoid a multi-level basement (which is incredibly expensive due to excavation).

Profit-Driven Logic: Zero-Profit / Non-Speculative. La Borda is a "right-to-use" (grant-of-use) cooperative on public land. Residents never "own" their units; they own a share of the cooperative. Residents do not own their apartments and cannot sell them. They pay an initial "entry contribution" (approx. €15,000–20,000) and a monthly "use fee" (cost-rent) that covers the mortgage and maintenance. If a resident leaves, they get their entry fee back (inflation-adjusted), but no profit from market appreciation.

Debt Reliance: High but "Socially Secured." The project relied on a €2.5 million loan from Coop57. This debt is serviced by the monthly "use fees" of the 28 units, making it a stable, long-term financial instrument rather than a speculative risk.

The "De-Commodified" Domesticity

Living Space Impact:

- **Shared Infrastructure:** To keep private units affordable, the residents "outsourced" several functions to common areas: a 100m² communal kitchen/dining room, a laundry room, two guest rooms, and a multipurpose space.
- **Modular Growth:** The 28 units (40, 60, and 75 m²) are designed on a modular grid. As families grow or shrink, the "Timber Democracy" allows for some internal walls to be moved or modules to be reallocated, though this is socially rather than physically "liquid" so far. To keep costs low, the apartments were delivered with basic finishes, allowing residents to customize their spaces over time as their own finances allowed.
- **Energy Justice:** By using passive design (no heating/cooling needed) and centralized energy, the project eliminated "energy poverty" for its residents, ensuring that housing costs remain predictable for decades.

Social

Location : Barcelona, Spain.
Land Type: former industrial plot
Site Area: 1,000 m²
GFA: 3,000 m²



Romeo and Juliet apartment buildings Hans Scharoun



wall design

social/ urban form

Financial Driver: Post-War Social Housing Subsidies. The project was financed under the German "Social Housing" (Sozialer Wohnungsbau) laws, which provided low-interest state loans to private developers in exchange for rent-controlled units.

Profit-Driven Logic: Regulated Private Investment. The developer was a private entity, but the "profit" was capped by the state to ensure affordability for the post-war middle class. The logic was long-term stability and urban prestige rather than short-term speculative gains.

Debt Reliance: High (State-Backed). Reconstruction in 1950s Germany relied heavily on government-guaranteed loans to jump-start the housing market after the total collapse of the private banking sector during the war.

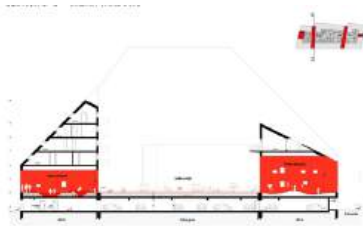
Architectural Form: Organic Expressionism. * **Romeo:** A "jagged," pointed tower with sharp angles and protruding balconies that look like a ship's prow.

- **Julia:** A softer, curving "horseshoe" or crescent-shaped block that creates a semi-private courtyard.

The "Living Landscape." Non-Parallel Walls: Scharoun believed that the human soul is not "square." The rooms have oblique angles to create a sense of dynamic movement and flow.



Location : Bordeaux, France
Land Type: urban regeneration zone
Site Area: 2.5 hectares
GFA: 23,000 m².



Bundled Assets

Ilot Queyries is a landmark project in Bordeaux's Bastide Niel district, designed as a "test-bed" for MVRDV's larger masterplan. It represents a **Hybrid Private-Social model**, where high-density private development is used to deliver a significant volume of social housing and high-quality public space.

Ilot Queyries, France. MVRDV_ porous ground floor



Financial Driver: Private-Public Hybrid. The project was initiated by **Kaufman & Broad** (a major French real estate developer) and **ADIM** (a development subsidiary of the construction giant VINCI). It operates under the French **ZAC (Zone d'Aménagement Concerté)** framework, which mandates a specific percentage of social housing in exchange for development rights.

Profit-Driven Logic: Speculative Cross-Subsidization. The profit is generated by the sale of **145 private market-rate units** and a high-end rooftop restaurant. This profit margin helps offset the costs of the **163 social housing units** integrated into the same block. The logic is "Social Diversity as a Value-Add"—creating an attractive, mixed-income neighborhood to increase the long-term asset value of the site.

Debt Reliance: High (Commercial Loans). As a private development by Kaufman & Broad, it follows a typical commercial debt-to-equity model. The developer borrows heavily to finance construction and repays the debt through the "off-plan" sale of units (VEFA).

Hybrid

environmental

urban form

Architectural Form: The "Sloping Courtyard." The building is an irregularly shaped courtyard block. Its most striking feature is the **45-degree slanted roofs**, calculated parametrically to ensure every unit—and the central courtyard—receives the maximum possible sunlight and ventilation regardless of its position in the block.

Living Space Impact: The Porous Micro-Climate.

- **Porous Ground Floor:** The building is punctuated by "massive portals"—large, colorful openings at the street level that allow the public to see into (and sometimes access) the lush, red-walled interior courtyard.
- **The Elevated Park:** The 5,200 m² courtyard is raised one story above the ground, concealing a massive parking garage beneath it. This keeps the ground level "active" and car-free.



volumetric space

social/ urban form

Location : Motosu, Gifu Prefecture, Japan.
Land Type: a satellite residential area
Site Area: one block in 4.7 hectares
GFA: 5,700 m²

Gifu Kitagata Apartment Homestead SANAA



Public

Unit Granularity

The **Gifu Kitagata Apartment Homestead** is a masterclass in **Spatial Liquidity**. In versus to financial "liquidity" of ownership, SANAA focuses on the "liquidity of the wall"—breaking down the traditional, solid apartment into a transparent, flexible screen.

Financial Driver: Public Infrastructure Investment. Funded by the Gifu Prefectural Government as a "Model Housing Project" to modernize Japanese social housing and move away from the rigid "nDK" (Number of rooms + Dining/Kitchen) post-war standards.

Profit-Driven Logic: Zero-Profit / Social Welfare. The logic is based on **social return**, not financial return. The prefecture provides subsidized housing to low-income residents, elderly citizens, and single parents. The "profit" is the revitalized social fabric and the architectural prestige of the region.

Debt Reliance: Low (Government Funded). Financed through public bonds and taxes. There is no "exit strategy" or need to pay back private mezzanine lenders; the building is a long-term public asset.

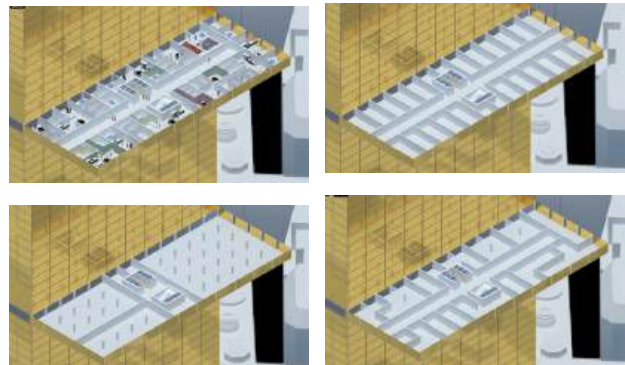
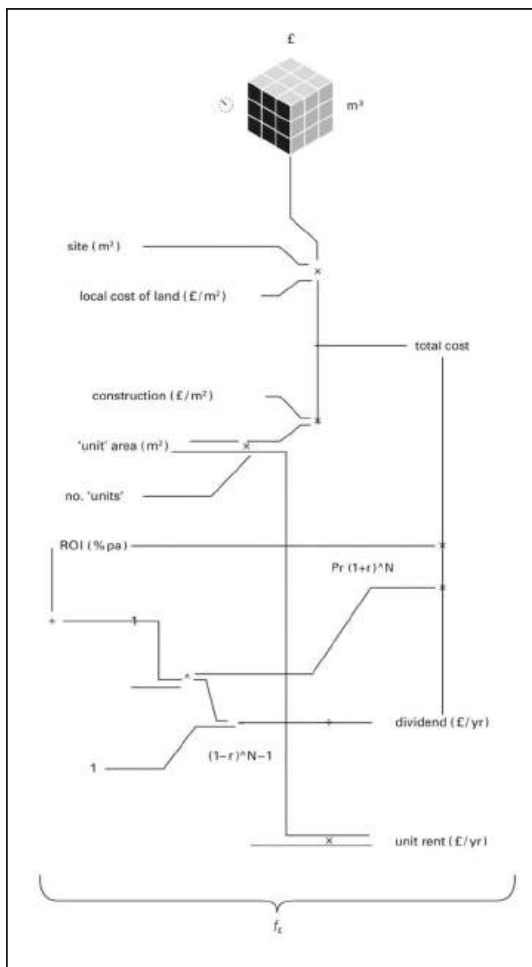
- **Architectural Form: The "Thin Slit" / Linear Block.** A incredibly thin, long, 10-story block (only 2.6 to 5 meters wide in some sections). The building is essentially a vertical "grid" of rooms connected by an external gallery and private terraces.
- **Living Space Impact: The "Room-as-Unit" Model.**
- **The Terrace-Room Sequence:** Each apartment is a sequence of "Room - Terrace - Room." This allows residents to use the outdoor terrace as an extension of the living room, effectively "liquefying" the boundary between inside and outside.
- **Non-Hierarchical Living:** By making every room roughly the same size and transparent, the architecture refuses to dictate how a family should live (e.g., which room is for sleeping vs. eating), allowing the space to adapt to different household structures.

The Ingot



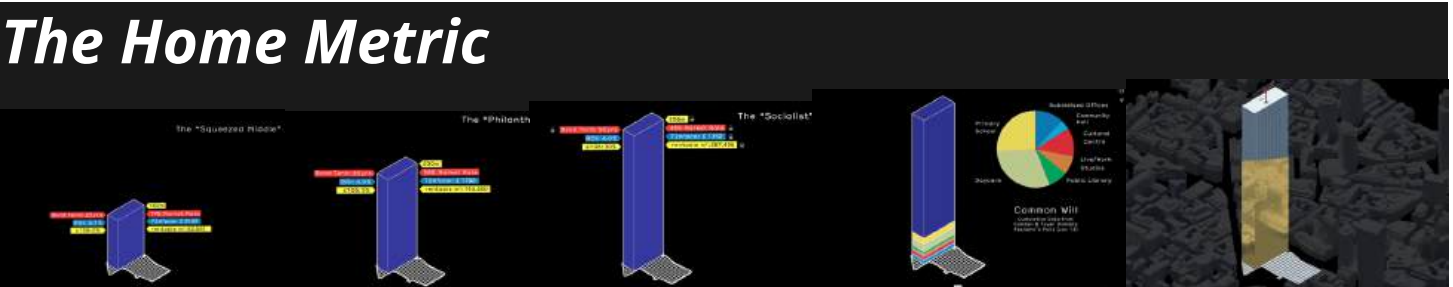
The Ingot project is a proposal for a residential tower in the City of London designed by Jack Self. It serves as a critique of neoliberal property markets, aiming to prove that architecture can be "financially profitable" while simultaneously liberating occupants from the "soul-crushing burden of debt".

Jack Self proposes a model of "**derivative architecture**" where the architect acts as a **financier** rather than just a designer. The project bypasses traditional property developers to eliminate their typical 20% profit margin, which Self identifies as a primary barrier to affordability. Instead of mortgages, the project is funded by **50-year long-term bonds**, similar to how universities raise equity for new buildings.



- **The Model:** Residents do not own the property but retain **usufructuary rights** (the right to use and enjoy the space).
- **The Security:** To hedge against property market crashes, the building's facade is electroplated with **170kg of gold**, making the structure itself a fungible commodity outside the real estate market.
- **Target:** Workers on the **living wage** who are currently priced out of Central London (where rent can exceed 100% of their income).
- **Assumptions:**
 - Inhabitants will accept **never owning** the property in exchange for low, stable rent.
 - Macroeconomic growth depends on low-paid workers living near their workplaces.
 - The architect can deliver the project on time and within budget to maintain "market trust".

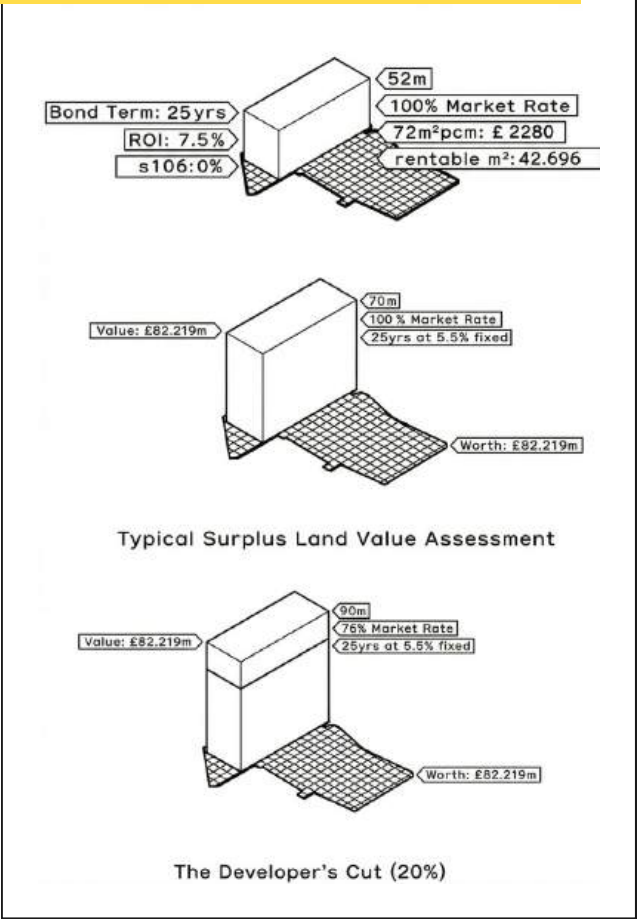
Self uses a parametric algorithm (f_c) to determine the building's form based on financial parameters rather than aesthetic ones.



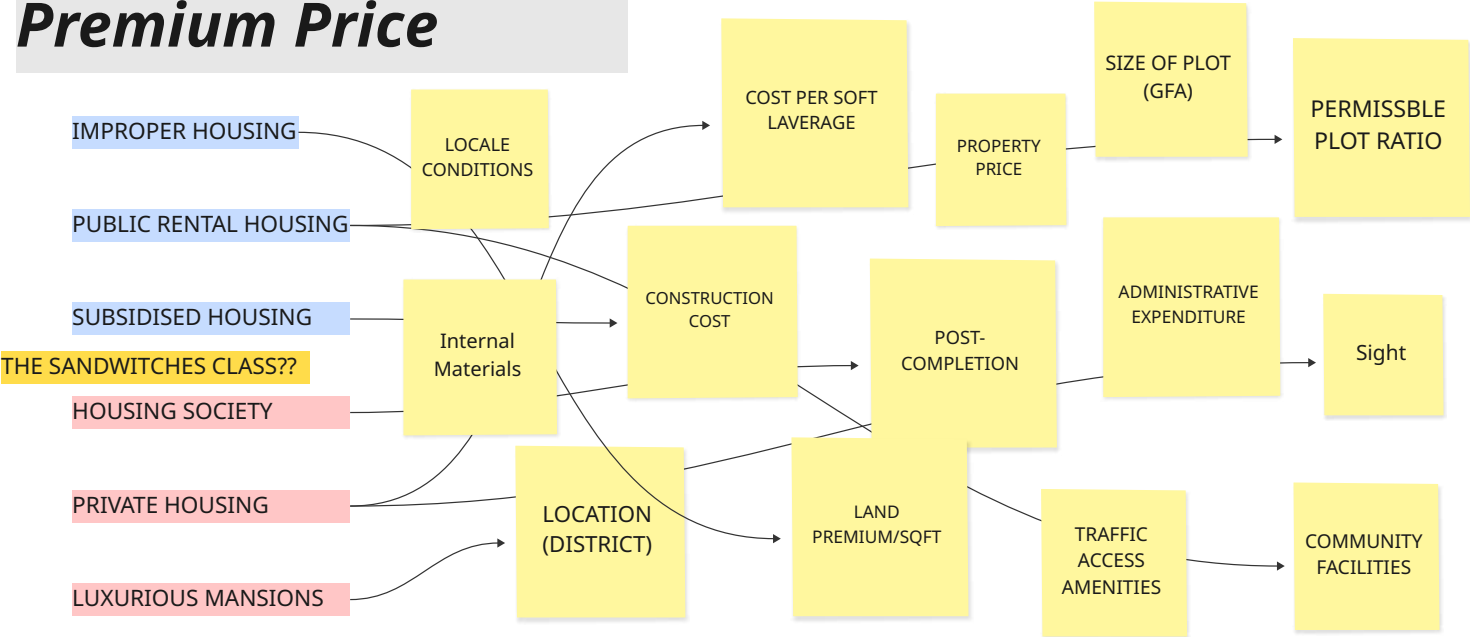
Factor	Jack Self's Quantification & Impact
Surplus Land Value	Land Cost + Construction + Demolition subtracted from Market Value.
Developer Profit	Standard 20% margin is removed to lower end-user costs.
Bond Repayment	Fixed at 4.5% per annum (2.5% interest + 2% principal).
Unit Rent	Set at 46% of market rate to be affordable for living-wage workers.
Space /Massing	Determined by the fixed ratio between construction cost and required rental revenue.

The primary factors underpinning any development are space, time and money. The algorithm attempts to integrate these fiscal, spatiotemporal and juridical parameters into the core of architectural design - pushing the logic of capitalist aesthetic expression and programmatic resolution present in the work of architects such as Ludwig Hilberseimer to their most logical or illogical extreme.

Potential Building Height as social weapon

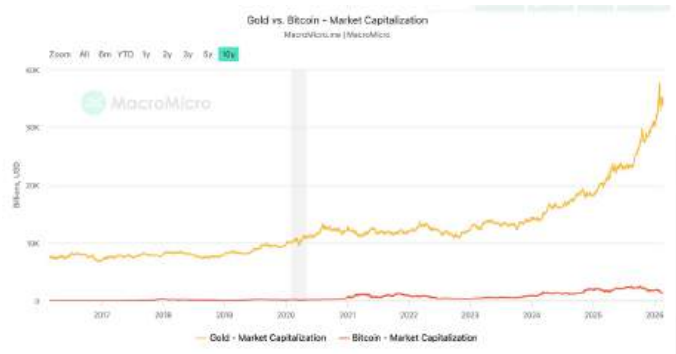


Premium Price



Land in HK are owned by the government. When a plot of land is to be developed, irrespective of scale, a premium has to be paid to the government as the developed value of the plot has increased.

Retaining Value



BITCOIN as
alternative
to GOLD

precious
metals

Material

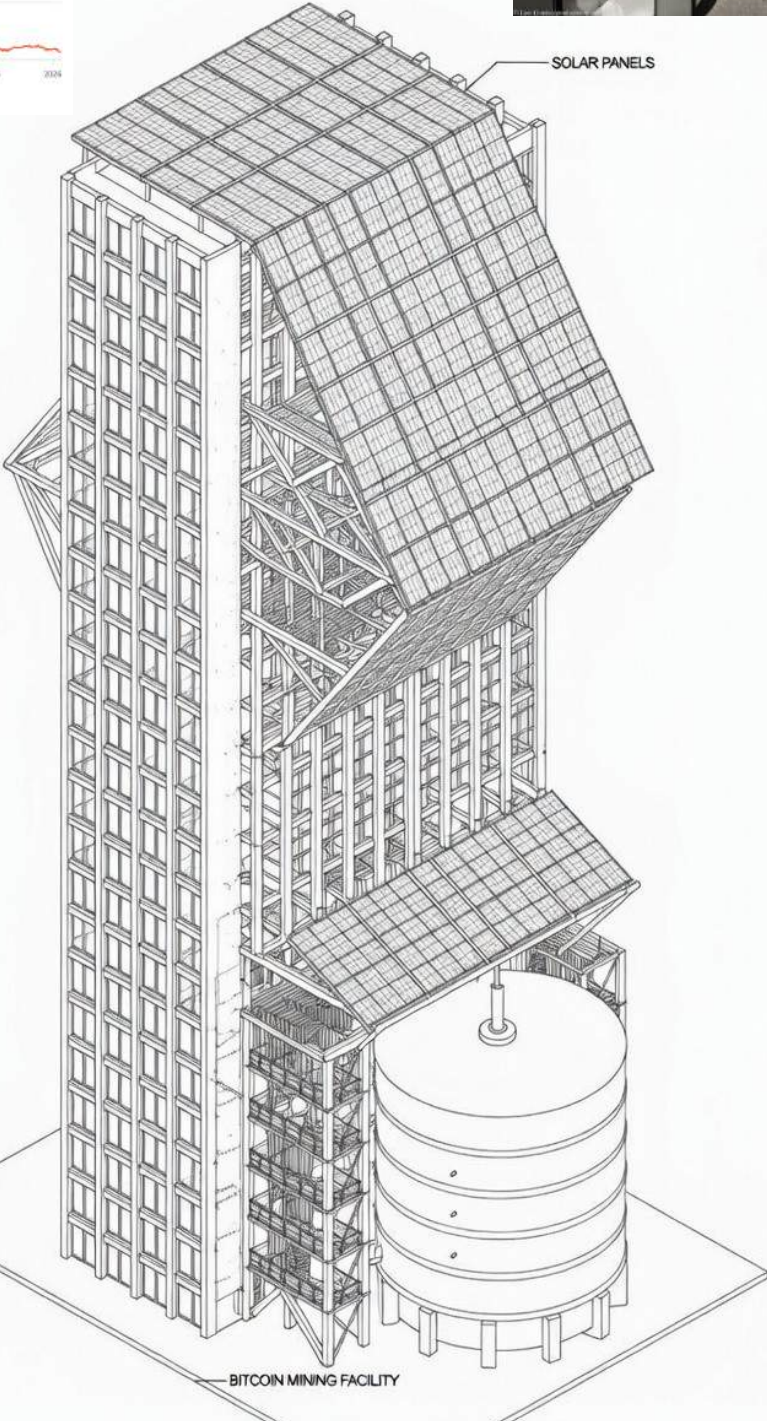


The Debt-Free Horizon. The project is designed to be a "non-profit residential development." It uses a model where the residents' monthly payments are used to pay off the initial construction loan.

The Profit Logic: De-commodification. Once the debt is retired (after 20-30 years), the "rent" for residents drops significantly to cover only maintenance. There is no landlord extracting profit. The building is an "Ingot" because it holds a fixed, unmovable value for the community, rather than a fluctuating value for a market.

Feature	Financial Wealth (111 West)	Energy Wealth (Super Liquid House)
Asset Type	Speculative (Market-driven)	Functional (Utility-driven)
Depreciation	Varies with economy	Follows Laws of Thermodynamics
Liquidity	High (in a boom) / Low (in a bust)	Constant (People always need energy)
Storage	Bank Ledger / Deed	Physical Battery / Thermal Mass / Tokenized Grid

Energy as
wealth
storage



	 Column 1	 Column 2	 Column 3	 Column 4	 Column 5	
1	Group	Variable	Value	Unit	Notes / Logic	
2	A	SITE & POLICY CONSTRAINTS				
3	A1	Site Area	15,000	sq.ft.	[INPUT]	
4	A2	Baseline Plot Ratio (PR)	12.0	Ratio	[INPUT] (Standard high-density)	
5	A3	Site Coverage (SC)	60%	%	[INPUT] (Max permissible)	
6	A4	Max Permissible GFA (Baseline)	180,000	sq.ft.	(A1 * A2)	
7	B	TARGET DEMOGRAPHIC (HK Living Wage)				
8	B1	Target HH Income (HK Living Wage)	21,000	HKD/month	[INPUT] (e.g., District median)	
9	B2	Affordability Threshold	35%	%	[INPUT] (Max % of income for rent)	
10	B3	Target Ingot Monthly Rent	7,350	HKD / month	(B1 * B2)	
11	C	CONSTRUCTION & FINANCE PARAMETERS				
12	C1	Average Unit Size (Gross)	480	sq.ft.	[INPUT] (e.g., ~45 m2)	
13	C2	Construction Cost per sq. ft.	3,800	HKD/sq.ft.	[INPUT] (Standard HK mid-rise)	
14	C3	Demolition/Preparation Cost	25,000,000	HKD	[INPUT] (Fixed)	
15	C4	Institutional Bond Period	50	Years	[INPUT] (Self's university model)	
16	C5	Proposed Bond Annual Coupon Rate	4.0%	% per annum	[INPUT] (2% Interest + 2% Principal)	
17	C6	Development Contingency	5%	%	[INPUT]	
18	D	INGOT MODEL "MAGIC" VALUES				
19	D1	Standard Developer Profit Margin (Removed)	0%	%	Ingot eliminates typical 20% margin	
20	D2	Public Benefit GFA Trade-off	15%	%	% of total GFA offered free to HK Gov	
21	D3	PR Relaxation Multiplier	1.25	Multiplier	The "recursive loop": The bribe logic. 15% Public GFA allows PR boost (e.g., from 12 to 15). (Linked to Design Model sheet)	
+						

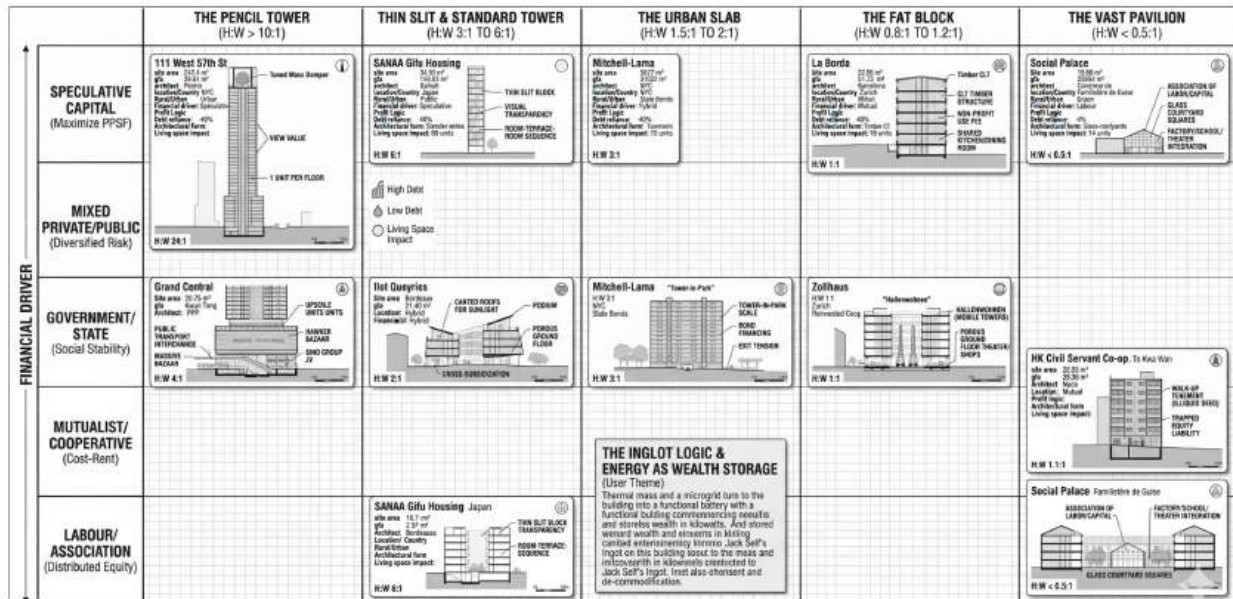
Non-Profit Land Assumption: In Hong Kong, land premium is the absolute barrier. For a non-profit Ingot model, your thesis must argue that the government provides the leasehold land at a nominal rate (e.g., \$HKD 1,000) *because* the project delivers free Public Benefit GFA (F8) and stable non-profit housing. Set E1 to 0 to test that scenario.

	Column 1	Column 2	Column 3	Column 4	Column 5	
1	Group	Variable / Formula	Value	Unit	Logic & Thesis Application	
2	E	CONVENTIONAL DEVELOPER MODEL (The Baseline Control)				
3	E1	Land Premium Estimate (Gov Lease)	1,200,000,000	HKD	[INPUT] or (A4 * market land rate)	
4	E2	Conventional Construction Cost	684,000,000	HKD	(A4 * Dashboard!C2)	
5	E3	Total Development Cost (TDC)	1,909,000,000	HKD	(E1 + E2 + C3) * (1 + C6)	
6	E4	Standard Developer Profit (20%)	381,800,000	HKD	(E3 * 0.20)	
7	E5	Total Market Capitalization	2,290,800,000	HKD	(E3 + E4)	
8	E6	Required Market Monthly Rent (per unit)	15,900	HKD / month	(E5 / (A4 / Dashboard!C1)) / 12 months / 10 years payback	
9	F	INGOT MODEL (Non-Profit Derivative Architecture)				
10	STEP 1: Quantify Affordability & Unit Count					
11	F1	Annual Target Ingot Rent (per unit)	88,200	HKD/year	(Dashboard!B3 * 12)	
12	F2	TDC Required Bond Payment (Annual)	0.04	Factor	(1 / Dashboard!C4) + (Dashboard!C5 - (1/C4))	
13	F3	Max Supportable Construction/Land Cost (per unit)	2,205,000	HKD / unit	(F1 / F2) Crucial metric. The model must deliver a home for this cost (including site).	
14	F4	Target Housing GFA	180,000	sq.ft.	[ITERATIVE LINK] or linked back to PR calculation below.	
15	F5	Number of Housing Units	375	Units	(F4 / Dashboard!C1)	
16	STEP 2: Capitalize Land & Construction					
17	F6	Non-Profit TDC (Land + Const + Demolition)	1,200,000,000	HKD	[INPUT] Total Bond Equity Required. (You must define land cost, perhaps via non-profit negotiation).	
18	F7	GFA Supportable by Bond TDC	315,789	sq. ft.	(F6 / Dashboard!C2) (Assumes 0 land cost to test feasibility, otherwise must include E1 here)	
19	STEP 3: Apply HK Policy Loop (Section 106 equivalent)					
20	F8	Public Benefit GFA Required	47,368	sq. ft.	(F7 * Dashboard!D2) (e.g., elderly care, non-profit school space)	
21	F9	Total Proposed Ingot GFA (Housing + Public)	363,157	sq. ft.	(F7 + F8)	
22	F10	Resulting Ingot Plot Ratio (PR)	24.2	Ratio	(F9 / Dashboard!A1). The financial model generated this PR. Compare it to the baseline A2.	
+						

Iterative Loop: Note that Section F (F4 and F10) forms a recursive loop. The Plot Ratio (F10) *generates* the Total GFA (F9), which *determines* the bond size (F6). In Excel, you can use "Goal Seek" to solve for the PR that delivers the exact TDC you can finance via bonds at the target "affordable rent."

The Architectural Finance Continuum

Height to Width Ratio VS Financial Drivers



Verticality = Speculative Liquidity (Easy to sell, hard to live in).

Horizontality/Bulk = Social Liquidity (Easy to live in, hard to sell).

The "Skin-to-Volume" Tax

Structural Liquidity vs. Financial Slenderness

The "Deep Plan" Opportunity

DERIVATIVE ARCHITECTURE

JACK SELF

If debt is indeed central to understanding, and thus combating, neoliberalism, it is because neoliberalism has, since its emergence, been founded on a logic of debt [which] reduces the future and its possibilities to current power relations... The principal explanation for the strange sensation of living in a society without time, without possibility, without foreseeable rupture, is debt. ¹

Maurizio Lazzarato

Credit must be viewed as far more than a financial arrangement, for it is nothing less than a fundamental dimension of our society, and in effect a new ethical system. ²

Jean Baudrillard

In the spring of 1973 a catastrophic collapse in the property markets wiped three quarters off the value of the London Stock Exchange. Lulled into complacency by steady capital gains during the 1960s, financial institutions had been unscrupulously lending to practically anyone wanting to build a house. The bubble quietly grew for more than a decade because the banks only recorded mortgages as revenue and equity. This created a debt blind spot at the centre of balance sheets – risk they couldn't see, and a scale of exposure they couldn't accurately gauge.

UNREAL ESTATES

MARK CAMPBELL

In 1980, two years after the artist Gordon Matta-Clark died, the city of New York began reappropriating the oddly shaped parcels of land – or 'gutterspaces' – he had assembled as his *Reality Properties: Fake Estates* (1973–4). After receiving a shoebox of loose photographs, receipts, deeds, maps and miscellaneous drawings, his executor had understandably failed to grasp the impetus of the work and as a result neglected to pay the requisite property taxes. 'Several property tax bills arrived at the loft,' Jane Crawford, Matta-Clark's wife, remembered of this inheritance. 'Having had no experience in paying property taxes, or even realising what they were, I ignored them... By 1980, I received several rather nasty letters saying the properties were being confiscated by the city for nonpayment of taxes. Again, I had no idea what they were talking about.'¹ In this context Crawford's misunderstanding is doubly understandable in that it first illustrates how it was only in retrospect that the artist's friends and wider art world realised the significance of what Matta-Clark himself considered his 'art pranks'. Secondly, it offers an example of the bureaucratic banality often associated with commercial property exchanges alongside the necessity for diligent financial maintenance, which he satirised in this project.

Fake Estates was purchased during two property auctions in October 1973 and originally consisted of 15 plots – 14 in the New York City borough of Queens and one on Staten Island. 'The idea of buying property in New York City for \$25 – \$75 was the American dream!' Crawford said, recalling the project's impetus. 'This was a myth that he could really get behind: America had so much land that there was enough for everyone.'² (A generosity that extended to the artwork, with Matta-Clark intending the property deed for each property to be signed over to whomever bought the work.) Like any

To prevent a cascading chain of failures, the Bank of England bailed out three-dozen of the smallest (so-called 'secondary') banks, saving the sector but failing to control inflation, which rapidly hit 25 per cent. Within a year, Britain's GDP had contracted by six per cent. This was a significant event in the history of British property busts, but it was also significant for another reason in that it marked the statistical uncoupling of three previously interrelated factors: house prices, household debt, and wage growth.

Before the crisis, wage growth and house prices had risen together (associated with general increases in post-war production), while household debt had fallen to historic lows (linked to cheap social renting). After the crisis, impacts on industrial output (the 1974 Oil Shock), policy change (the abolishment of fixed rents), and ultimately the economic reform that accompanied globalisation (the transition from industrial capital to financial capital), permanently stalled western wage growth – which still hasn't recovered. Meanwhile, household borrowing started to go up, and by 1975 it had effectively replaced wage growth. This is the origin of the UK's debt addiction, the beginning of the long-term property bubble that ended in 2007, and the foundations upon which a certain form of the neoliberal project was constructed.

For the past 40 years our 'lifestyle' has been funded by credit linked to household equity. The value of this equity is calculated by subtracting a home's total liabilities from its total worth. As property prices went up this number got bigger, since the debt on the property was still tied to its cost at the time of purchase. Banks issued personal loans based on a home's equity, believing that if the debtor defaulted they could sell the house and get back all of their money, which explains the connection between rising house prices and increased debt. However, in order to keep functioning, the system as a whole relied on infinite capital gains in the property markets, to perpetuate the extension of what was actually unsecured credit (Figure B). If that gain were to slow or stop, making a home worth less than the sum of its debt, everyone would be in deep trouble. And that's precisely what happened.

mythology, this value was symbolic rather than actual, and this promise of affordable surplus was negated by the unusability of these 'estates', which the artist described in a 1974 interview as 'so absurd as to ridicule the idea of function'. 'When I brought those properties at the New York City Auction, the description of them that always excited me the most was 'inaccessible'', he said, elaborating, 'What I basically wanted to do was designate spaces that wouldn't be seen and certainly not occupied. Property is so all-pervasive. Everyone's notion of ownership is determined by the use-factor.'³ This sense of determinacy was directly challenged by these 'gutterspaces', which were the 'artefacts of a spatial process' that provided the 'unseen punctuation in the text of physical architecture'.⁴ (In this way, *Fake Estates* further illustrates the artist's fascination with property and provides a 'preamble' to his 'use of this construction world as his raw material'.)

As the critic Pamela M Lee notes in her essential work on the artist, *Object to be Destroyed* (2001), this project was 'organised principally around a business transaction', with the act of purchasing such 'useless and irrelevant' property exposing the 'absurdity of real estate', together with the conflicted rationality of New York City's planning demarcations. In this way, the 'deadpan character' of the photographs, drawings and textual documentation that form the work refer to the 'rhetoric of bureaucracy, of contracts and property rights, deeds and lot numbers',⁵ a rhetoric that Lee concedes 'affords no visual pleasure'. (Indeed, despite its obvious debt to conceptual art, as an aesthetic operation, Matta-Clark's *Fake Estates* offers a less convincing meditation on the absurdities of empirical measurement than that offered by a number of other works, including his godfather Marcel Duchamp's seminal *3 Stoppages Étalon*, 1913 – a work produced by dropping three strings, each measuring one metre in length, from a height of one metre in order to generate a new system of measurement.) This rhetoric of bureaucracy was not only aesthetically limiting, but also conceptually arresting. 'Gordon didn't know what to do with it,' a fellow member of the anarchitecture group recalled. *Fake Estates* 'had become tedious to the point where he was letting it go'.⁶

Chapter 2:

Site and Conditions

ANALYSIS BY DISTRICT (PROJECTED DATA BASED ON TYPICAL MARKETS)

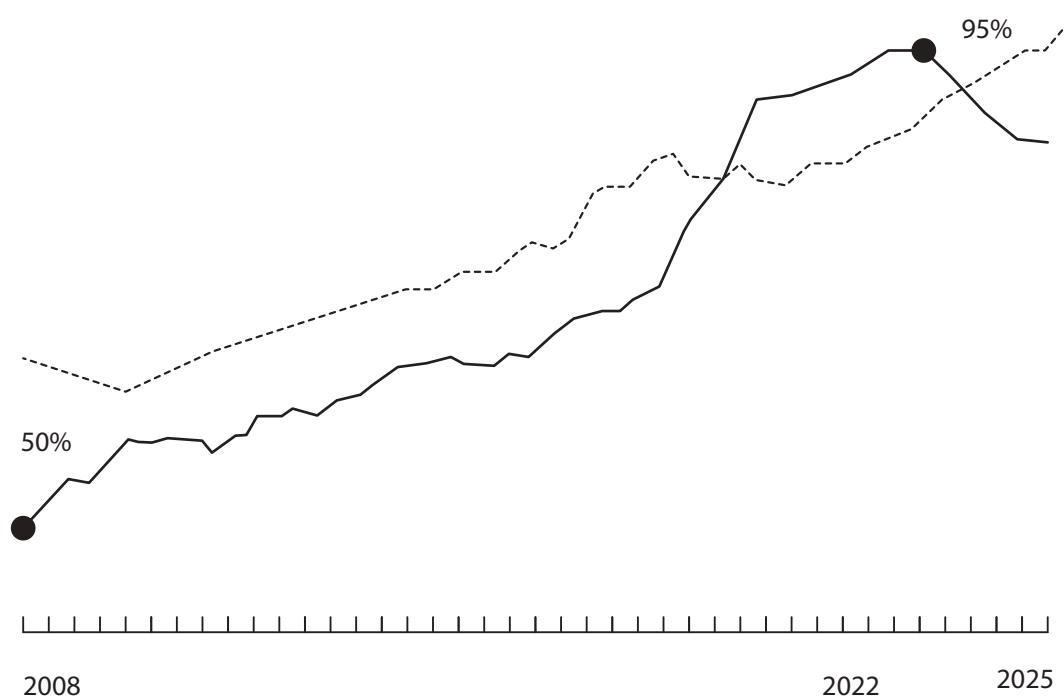
ANALYSIS BY DISTRICT (PROJECTED DATA BASED ON TYPICAL MARKETS)



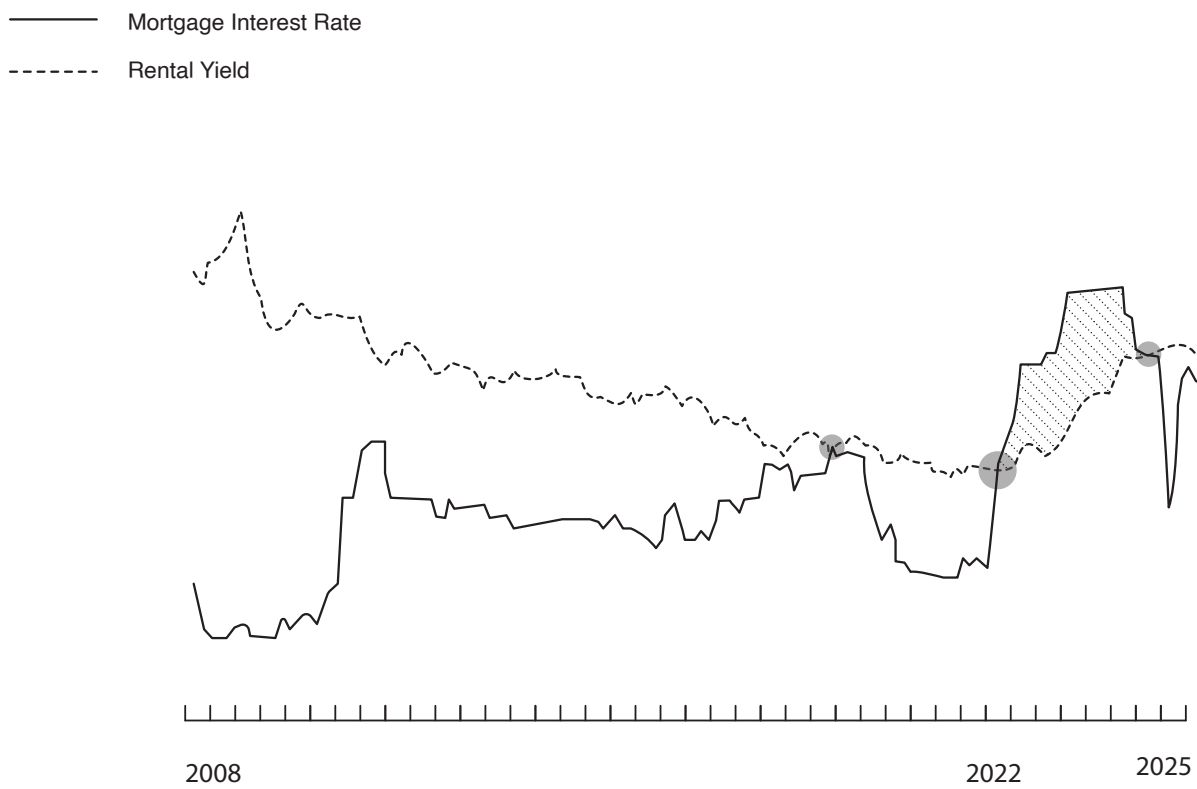
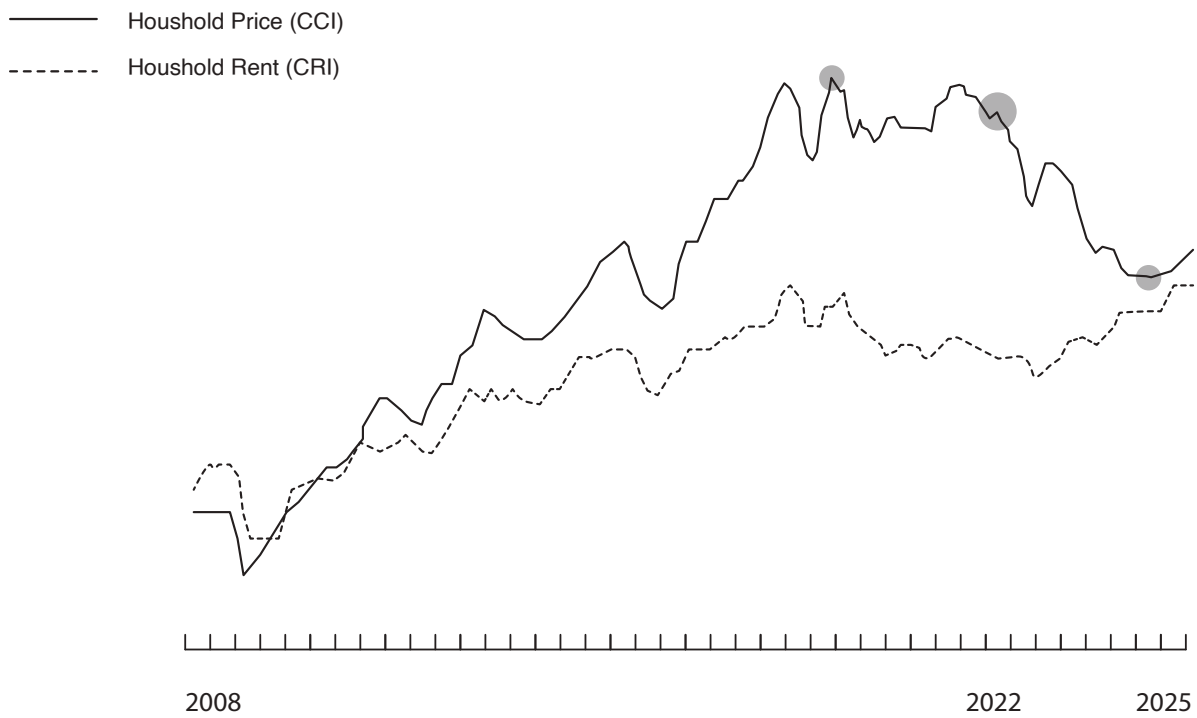
 >60% Burden
 50-60% Burden
 40-50% Burden
 30-40% Burden
 <30% Burden

—— Household debt to GDP

----- Average Wage



Household Debt in Hong Kong





History | “The Jefferson Gird”

“The Public Land Survey System” was proposed by Thomas Jefferson at the end of the 18th century, during the expansion of the US westward. It was an method to divide and **SELL THE OWNERSHIP** of large expanses of **NEWLY ACQUIRED TERRITORIES.** However you plan off the gird, you are within a larger grid which area can be directly correlated to price.

SAMPLE



比例尺 SCALE 1:1 000

米 10 0 10 20 30 40 50 米
metres



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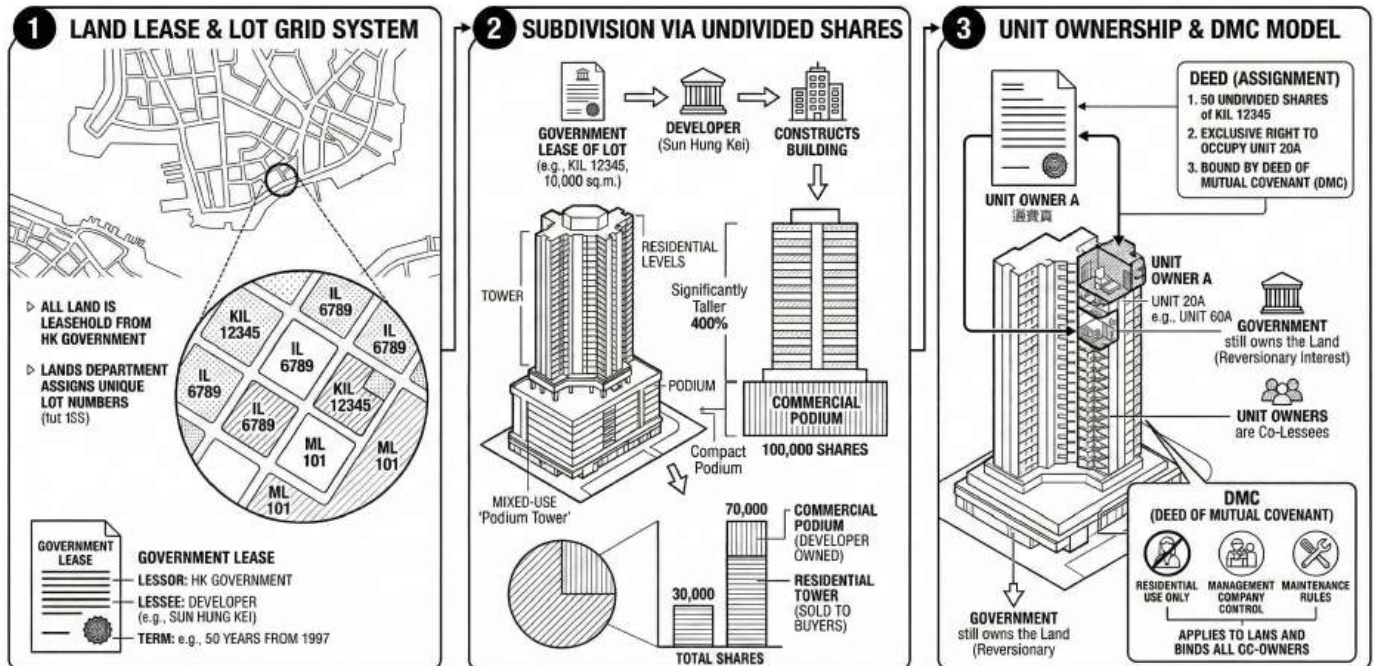
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Explanatory notes: This plan shows the graphical boundaries of different kinds of permanent and temporary land holdings with the topographic map in the backdrop. The land holdings as shown may include private lots, government land allocations, short term tenancies and other permitted uses of land. It must be noted that: (1) the information shown on this plan is subject to update without prior notification, (2) there may be time lag between an update and the related changes taken place, and (3) the graphical boundaries as shown are for identification purpose only and interpretation of their accuracy and reliability requires the advice from professional land surveyor. The Government of Guyana cannot be held responsible for any loss or damage, however arising, from the use of this plan or in reliance upon its correctness, completeness, timeliness or accuracy.

Context | Leased Land Lot in Hong Kong

Hong Kong's **LEASEHOLD SYSTEM** is developed from a colonial model. All land is owned by the Government (except certain cathedrals). When anyone "purchase" land in HK, you are actually leasing it (usually for 50 or 75 years + renewal). Since no one truly "owns" the land, Hong Kong is arguably a **SUBSCRIPTION MODEL**. In principal it is historically set up for tokenization of ownership. Layers of rights were made more granular through fractionanl ownership.

HONG KONG LAND & PROPERTY OWNERSHIP SYSTEM



Ownership | Financing Hong Kong Private Development

The Hong Kong model is defined by hyper-density and the podium-tower typology, where a high-rise residential tower sits atop a commercial podium. Developers typically seek a 20% profit margin, making sites unviable if costs cannot be squeezed or densities increased.

This model optimizes space-saving and efficiency but often results in a “blurred zone” of anonymity and lack of privacy for residents.

For the past 40 years, this system has relied on rising property values to justify high household borrowing, essentially replacing wage growth with debt.

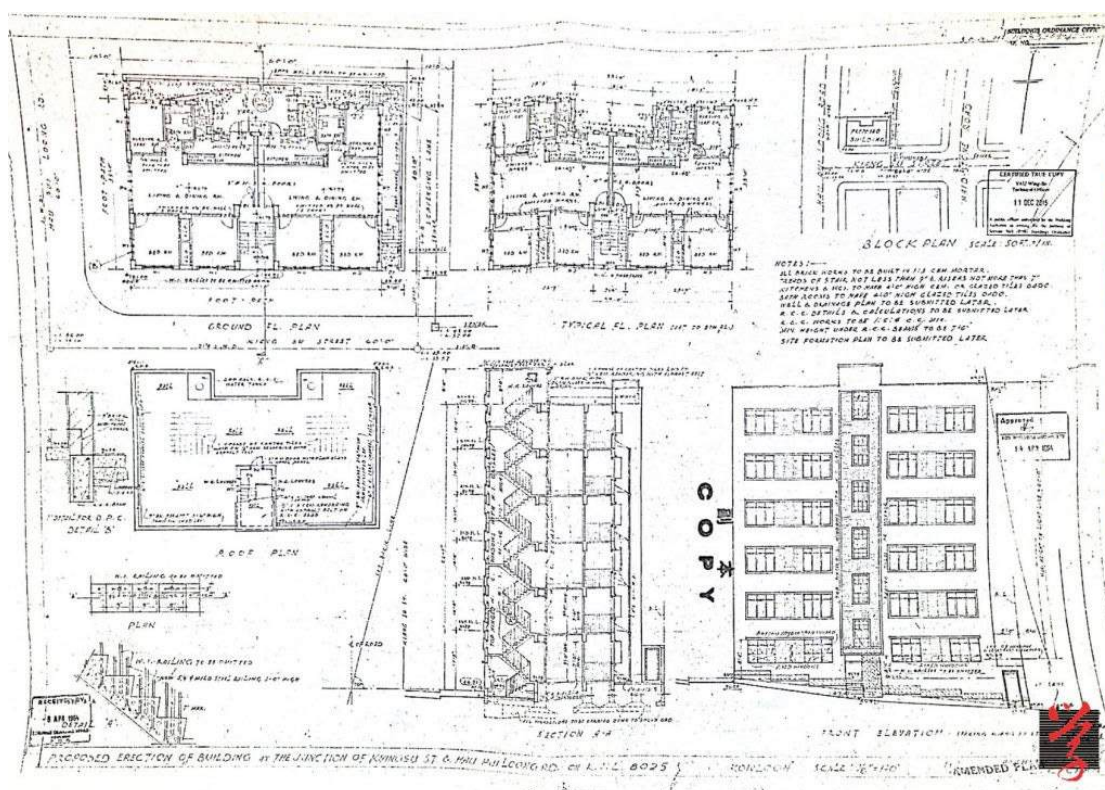
Context | **Civil Servants' Cooperative Building Society (CBS)**

The Civil Servants' Co-operative Building Society (CBS) scheme was introduced by the Hong Kong Government in 1952 to provide affordable housing for middle-ranking civil servants. To fund these projects, the government granted land to newly formed co-operatives at a massive discount—usually just one-third of the normal market price. The government also provided low-interest construction loans, which the civil servants paid back slowly through monthly salary deductions. However, the ownership model of these buildings is highly restrictive and unique. Individual residents do not actually own their specific apartments. Instead, the Co-operative Society owns the entire building and the land. **Residents simply own shares in the society, which gives them the legal right to live in a specific flat.** This collective ownership creates a massive legal trap today. Because the land was originally bought at a steep discount, owners cannot easily sell or redevelop their aging buildings. To sell on the open market, the co-operative must be legally dissolved, and the residents must pay the remaining two-thirds of the **“land premium”** back to the government at today's sky-high market rates. As a result, these elderly residents are land-rich but cash-poor, completely trapped in decaying buildings they cannot afford to redevelop.

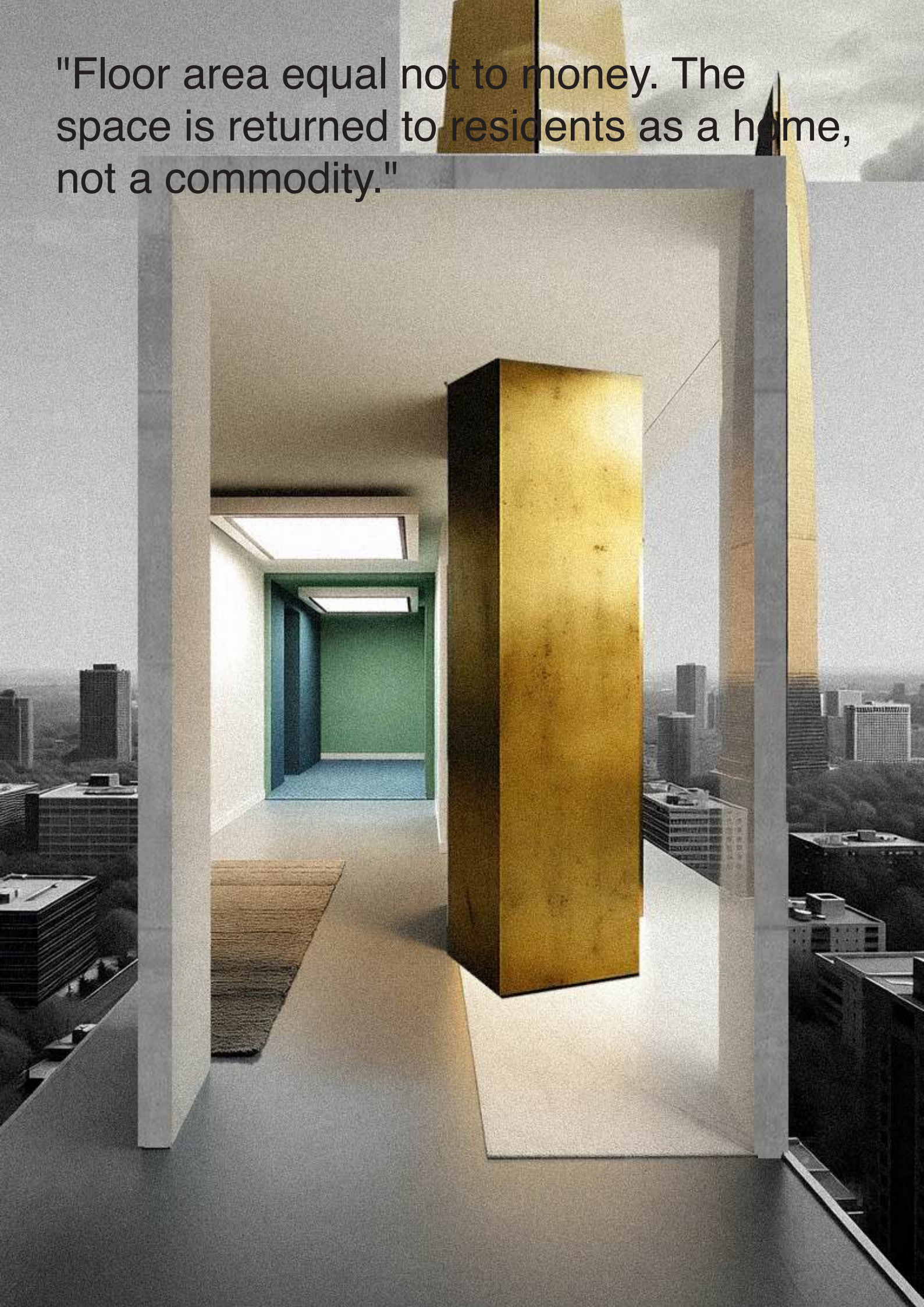
公務員建屋合作社計劃



時期	合作社項目
1945	二戰結束，中國國共內戰開始，大量難民開始湧港
1950	香港人口急增至約 200 萬，房屋需求暴增， 華員會 倡導「公務員合作社建屋計劃」
1956	首個公務員合作社「 西環高級公務員宿舍 」（現改建為 寶翠園 ）落成
1981	首個公務員合作社屋邨「 華員邨 」落成
80 年代中期	計劃終結，期間成立了共 258 個合作社，提供約 5000 個單位
1987	政府 制定機制 容許合作社有條件 取消業權轉讓限制 ，包括全體社員同意解散合作社
1993	政府 放寬機制 至 7 成半社員同意便可解散合作社
2016	186 個合作社已經解散，12 個轄下樓宇被 私人發展商 重建
2020	市建局提出 首個公務員合作社重建收購項目 ，在九龍城區啟動兩個試點，分別為「 馬頭圍盛德街/馬頭涌道發展項目 」（CBS1）以及「 土瓜灣靠背壟道/浙江街發展項目 」（CBS2）



"Floor area equal not to money. The space is returned to residents as a home, not a commodity."



Chapter 3:

Urgency

Urgency | **The 100-Year Bond and the Elasticity of Time**

The temporal framework of Super Liquid Housing operates on two vastly different scales: the immediate crisis of the present and the deep time of the architectural asset. The project is driven by the immediate urgency of the sandwich class, who are currently suffering a loss of place attachment due to transient rental markets.

However, the proposed solution operates on a century-long timeline. The core mechanism is the 100-year spatial bond. When a resident enters the system, they stake a 10% token deposit, leaving the remaining 90% as a century-long debt attached to the volumetric right, not the individual. While a century-long debt cycle may initially appear abstract, it borrows from the economic reality of University Century Bonds (utilized by institutions like Oxford to fund campus expansion).

The lifespan of the project is therefore continuous. The gold-backed Infrastructure Vault is designed for material permanence, acting as an enduring anchor in a volatile city. Conversely, the lightweight living modules operate on highly accelerated, elastic temporalities. Over a 50-year human lifecycle, a resident might expand their unit into the communal grid to accommodate a growing family, and later contract their spatial footprint to sublet to students as they age in place. The time of this project is not static; it breathes with the demographic reality of its inhabitants.



Urgency | **Micro-Temporalities and Human Lifecycles**

While the Vault operates in deep time, the volumetric living units operate on highly elastic, accelerated micro-temporalities. The lifespan of the project is not a singular line from construction to demolition, but a continuous, breathing cycle of expansion and contraction dictated by human life stages. The architecture is designed to respond to a 50-year human lifecycle. At Year 1, a young resident stakes their tokens and occupies a minimal core unit, leaving the majority of their designated spatial grid to the communal pool. By Year 20, as their family structure expands, their spatial boundary flexes outward, absorbing communal space to build a nursery. By Year 50, as the resident ages and children depart, the timeline contracts. The architecture allows them to “de-stake,” shrinking their volumetric footprint and subletting the released space to a new generation of youth. In this model, the architecture never truly “finishes.” The anticipated duration of existence for the Vault is indefinite, but the internal configuration changes daily, monthly, and yearly. The urgency of the project, therefore, is not merely to build a static shelter for today, but to establish an algorithmic and spatial clockwork that can continuously recalibrate itself to the demographic and economic realities of the next century.

Chapter 4:

Agents

Agents | **Dismantling the Traditional Triad**

Historically, the profession of architecture has operated within a rigid, top-down hierarchy of agents: the Client (who holds the capital), the Architect (who translates capital into form), and the User (who passively consumes the form). In the hyper-commodified context of Hong Kong, this triad is heavily skewed. The Developer acts as the ultimate client, dictating spatial minimums to maximize gross floor area, while the sandwich class resident is reduced to a deeply leveraged consumer with zero agency over their built environment. Super Liquid Housing entirely dismantles this traditional hierarchy, replacing it with a decentralized, multi-agent ecosystem. The significance of this project lies in its radical re-assignment of usership, shifting power dynamics away from monopolistic developers and empowering a new network of active stakeholders.

In this thesis, the architect transitions from a “form-maker” to a “spatial economist” and “protocol designer.” The role of the architect is to design the rules of engagement between the competing agents. The most critical intervention by the architect in this model is not the facade, but the Social Constitution—the hardcoded rules that protect the residents from the yield-seeking behavior of the retail investors.

The project’s primary users are the sandwich class. Under an ideal financial model, they are no longer passive debtors; but “Occupants” of voids.

In this section, an architectural white paper is written as initial design tool to develop a partial ownership and fractional financial model.

Hong Kong's 'sandwich class' flats are in the spotlight as buyers snap up Tsing Yi abode for fear of missing out on property bull run

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Sandwich Class Housing Scheme

Properties developed under the Sandwich Class Housing Scheme:



Bel Air Heights

Completion Year Location
1999 Diamond Hill



Cascades

Completion Year Location
1998 Homantin



Hibiscus Park

Completion Year Location
1998 Kwai Fong



Highland Park

Completion Year Location
1998 Kwai Chung



Marina Habitat

Completion Year Location
1998 Ap Lei Chau



Park Belvedere

Completion Year Location
1998 Ma On Shan



Radiant Towers

Completion Year Location
1998 Tseung Kwan O



Sunshine Grove

Completion Year Location
1998 Shatin



The Pinnacle

Completion Year Location
1998 Tseung Kwan O



Tivoli Garden

Completion Year Location
1995 Tsing Yi



**HONG KONG
HOUSING SOCIETY**
香港房屋協會

Housing Society as provider of affordable housing to sandwich class resident.

SUPER LIQUID HOUSING

Redefining Spatial Politics & Ownership in Hong Kong

*Architecture Thesis White Paper
Proposed Financial & Spatial Model*

EXECUTIVE SUMMARY

The "Super Liquid Housing" project addresses the systemic failure of the housing market for Hong Kong's "sandwich class"—the demographic earning too much for public housing but priced out of the private market. Trapped in a cycle of heavy debt and renting, this demographic faces a total loss of place attachment and spatial autonomy. This thesis proposes a radical decoupling of *financial asset value* from *spatial use value*. By establishing a permanent superstructure backed by tangible assets (gold/rare metals) and tokenizing its liquidity, the project introduces a 100-year spatial bond. This mechanism redefines the concept of home: spatial configurations become entirely expandable, property speculation is eliminated for the resident, and housing is returned to its primary function as a living space rather than a financial commodity.

1. The Context: The Sandwich Class Crisis

In high-density urban environments like Hong Kong, real estate relies on a fundamental fusion of the structure, the land, and the volumetric air rights into a single commodified asset. For the sandwich class, purchasing a home requires committing to decades of crippling debt, forcing many to abandon ownership and choose the rental market. However, renting strips residents of spatial autonomy and permanence, creating a transient population disconnected from communal development.

Traditional cooperative housing fails because management structures are rigid, and the spaces offer little financial incentive or liquidity to sustain operations without compromising affordability. **Super Liquid Housing** introduces a financial architecture that injects massive liquidity into the housing model without passing the speculative burden onto the resident.

2. The Architectural Core: The Vault & The Units

The physical manifestation of this economic model relies on a distinct separation of building components, inspired heavily by Jack Self's *Ingot* project.

THE INFRASTRUCTURE VAULT (THE ASSET)

The developer constructs a permanent, heavy superstructure. To clarify, this does not imply literal solid gold columns; the physical structure is engineered from hyper-durable, standardized materials such as high-grade steel and carbon-infused concrete to ensure a century-long lifespan. However, to divorce its value from the volatility of localized real estate speculation, the core material value of this vault is financially hedged against a reserve of gold or rare metals held by the Housing Trust. This permanence ensures that the superstructure itself retains intrinsic value over centuries. It acts not as a building, but as a physical vault of value—a highly liquid asset pool.

THE EXPANDABLE VOLUMETRIC UNITS (THE SPACE)

Within this permanent infrastructure vault, living spaces are deployed as lightweight, adaptable volumetric units. These units do not carry the structural or land-cost burden of the vault. Floor area no longer equates directly to monetary value. The space is returned to the resident purely as a *home*. Because the structural asset is secured and the physical boundaries are unbound by traditional property lines, the volumetric unit is completely free to grow, adapting and expanding based on human lifecycle needs.

"Floor area equal not to money. The space is returned to residents as a home, not a commodity."

3. The Financial Flow Engine

To realize this, the traditional developer-to-buyer pipeline is replaced by a closed-loop tokenized ecosystem mediated by a central Housing Trust.

Stakeholder	Role & Input	Aims & Returns
The Developer	Builds the gold-backed infrastructure vault and immediate connectivity.	Sells the complete structure to the Housing Trust. Gains immediate exit liquidity and standard profit margins.
The Housing Trust	Holds the physical asset. Issues Real World Asset (RWA) tokens and spatial bonds.	Acts as the liquidity pool and buffer. Absorbs debt, manages fractionalization, and resets spatial leases.
Retail Investors	Purchases tokens for financial yield. Has no physical ownership of spatial units.	High mobility and liquidity. Earns yield from resident "rents" (debt service). Gains governance rights over communal spaces if tokens are locked up.
The Residents	Pays 10% entry deposit (staking) and services a 100-year spatial bond.	Gains lifelong spatial autonomy. Low entry cost. 50% private / 50% communal space. Recovers un-inflated deposit upon exit (no profit).

THE 100-YEAR SPATIAL BOND & THE UNIVERSITY PRECEDENT

When a resident enters the system, they stake a token equivalent to a 10% deposit. The remaining 90% is structured as a 100-year long-term bond. Crucially, **this debt is attached to the spatial volumetric right, not the individual resident**. The resident's monthly "rent" is actually the servicing of this bond.

While a century-long debt cycle may initially appear burdensome for an individual, it operates on the same financial logic as a **University Century Bond**. Elite institutions (such as Oxford or Rutgers) issue 100-year bonds to fund physical campus expansion because the institution outlasts any single human lifespan. In this architectural proposal, the Housing Trust acts as the enduring institution, and the permanent Armature serves as the "campus." The retail investors purchase these bonds because they trust the 100-year viability of the gold-hedged asset, while the resident benefits from drastically suppressed monthly payments spread over a century, providing unprecedented affordability.

4. Spatial Mechanics: The 50/50 Rule & Human Lifecycles

The token staked by the resident grants them a volumetric allocation within the vault. However, to foster community and prevent the privatization of the entire infrastructure, a strict 50/50 ratio is enforced:

- **50% Private Ownership:** The enclosed, personal living space (the volumetric unit).
- **50% Communal Space:** Shared public space immediately adjacent to or within their designated structural zone, managed by the spatial right holders.

Because the architecture is unbound and the debt is attached to the volume, residents can dynamically adapt their spatial boundaries over time. This generates several distinct lifecycle scenarios:

Scenario A: The Expanding Nucleus (Growing Family)

The Catalyst: A young couple who initially staked tokens for a modest volumetric unit are expecting children and require more space.

The Action: Rather than selling and moving to a new district, they incrementally expand their private unit boundary into their 50% communal allocation. Their structural footprint within the vault remains identical, but the ratio of private to shared space shifts. They retain their spatial autonomy and community ties without taking on a new, highly inflated mortgage.

Scenario B: Multi-Generational Consolidation (Family Structure Change)

The Catalyst: An aging parent requires daily care but desires independence. The core family wishes to integrate them without severely compromising either party's privacy.

The Action: Instead of forcing the parent into an institutional care facility, the family stakes an additional 10% deposit to acquire an adjacent volumetric unit within the vault. The architecture allows them to merge the two units. By combining their 50% communal allocations, they create a shared, semi-private intergenerational courtyard between the two private living quarters. The family structure changes smoothly without the friction of traditional real estate transactions.

Scenario C: Aging in Place & The Micro-Economy

The Catalyst: An elderly resident's children have moved out, leaving them with excess space that is difficult to maintain and financially burdensome to heat and clean.

The Action: Rather than selling the home and facing displacement from their long-term community, the resident retracts their private volumetric boundary by 40%. They release this 40% spatial right back to the Housing Trust. The Trust absorbs that portion of the bond, immediately lowering the resident's monthly "rent" (bond servicing). The Trust can then re-issue that fractional volume to a young university student. The elderly resident ages in place with reduced financial strain, while fostering a micro-economy and intergenerational social connection.

5. The Exit Liquidity Mechanism

A central challenge in traditional housing is the exit. When a resident wishes to leave Super Liquid Housing, they do not sell their flat on an open market. Instead, they "un-stake" their token.

The Housing Trust pays back their initial 10% deposit (adjusted for inflation, but zero speculative profit). Because the 100-year bond is attached to the volume, the Trust absorbs the remainder of the debt back into the liquidity pool. The space immediately reverts to a liquid asset, ready to be fractionalized, merged, or issued to a new resident on a new bond cycle. This protects the resident from negative equity while ensuring the building remains continually occupied and financially viable.

6. Urban Governance & Retail Tokenization

While residents manage the micro-spatial politics, retail investors drive macro-urban evolution. Retail investors hold tradable tokens backed by the trust. They seek profit (from bond yields) but are given a unique urban hook: **Governance through Lock-up.**

If retail investors choose to lock up their tokens for an extended period, they gain voting and developmental rights over the macro communal spaces. This creates dynamic urban scenarios:

Urban Evolution Scenario: The Infrastructure Bridge

A consortium of retail investors notices high foot traffic near a specific vault block. They pool their locked tokens to propose and fund a new infrastructure sky-bridge connecting the vault to an adjacent MTR station, alongside a coffee shop in the communal zone. The intervention increases the foot traffic and desirability of the block, thereby increasing the overall yield of the liquidity pool. The investors profit from the commercial venture, while residents benefit from improved urban connectivity without bearing the cost.

7. Regulatory Friction: The "Hardcoded" Void & Statutory Compliance

A critical vulnerability in traditional adaptable architecture is the rigidity of statutory building codes. Hong Kong enforces some of the strictest building regulations globally, particularly concerning natural light, ventilation, and fire safety. In a standard development, any alteration to a unit's boundary requires a massive and costly resubmission to the Buildings Department (BD). If residents and retail investors are continuously shifting spatial boundaries, how does the building remain legal?

The solution lies in the dimensional logic of the infrastructure vault. The superstructure is not merely a physical and financial asset; it acts as a **Statutory Buffer**.

PRESCRIPTIVE ZONING & PNAP APP-130

To bypass the need for constant regulatory resubmission, the legal requirements for light, air, and egress are structurally "hardcoded" into the vault itself.

According to the Hong Kong Buildings Department's Practice Note for Authorized Persons (PNAP APP-130), habitable rooms must satisfy specific Unobstructed Vision Area (UVA) formulas and cross-ventilation criteria. Rather than relying on the volumetric units to calculate these angles, the permanent vault is designed with fixed, un-fillable volumetric voids.

These permanent voids guarantee that no matter how the volumetric unit expands, contracts, or fractionalizes within its designated structural bay, it will automatically inherit the necessary light and air angles.

8. The Urban Testbed: To Kwa Wan CBS Redevelopment

To ground this economic theory in physical reality, the project targets the **Civil Servants' Co-operative Building Society (CBS)** schemes in To Kwa Wan (Ma Tau Kok) as the primary urban testbed.

Built between the 1950s and 1970s, these low-density CBS buildings represent a failed 20th-century cooperative model. Today, they are paralyzed by aging infrastructure, elderly residents aging in place without elevators, and a legal gridlock: to redevelop the site, 100% of owners must agree to dissolve the co-op and pay a massive "land premium" (補地價) back to the government before selling to a private developer or the Urban Renewal Authority (URA).

The **Super Liquid Housing** model serves as the perfect bureaucratic and architectural intervention for this paralysis:

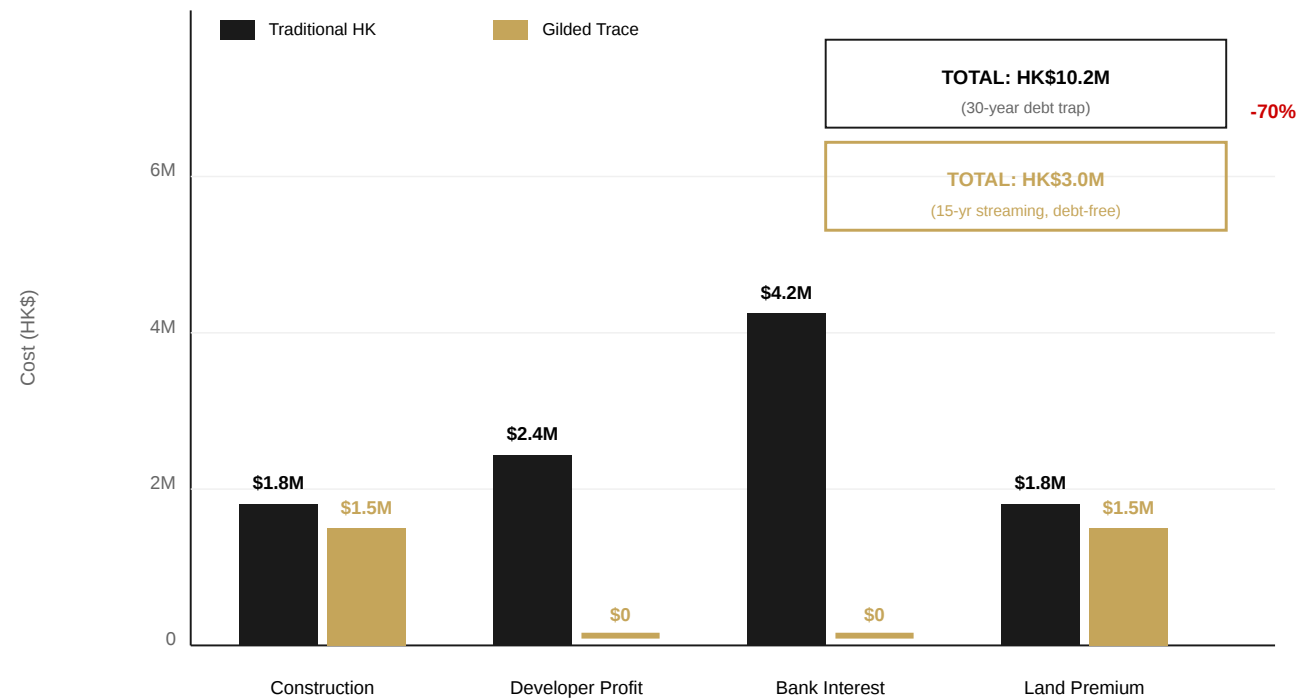
- **Unlocking the Land Premium:** The tokenized Housing Trust acts as the central capital pool, absorbing the exorbitant land premium required by the government, effectively unlocking the site for high-density redevelopment without displacing the original owners.
- **Transitioning the Residents:** The aging civil servants surrender their existing rigid flats in exchange for staked tokens in the new Armature. Rather than being bought out and displaced, they transition into new Expandable Volumetric Units on the same site.
- **Aging in Place (Scenario C):** Because they no longer need large family flats, these original residents utilize the architecture's elasticity to shrink their spatial footprint. The newly freed volume, combined with the radically increased density of the new armature, is injected with retail-backed productive infrastructure (micro-economies, student housing, public amenities), injecting immediate liquid capital and youth back into the aging To Kwa Wan district.

9. Conclusion

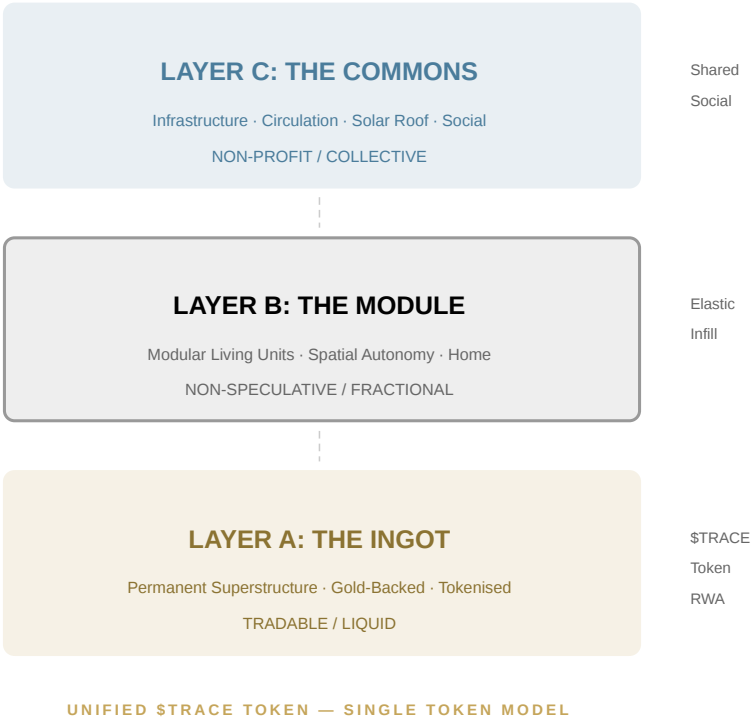
Super Liquid Housing fundamentally rewrites the spatial politics of Hong Kong. By displacing the financial burden from the sandwich class resident to a massive, tokenized liquidity pool backed by tangible assets, the architecture is freed from the constraints of commodity. The building becomes a living, breathing organism—expanding and contracting to fit the demographic reality of its inhabitants, driven by a sustainable engine of retail investment and permanent structural value.

In this paradigm, the resident is no longer a debtor trapped in a concrete box; they are active participants in a hyper-flexible, democratic spatial framework.

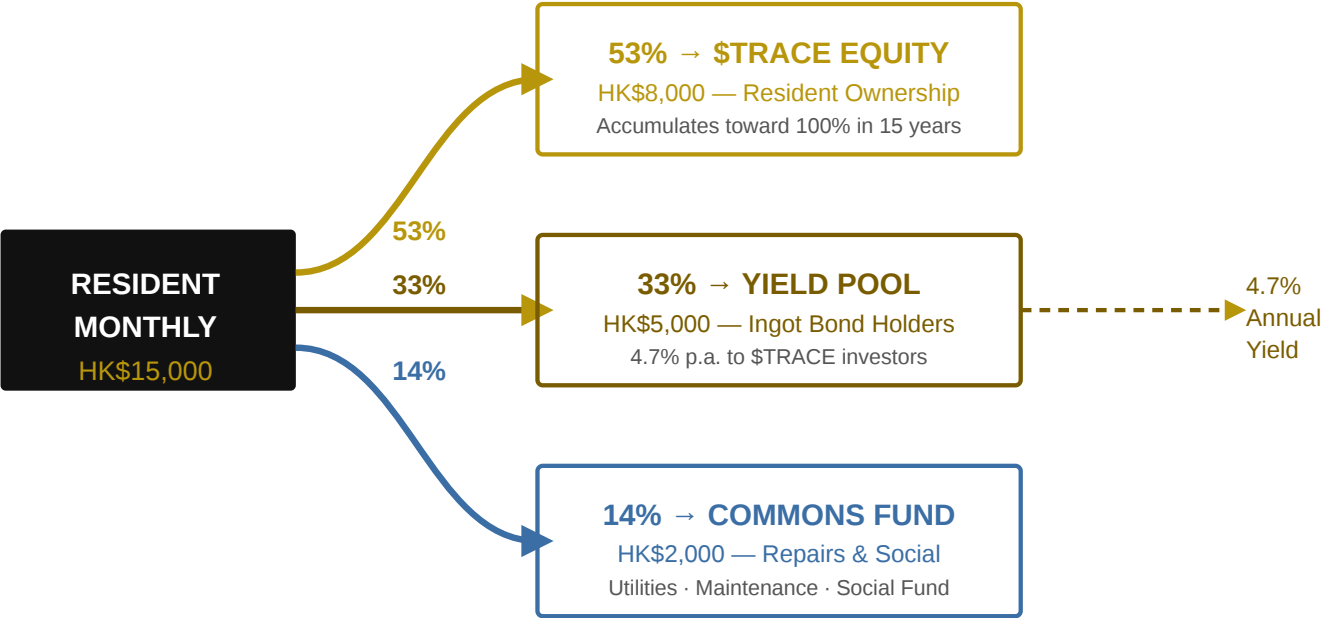
WHERE DOES YOUR MONEY GO?



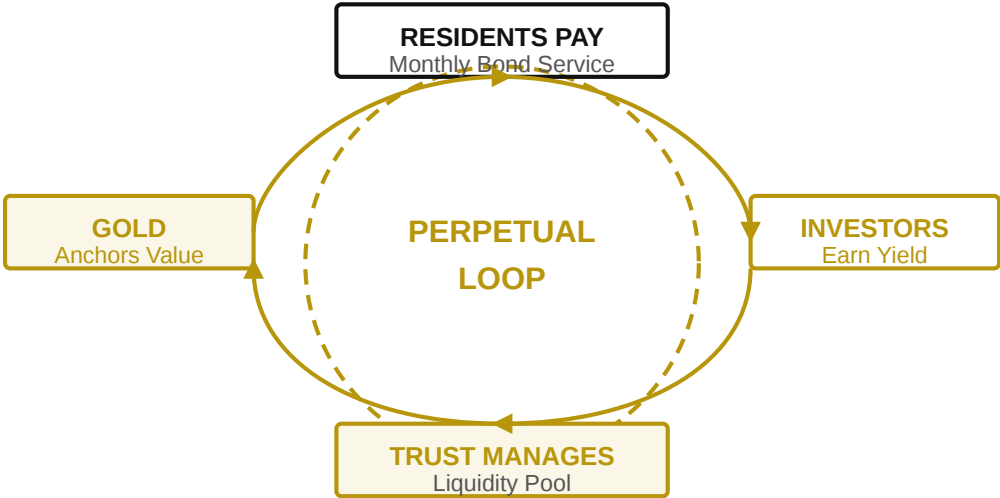
SCAFFOLD & INFILL



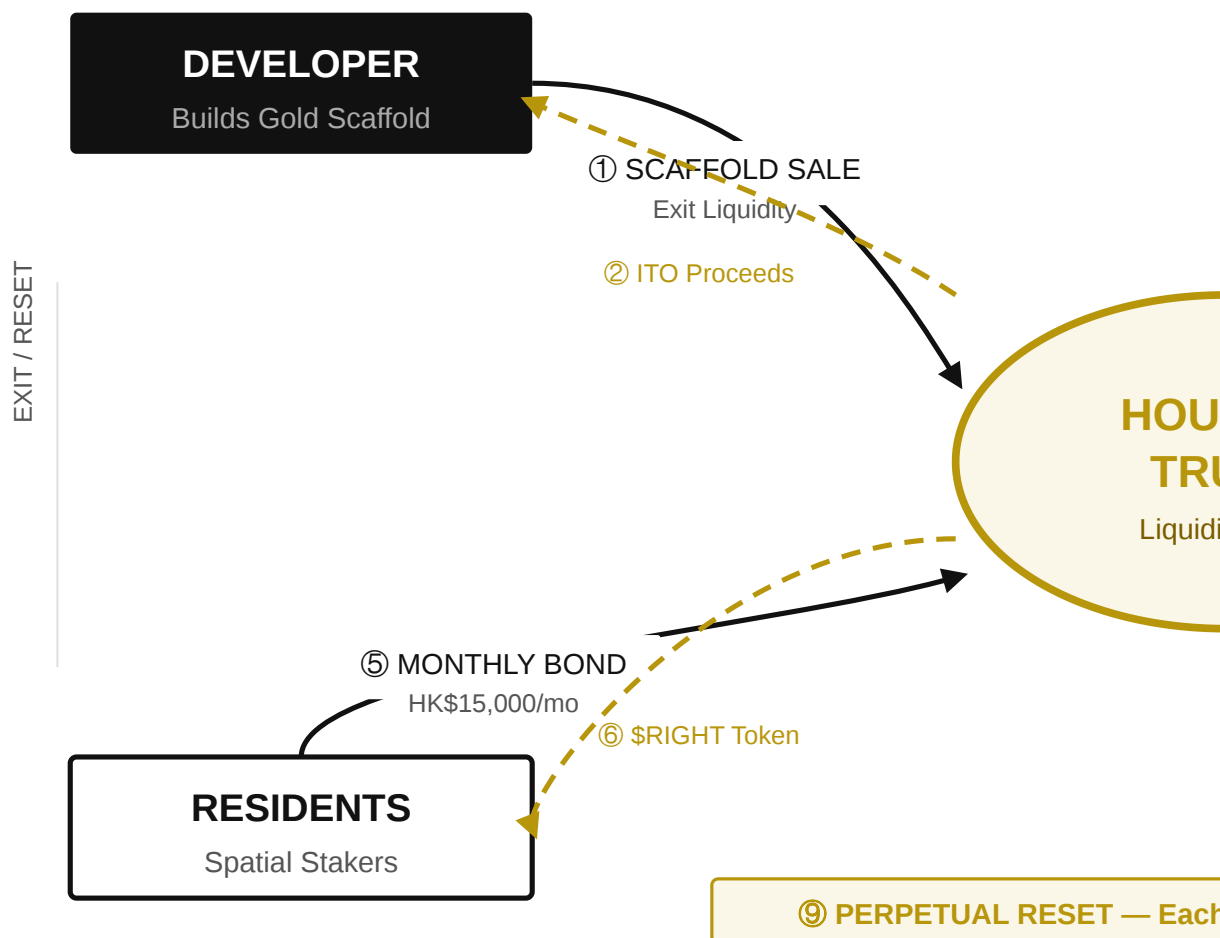
DECOUPLING ASSET FROM HOME



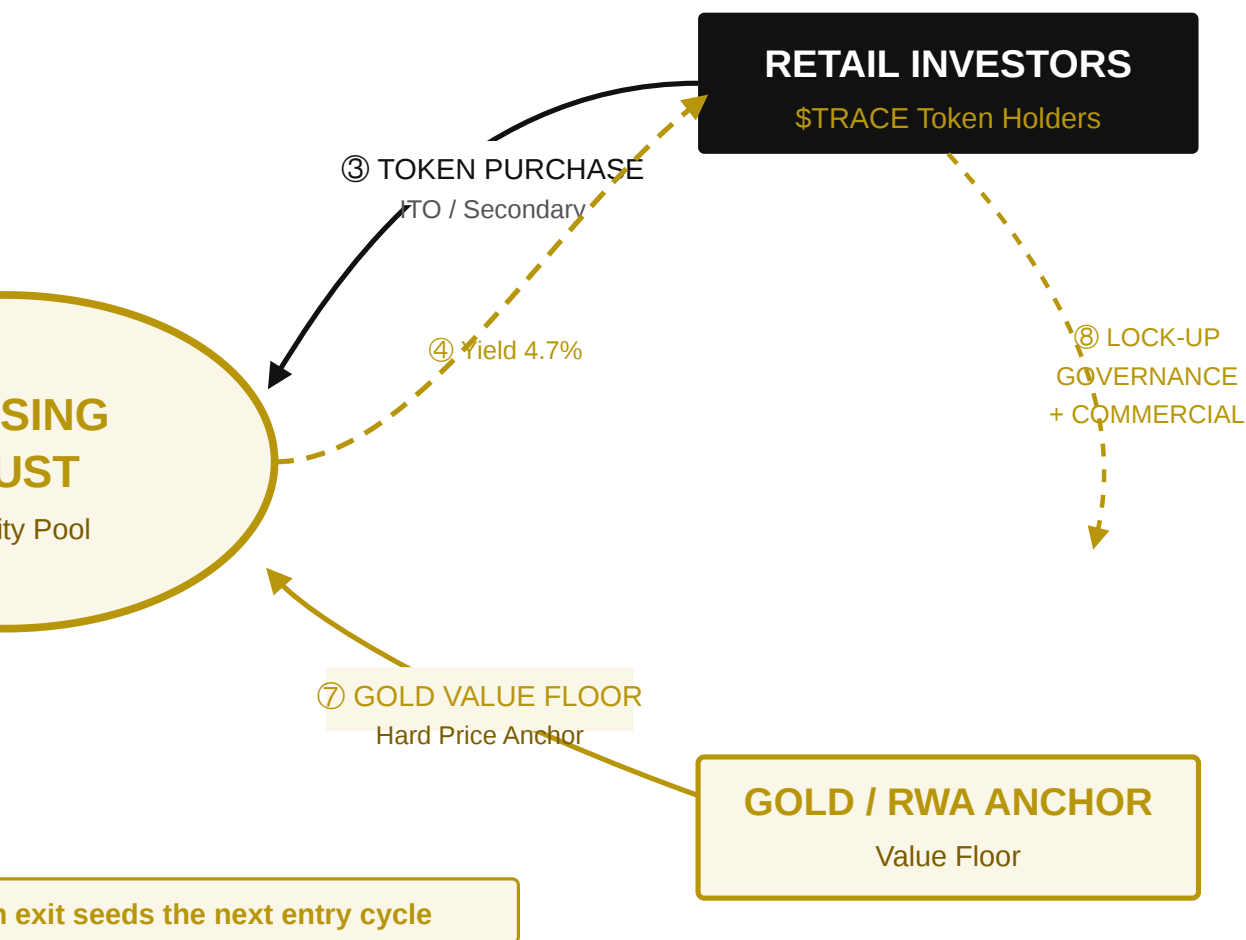
Why the model is self-sustaining



9-STEP PERPETUAL CYCLE · UNIFIED TOKEN



- ① **SCAFFOLD SALE** — Developer builds gold-backed superstructure, sells to Housing Trust at standard profit margin
- ② **GENESIS ITO** — Trust pays Developer via Initial Token Offering proceeds. Retail investors buy \$TRACE → capital enters Trust
- ③ **TOKEN PURCHASE** — Ongoing: retail investors buy \$TRACE on ITO or secondary market → continuous capital inflow
- ④ **YIELD DISTRIBUTION** — Trust distributes ~6.0% annual yield to \$TRACE holders from resident bond payments (after 2–5% op fee)
- ⑤ **MONTHLY BOND** — Resident pays HK\$15,000/mo: 53% equity accumulation + 33% yield pool + 14% commons maintenance
- ⑥ **SPATIAL RIGHT** — Trust grants resident volumetric allocation within scaffold. Debt attached to VOLUME, not individual
- ⑦ **GOLD VALUE FLOOR** — Token price anchored: $\text{Floor} = \text{Gold Spot} \times \text{Au Weight} \div \text{Token Supply}$. Cannot fall below material value
- ⑧ **COMMERCIAL SECOND LOOP** — Locked investors vote to fund commercial interventions → revenue flows back → yield increases → attracts more investors
- ⑨ **PERPETUAL RESET** — Resident exits → Trust refunds deposit (inflation-adjusted, zero profit) → absorbs bond → space re-issued → cycle continues



KEY FINANCIAL FORMULAS

Core calculations underpinning the model

Token Floor Price

$$= (\text{Gold Spot} \times \text{Scaffold Au Weight}) \div \text{Total Token Supply}$$

Annual Investor Yield

$$= (\text{Residents} \times \text{Monthly Bond} \times 33\%) \div \text{Total Token Value}$$

Resident Entry Cost

$$= 10\% \text{ Deposit (token stake)} + \text{HK\$15,000/mo bond service}$$

Traditional vs Gilded Trace Total Cost

$$\text{HK\$10.2M (30-yr mortgage)} \text{ vs } \text{HK\$3.0M (15-yr bond)} = -70\%$$

Optimal Commodity Ratio

$$25\text{--}30\% \text{ of unit cost in gold/metal} \rightarrow \text{yield} > \text{HK Gov Bond rate } (\sim 4\%)$$

Stakeholder	Role & Input
Developer	Builds gold-backed scaffold
Housing Trust	Holds asset. Issues \$TRACE + spatial bonds
Retail Investors	Buy \$TRACE for yield. No spatial ownership
Residents	10% deposit + service 100-yr bond

Agents | Stakeholders

DEVELOPER

INPUT: Constructs gold-backed scaffold. Delivers permanent superstructure to Housing Trust.

RETURNS: Sells scaffold at ITO. Immediate exit liquidity + standard margin. No long-term exposure.

RISK: Construction risk only. No housing market exposure post-sale.

LOWEST RISK

RETAIL INVESTORS

INPUT: Purchase \$TRACE RWA tokens via ITO or secondary market. No physical unit ownership.

RETURNS: Yield from resident bond service (4.7% p.a.). Governance rights when locked. Token liquidity.

RISK: Token price anchored by gold floor + yield. Liquid exit via secondary market.

MEDIUM / LIQUID

RESIDENTS

INPUT: 10% deposit (token stake) + monthly bond service. Debt attached to volume, not person.

RETURNS: Lifelong spatial autonomy. Elastic 50/50 private/communal. Deposit returned on exit.

RISK: No speculative risk. No negative equity. No market transaction on exit.

LOWEST RISK

HOUSING TRUST

INPUT: Manages liquidity pool. Issues tokens + spatial bonds. Absorbs exits. Re-issues volumes.

RETURNS: Operational fee (2–5%) from bond service. Commercial space revenue. Perpetual asset.

RISK: Gold-backed scaffold = floor value. Diversified income from bonds + commercial.

MANAGED

Aims & Returns

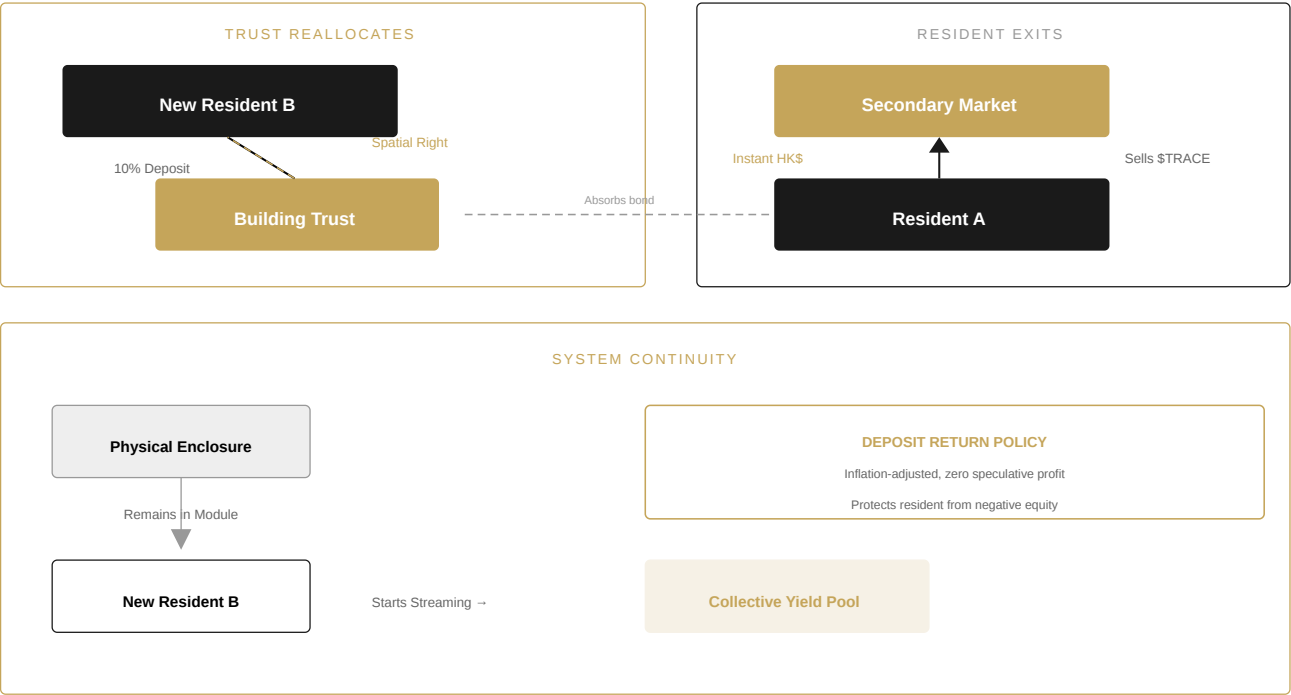
- Immediate exit liquidity via ITO proceeds + standard profit
- 2–5% op fee + commercial revenue buffer. Manages 100-yr bonds
- 6.0% yield + governance rights if locked up + commercial upside
- Lifelong spatial autonomy. 50/50 rule. Deposit returned on exit

THREE-LAYER VALUE:

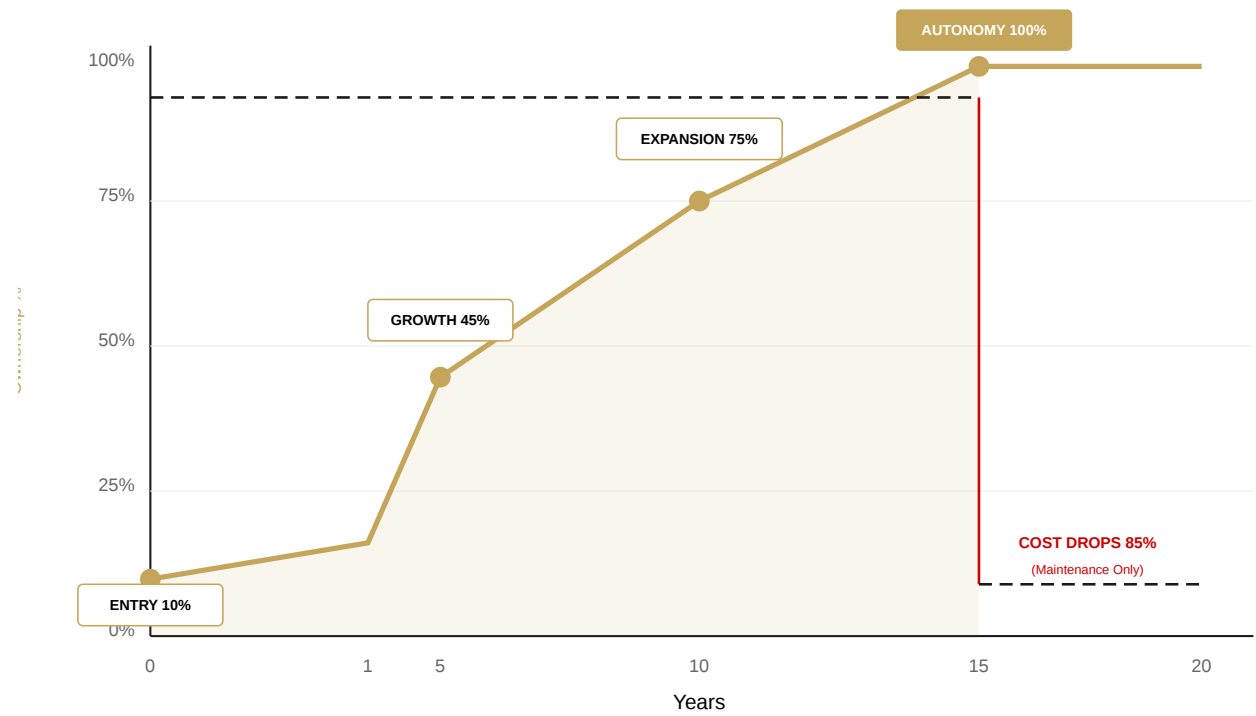
- ③ GOVERNANCE PREMIUM
Lock-up → voting → commercial upside
- ② BOND YIELD
Steady 6.0% from resident payments
- ① GOLD FLOOR
Unbreakable material baseline

D 9 — EXIT MECHANISM & SYSTEM CONTINUITY

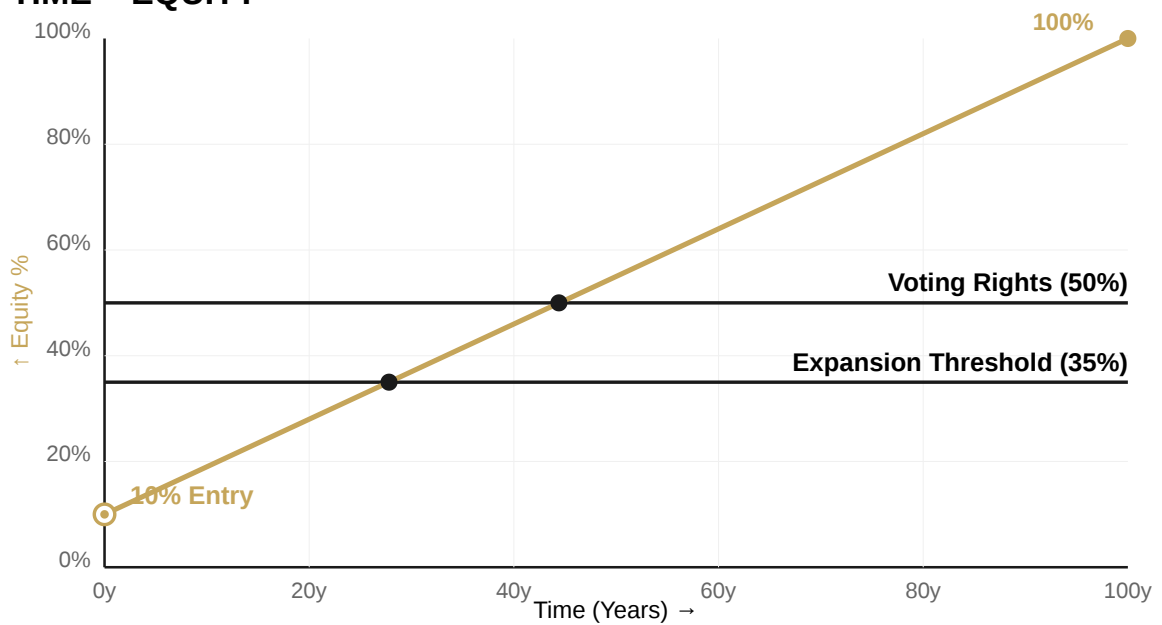
LIQUIDITY & PERPETUAL RESET

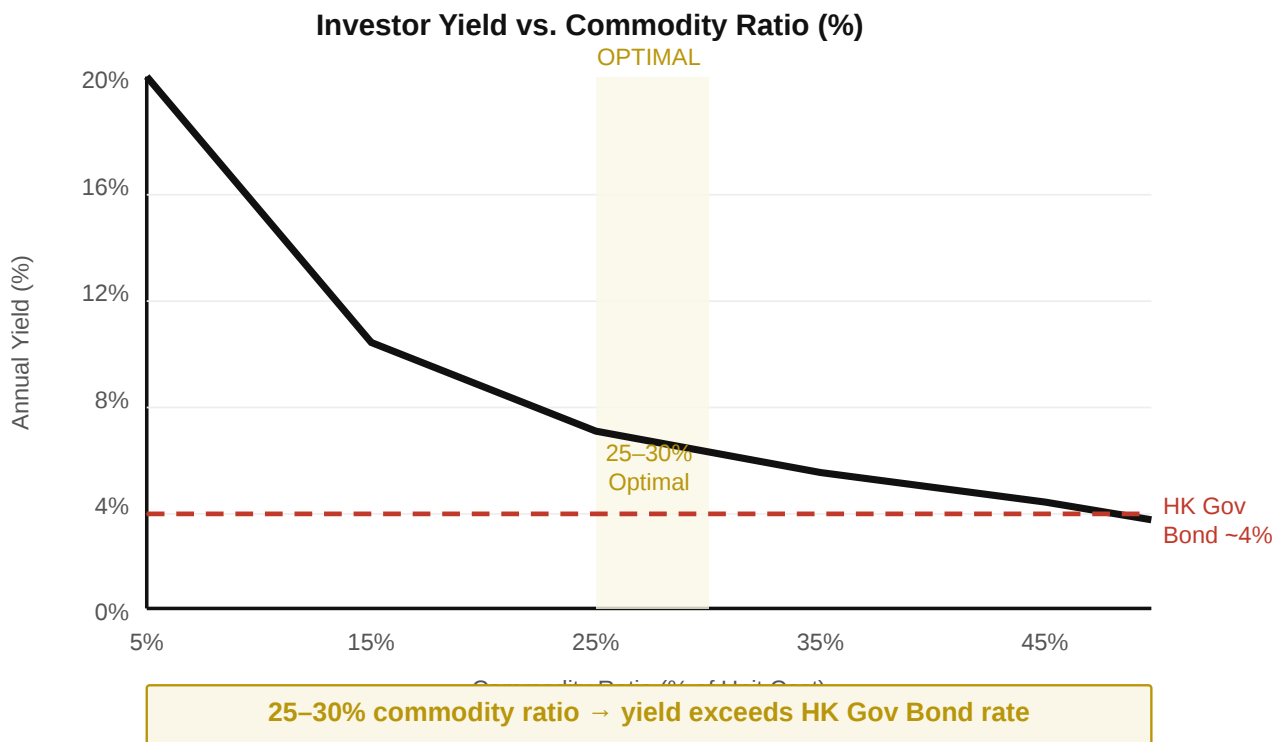


FROM ENTRY TO FULL AUTONOMY



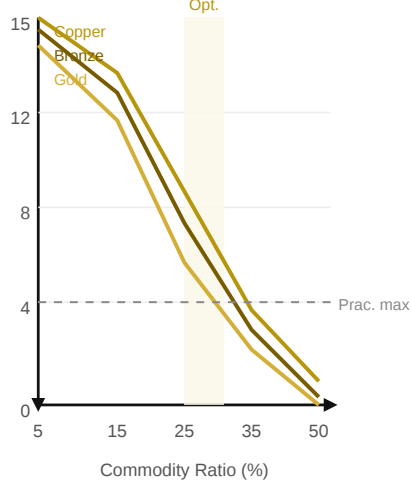
TIME = EQUITY



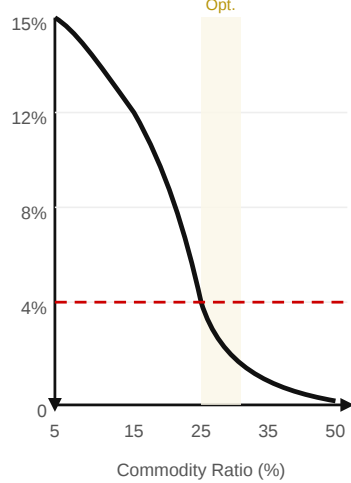


COMMODITY-TO-STRUCTURE RATIO OPTIMISATION

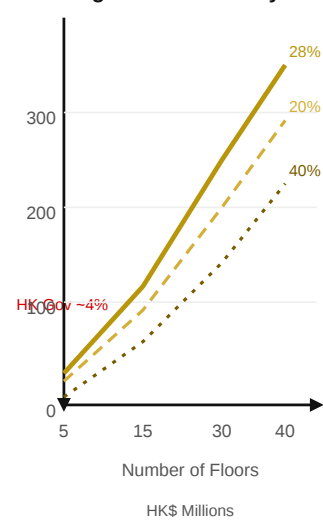
A: Material Thickness vs Ratio



B: Investor Yield vs Ratio



C: Height vs Commodity Value



D8 — THE SPATIAL THRESHOLD

FINANCE → SPACE MANIFESTATION

50% Private — enclosed personal living space (infill module)

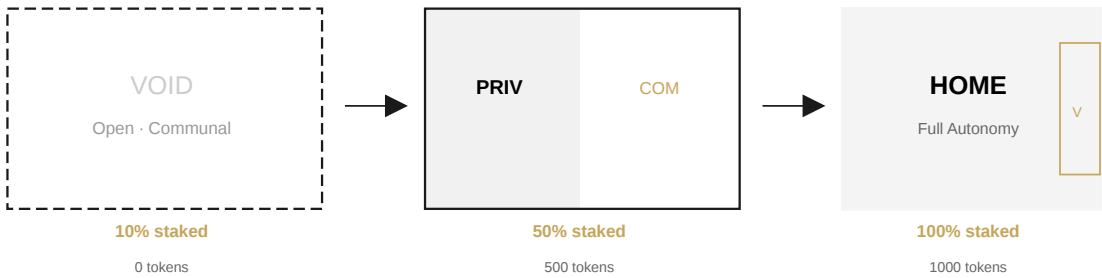
50% Communal — shared public space, managed collectively

Debt attached to the VOLUME, not the individual resident

→ Boundary is elastic: expands/contracts with lifecycle

~70% residents in equity phase sustain yield pool

~30% at full autonomy enjoy minimal cost (maintenance only)



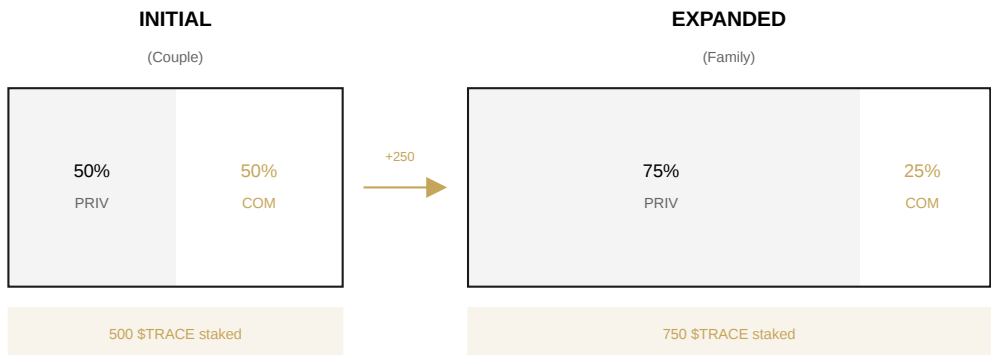
$$\text{Token Floor} = \text{Gold Spot} \times \text{Au Weight} \div \text{Supply}$$

Price anchored — cannot fall below material value

SCENARIO A — EXPANDING NUCLEUS

GROWING FAMILY

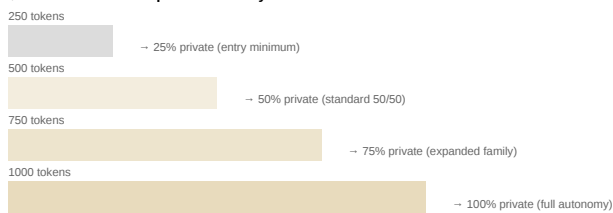
Young couple → expecting children → need more space



Boundary expands — no new mortgage

More tokens staked → larger private boundary → same gold-backed floor

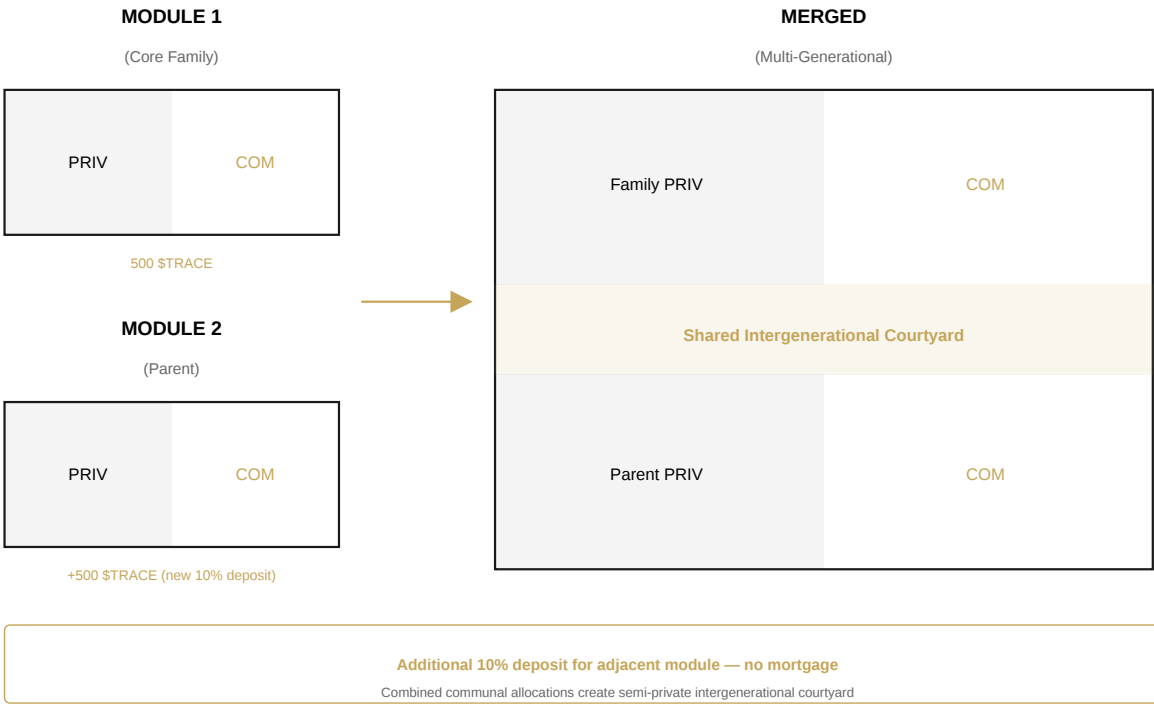
\$TRACE Staked vs. Spatial Boundary:



Token price unchanged — staking locks supply, may increase market price

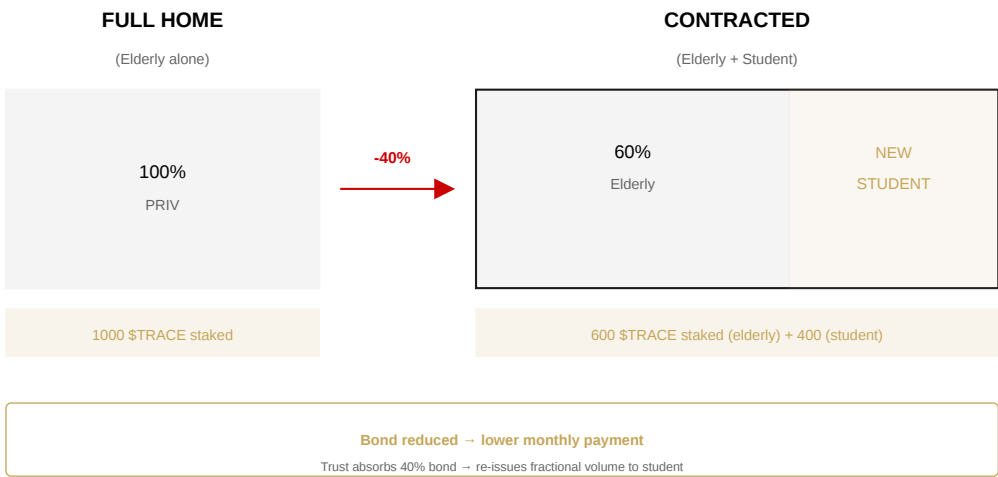
FAMILY STRUCTURE CHANGE

Aging parent → integrate without compromising privacy



CONTRACT · RELEASE · REDISTRIBUTE

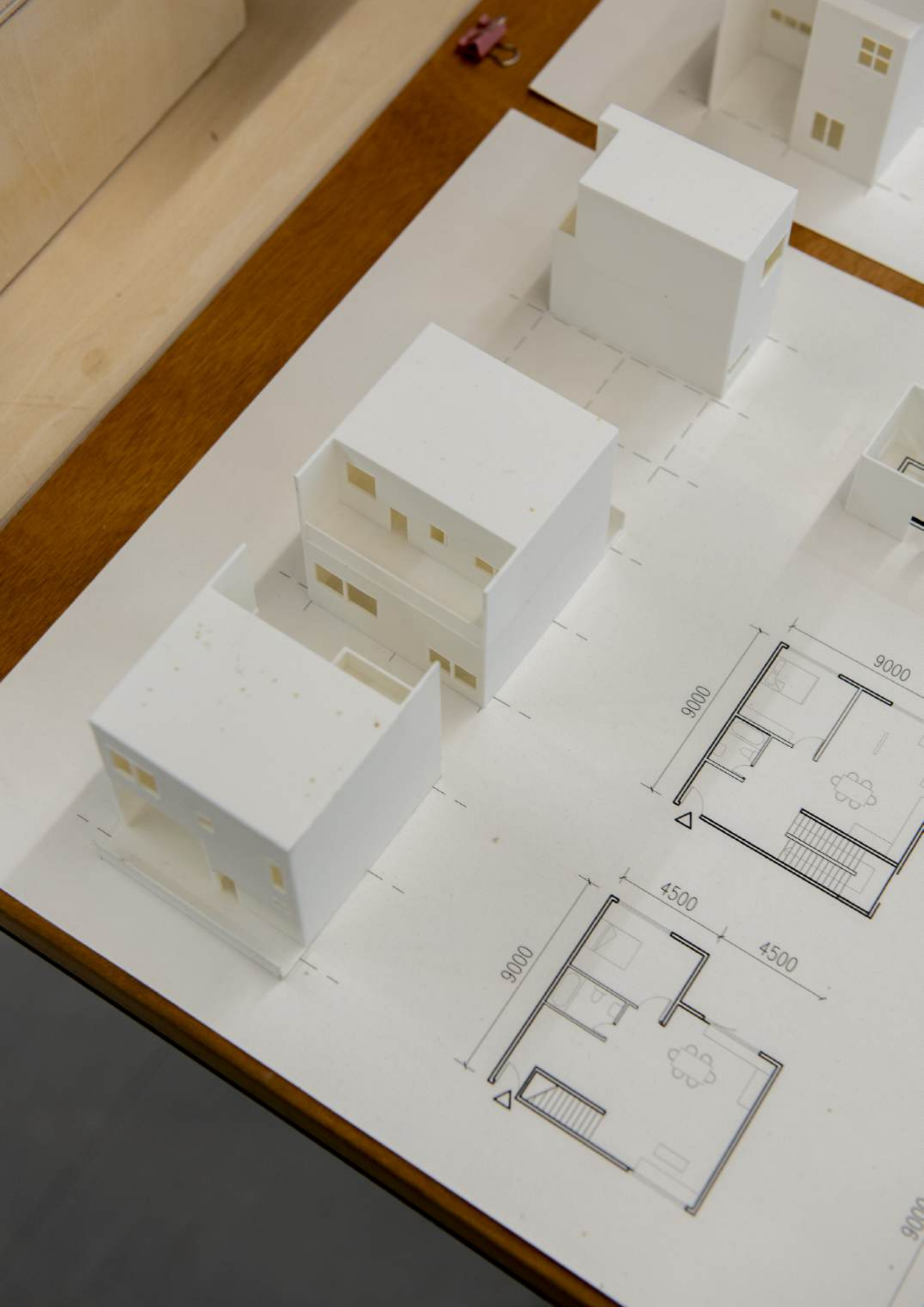
Children leave → contract space → release bond → lower payment



Liquidity requires the backing of robust and long lasting infrastructure.

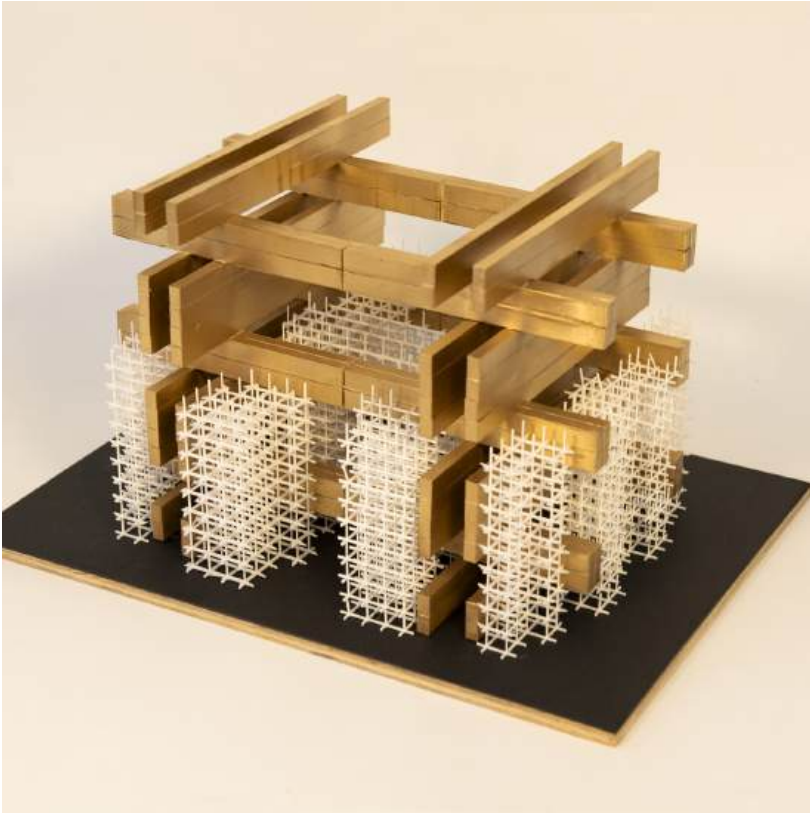
Chapter 5:

Methodology & Structure



Methods | **Phase 1: Prototyping**

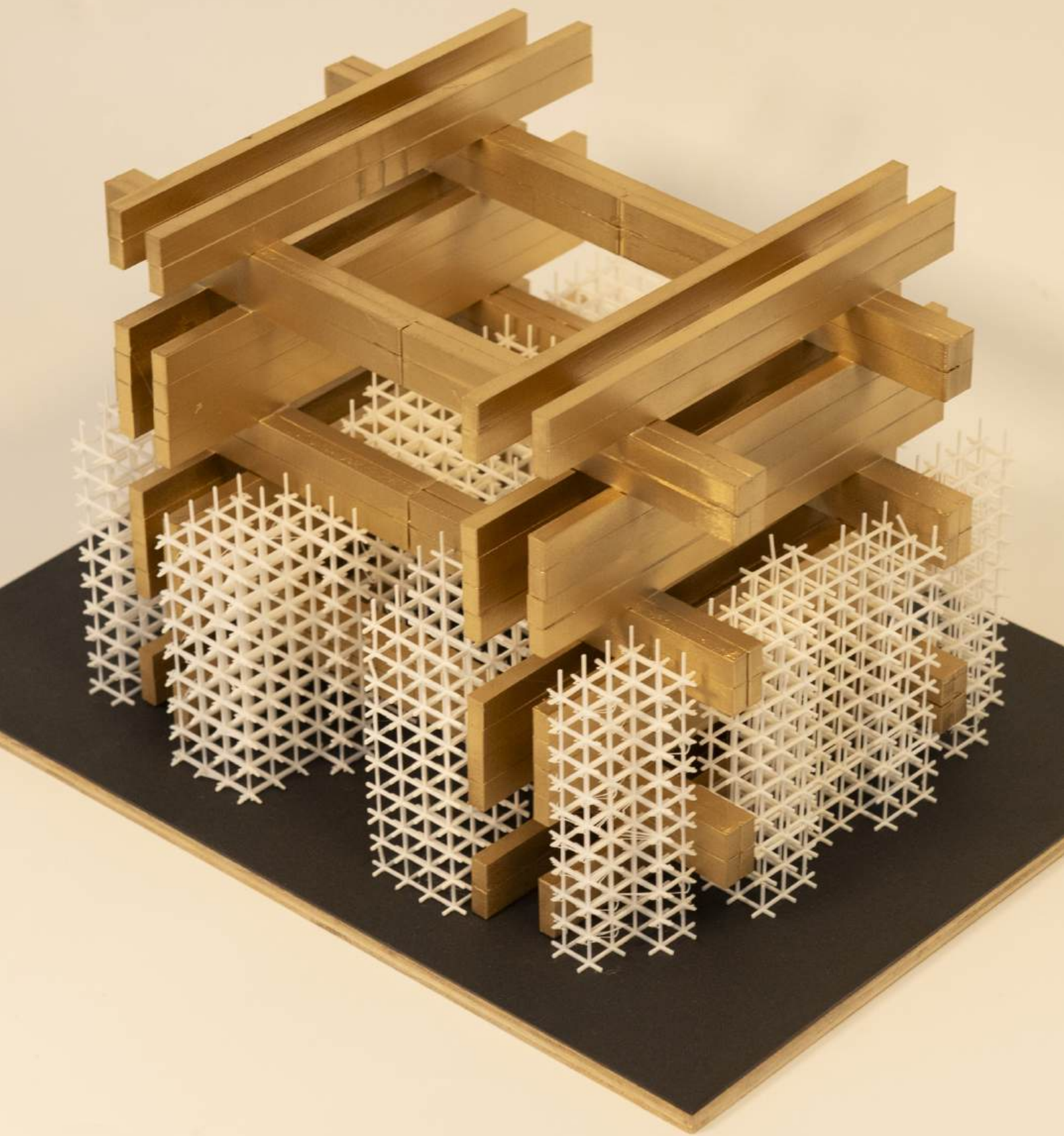
A system of hyper-durable infrastructural frames was explored with two prototypes that rules interconnectivity, degree of expansion, light and air compliance, and formal expression. The permanent infrastructure “Vault” not only acts as value storage, hedged against a gold-backed reserve to secure long-term liquid value, but also serves as framework for expandable modular living units deployed within system.

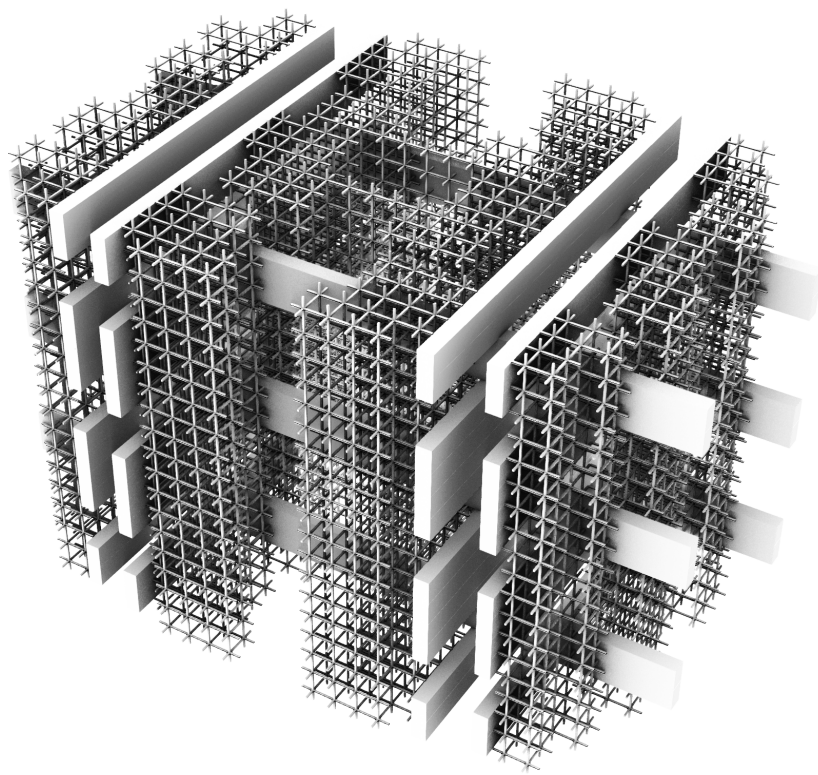


Prototype 1

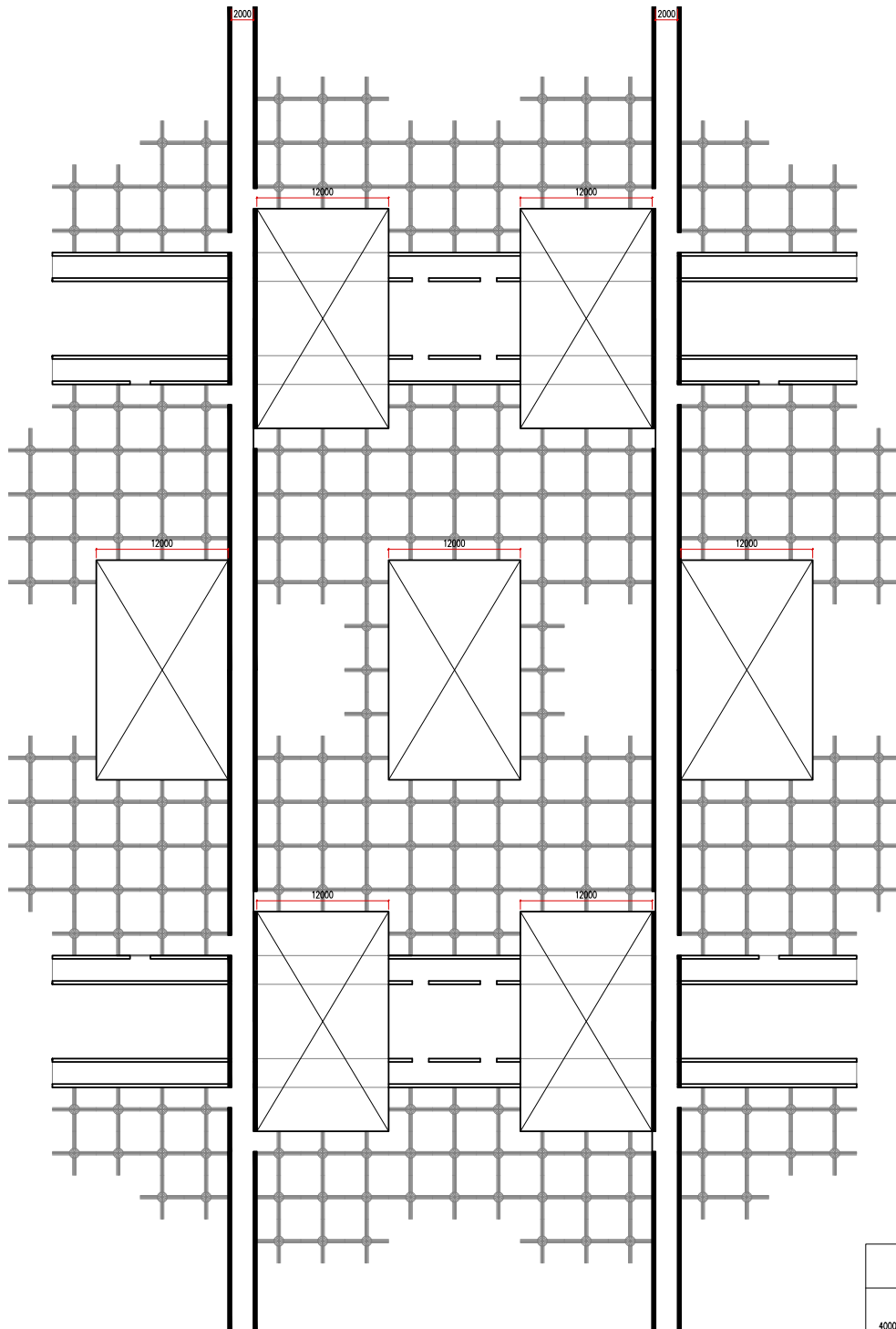
The Ingot Infrastructure

The key idea of this prototype is a durable mega infrastructure that provide horizontal connectivity across vertical residential blocks. The value of the building go to the infrastructure and the whole development is fractionalized and packaged into a commodity token for trading. Quality spaces are then returned to residents for modular expansion. The infrastructure at the same time act as spatial organizer that limits the extent of expansion and size of voids and air gap. The infrastructure itself is wide enough to provide communal spaces that form clusters every 4 floors.

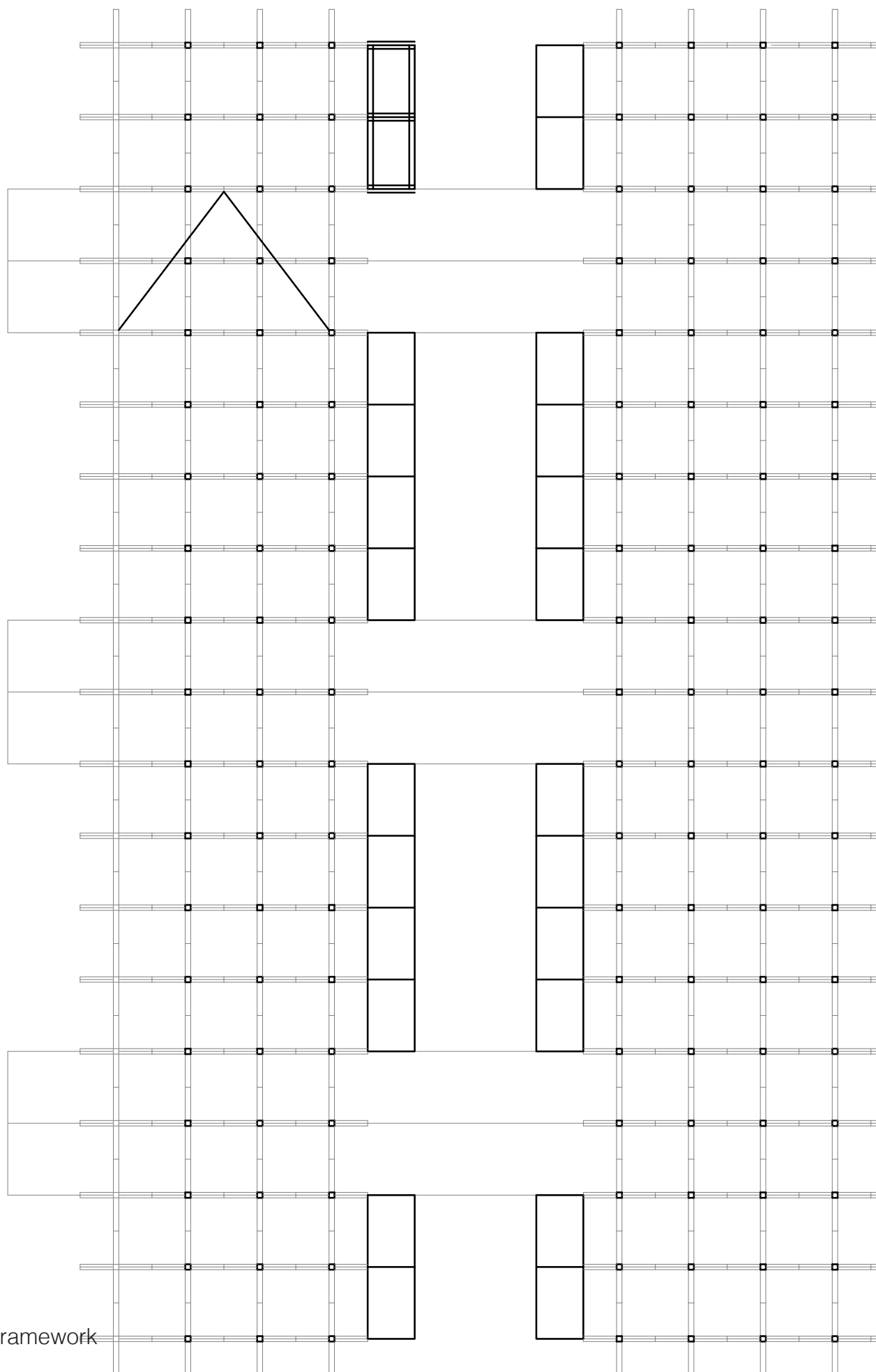




A frame system laterally braced by the infrastructure

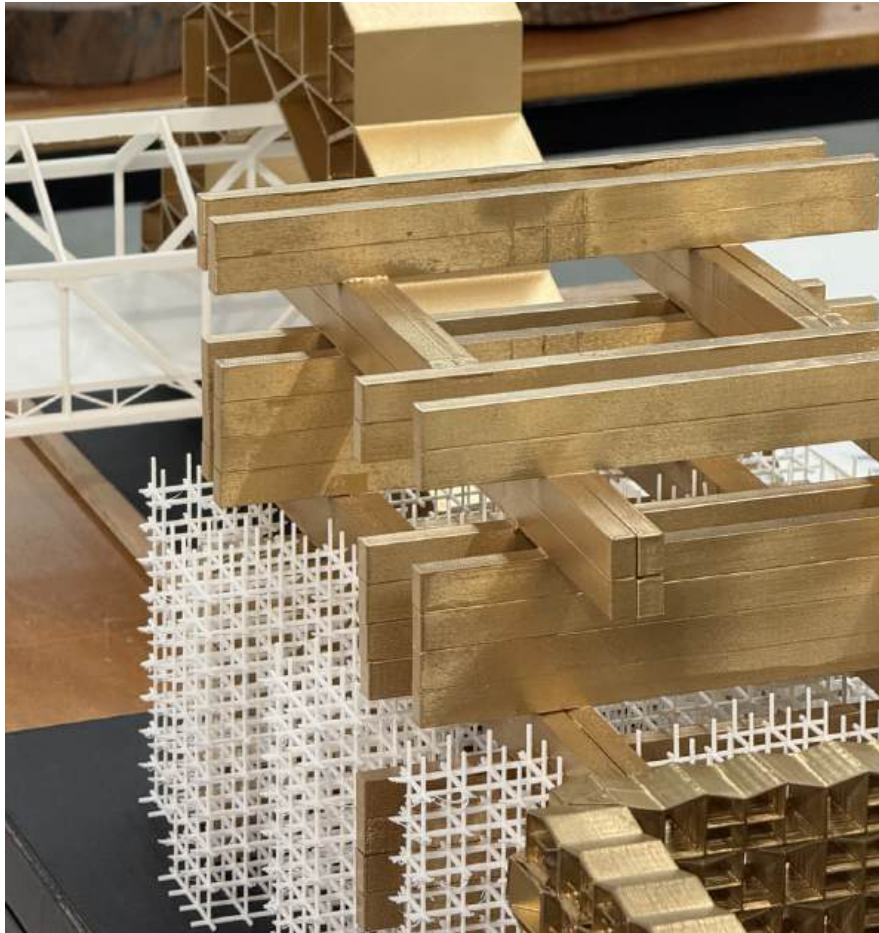


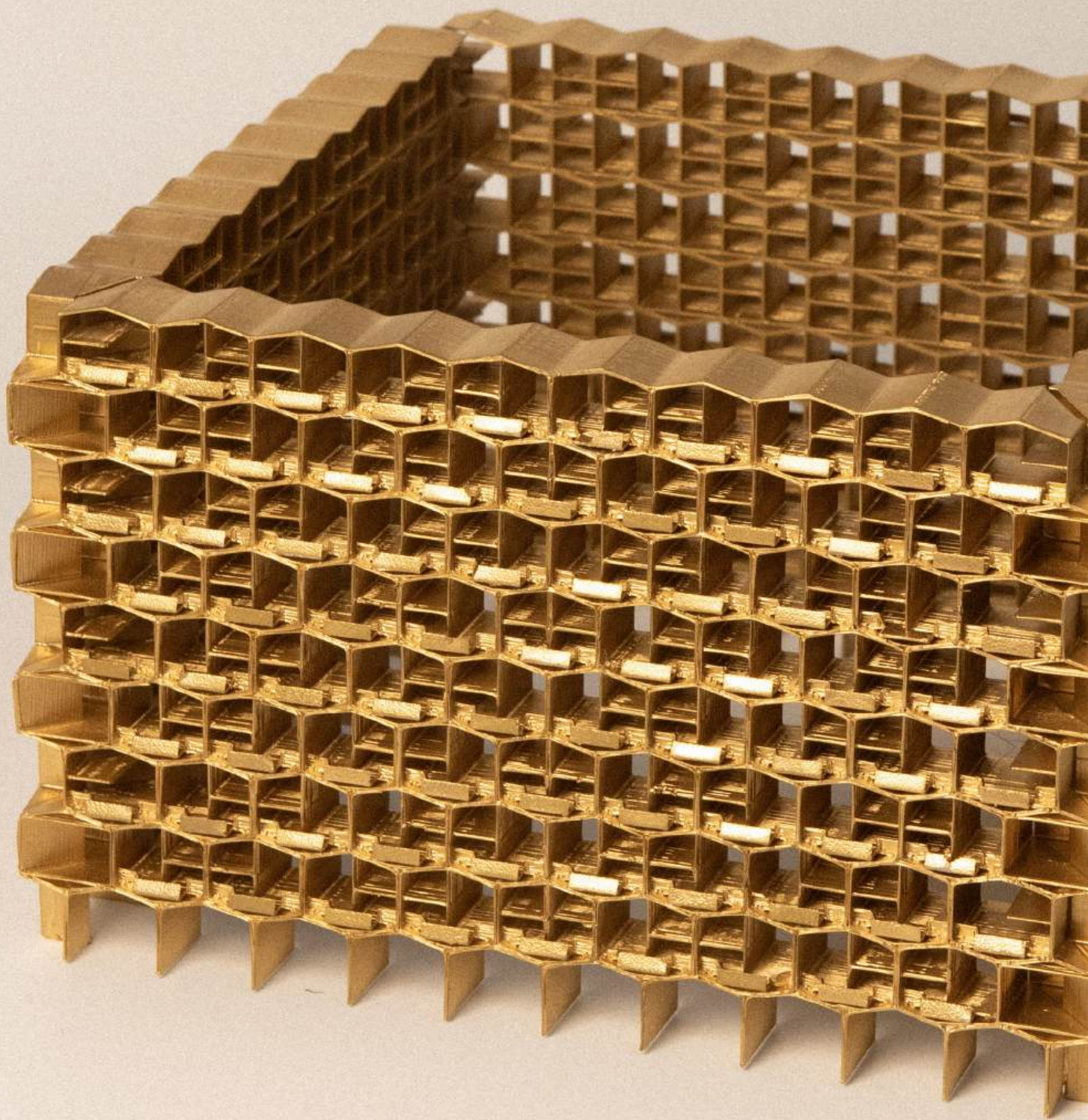
Void size are planned in the system



4.5m x 4.5 framework



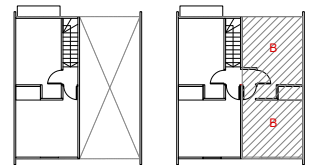
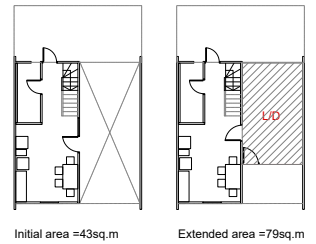
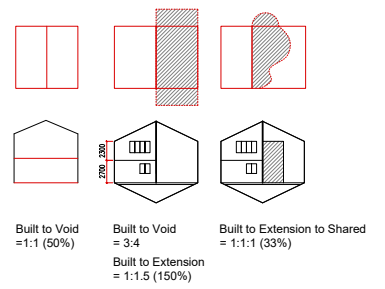
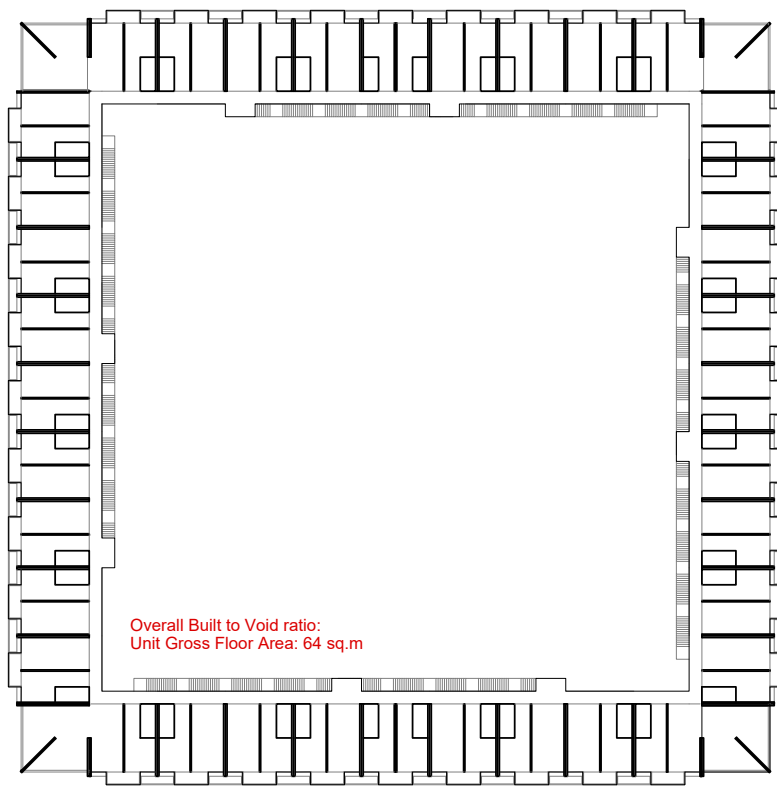




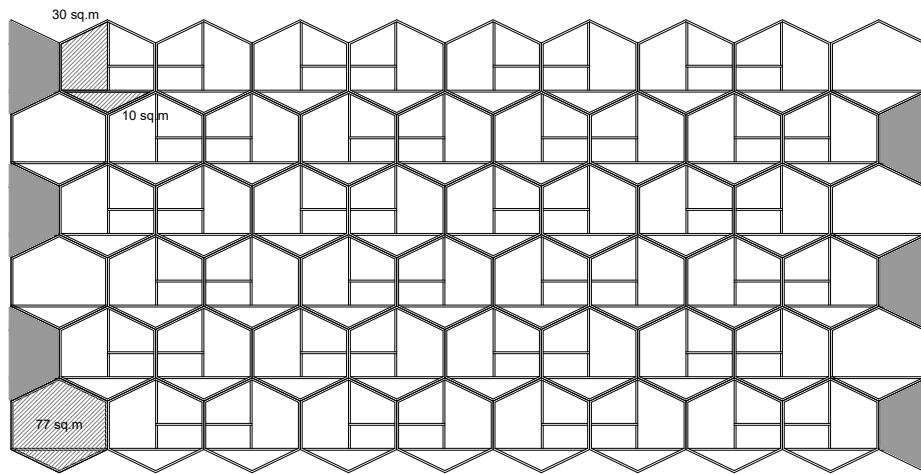
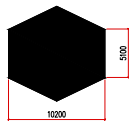
Prototype 2

The Ingot Infill

When the pricing of a space shift from the calculation of Gross Floor Area to Cross Sectional Area, volumetric spaces become a potential driver of module size. This prototype designed a hexagonal frame that is durable for 100 years. It first day provide void space is halfy built, borrowing idea of Half House by The Elemental. Voids act importantly as ventilator and communal space before expansion.

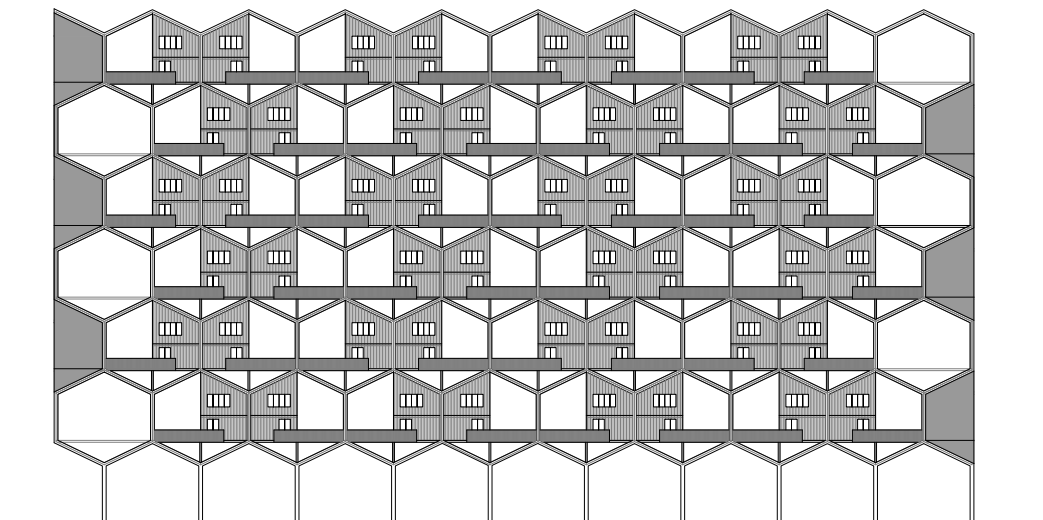
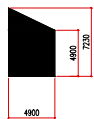


Plan and Spatial Logic of Owning Half and Expansion



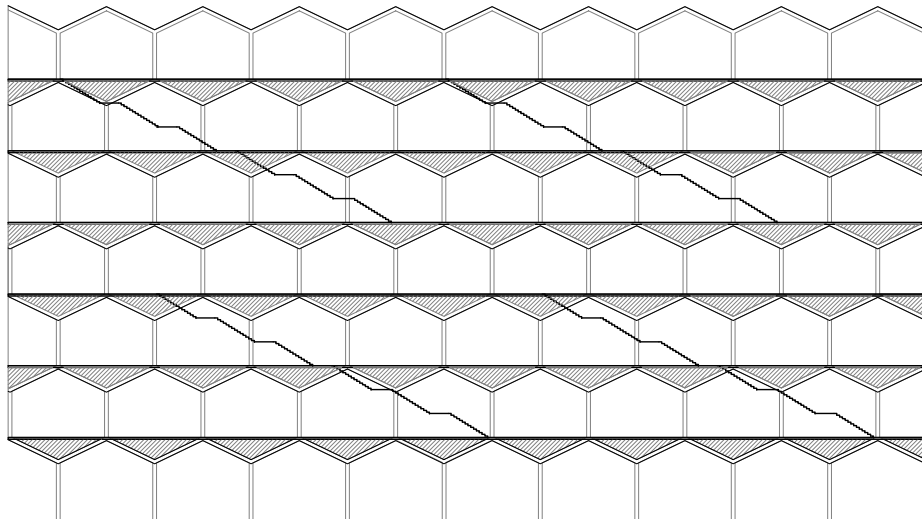
Overall Built to Void ratio:
Unit Gross Sectional Area: 77 sq.m

Section

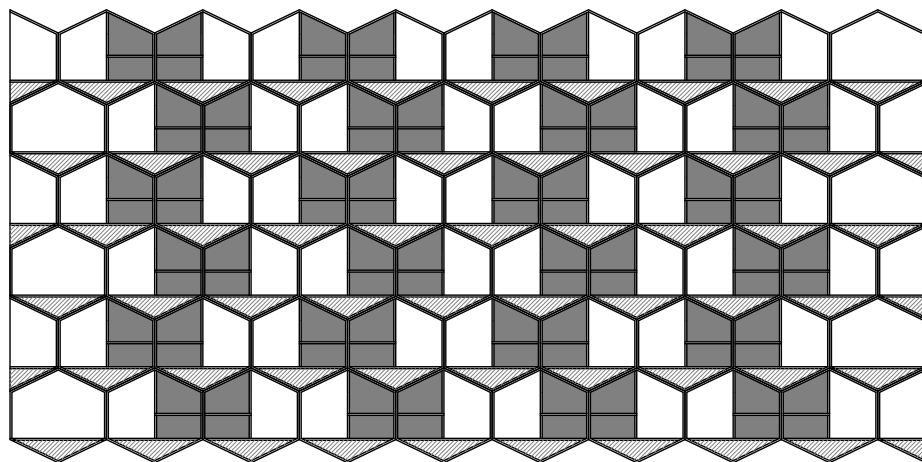
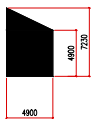


Elevation 1





Elevation 2



Unit Gross Sectional Area: 77 sq.m

Built / Unbuilt

Methods | **Phase 2: Urban Deployment**

After developing the idealized economic prototype, the methodology shifted to site-specific urban testing within the To Kwa Wan CBS scheme. This phase was crucial for determining how the Super Liquid protocol reacts when forced to negotiate with the physical realities of the city. To rigorously evaluate the architecture's scalability and resilience, the Vault was deployed across four distinct site conditions.

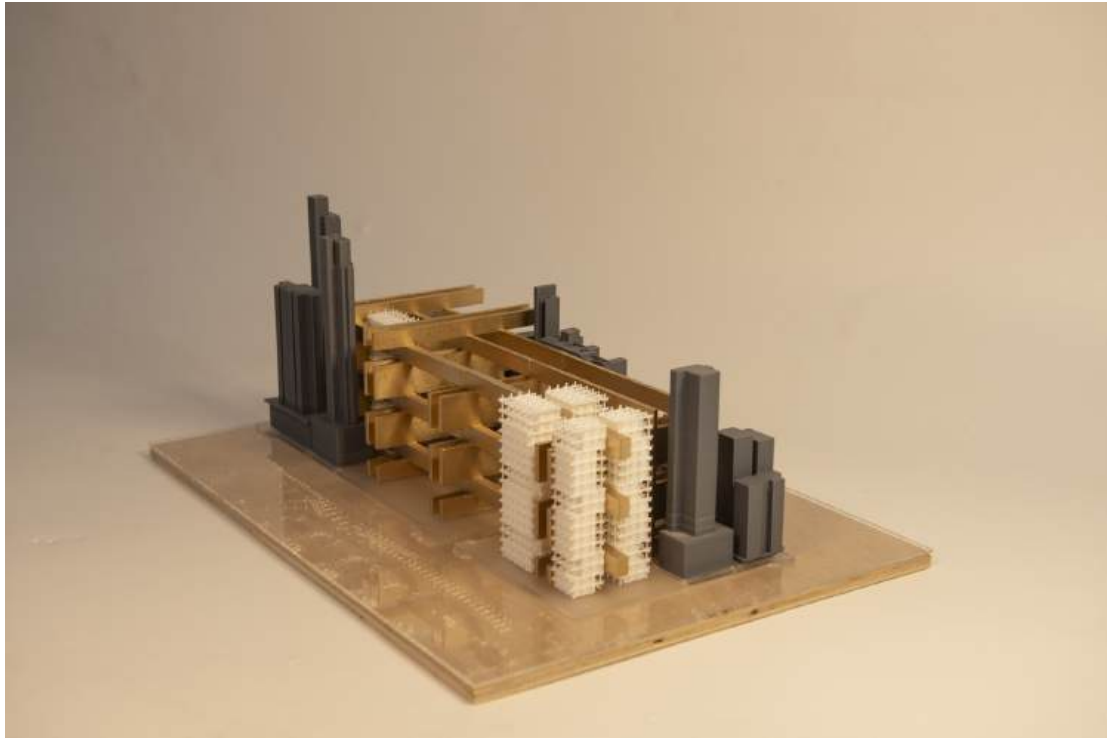
First, the full empty block scenario tested the protocol's maximum internal efficiency by optimizing communal voids without the friction of existing neighbors.

Second, the half urban block scenario tested symbiotic insertion, forcing the structural grid to dynamically adapt its hardcoded light voids (UVA) to respect the party walls of adjacent buildings.

Third, the across the road condition visualized networked governance, simulating how retail investors might fund floating infrastructure like skybridges to connect multiple Vaults.

Finally, the ground floor condition resolved the critical civic interface, mediating the architectural friction between the high-yield commercial streetscape and the secure residential sanctuary above.

By testing the abstract prototype against these grounded realities, this methodology proves the tokenized financial protocol can elastically adapt to Hong Kong's constrained physical fabric.



The Urban Testbed



The Civil Servants' Co-operative Building Society (CBS) scheme was a housing benefit program introduced by the Hong Kong Government in 1952. Running until the mid-1980s, the program was designed to encourage local, middle-ranking civil servants to build and own their homes, helping to stabilize the civil service during a period of rapid population growth and housing shortages. The building design follows Tong Lau's typology.

公務員合作社



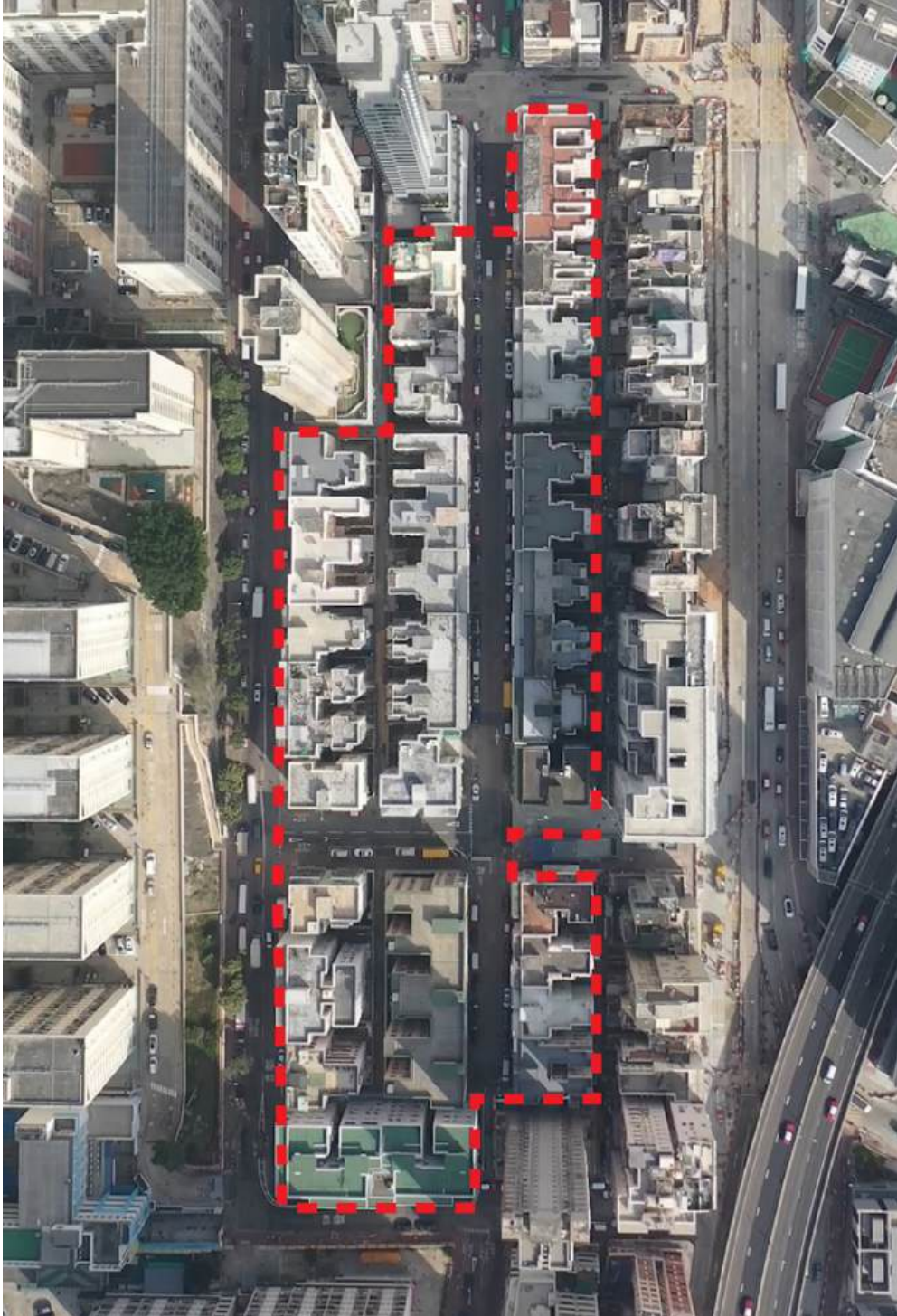


Collectivity

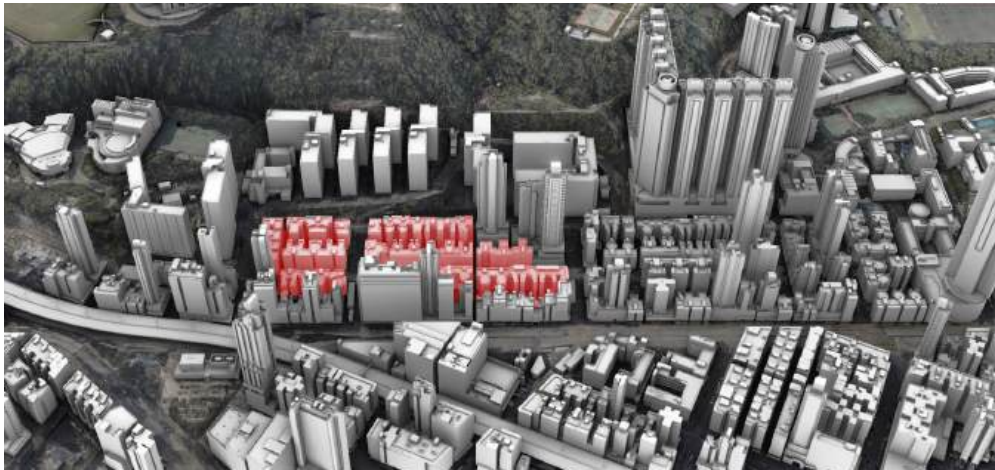


Materiality

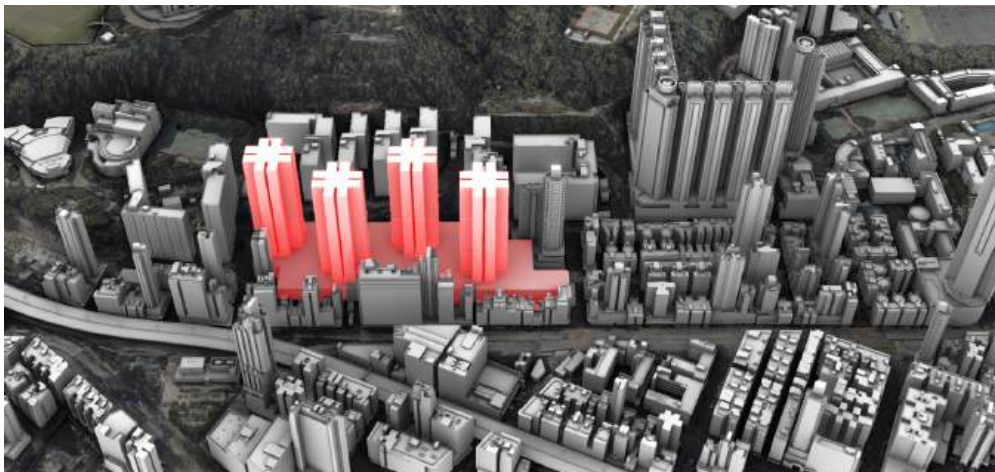




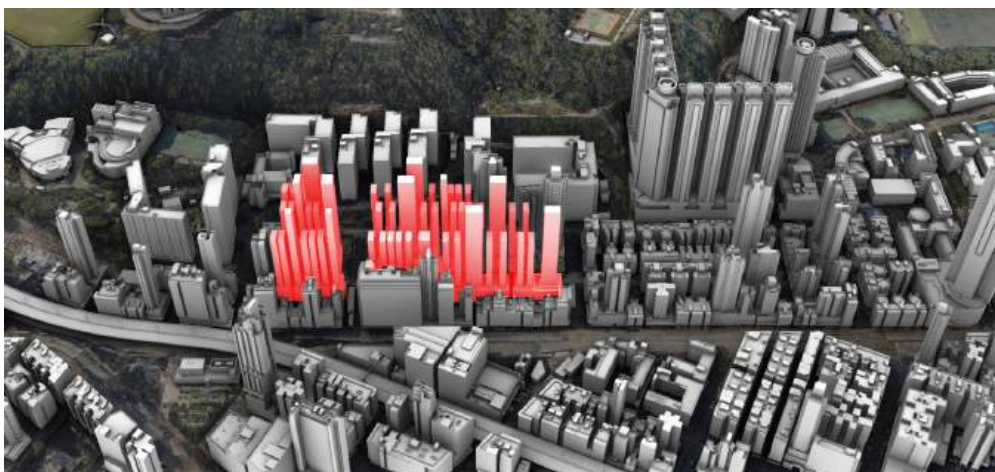
CBS Buildings to be Reveloped



existing buildings



private redevelopment



seperate development



Site Map 1:800

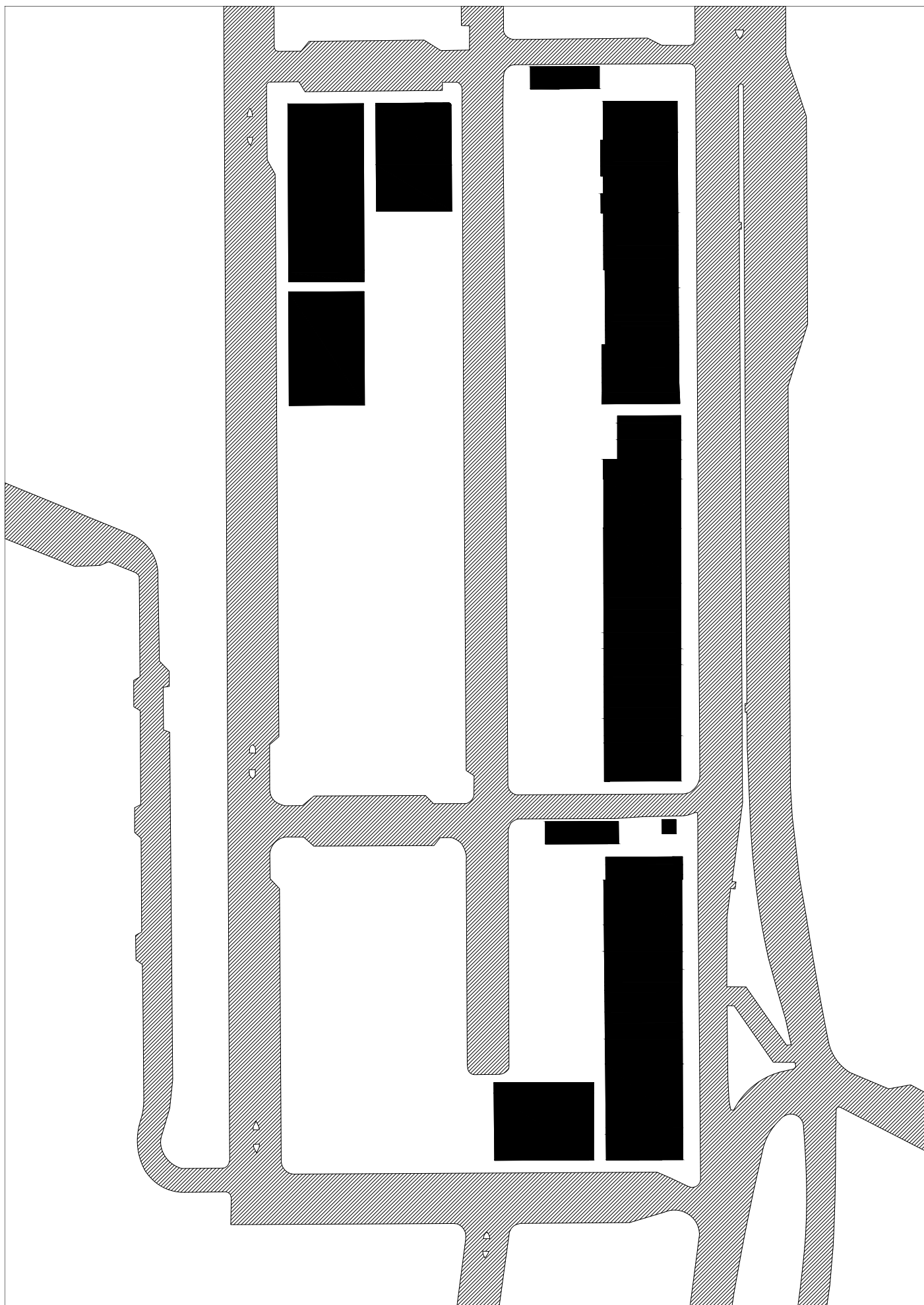
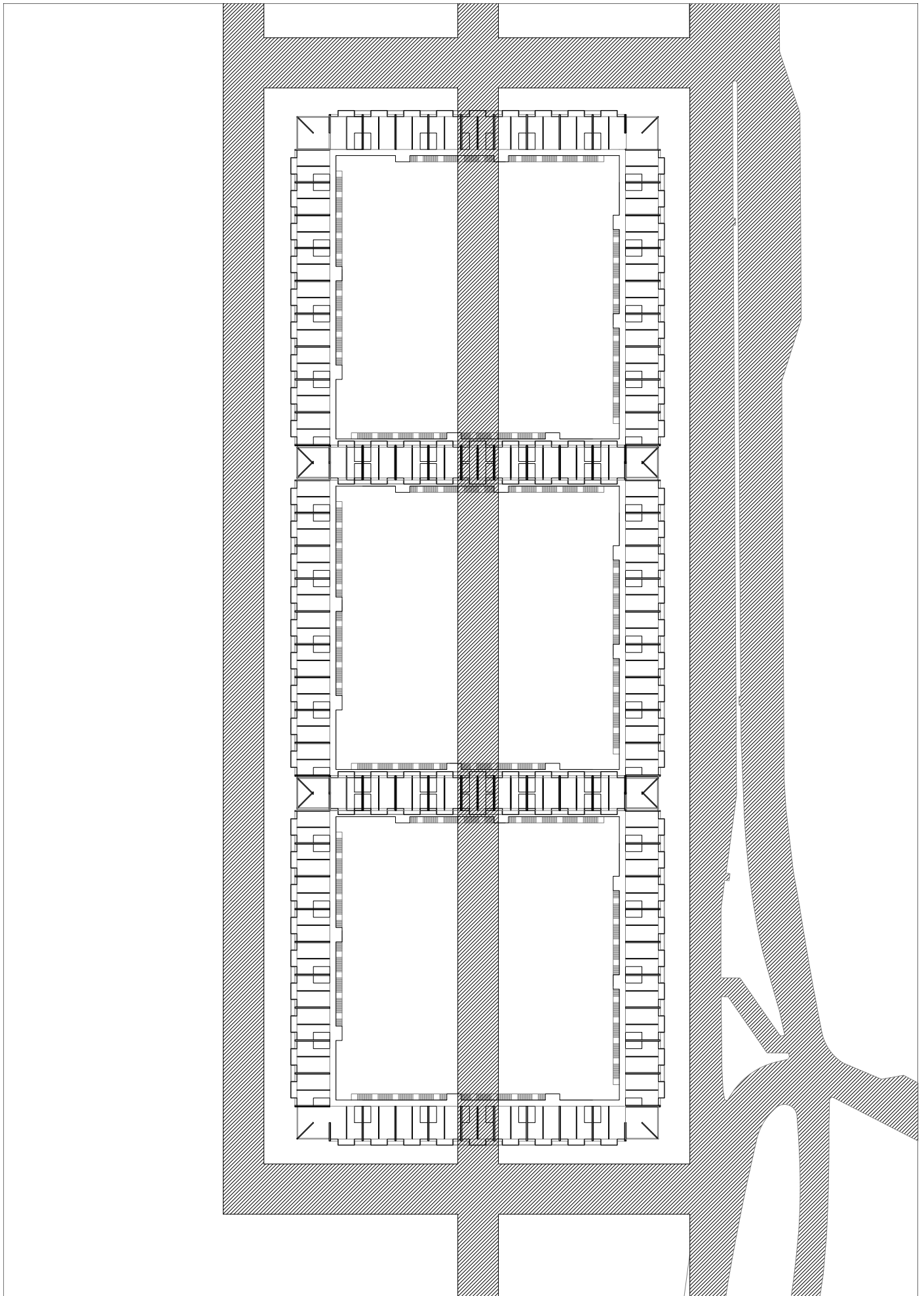


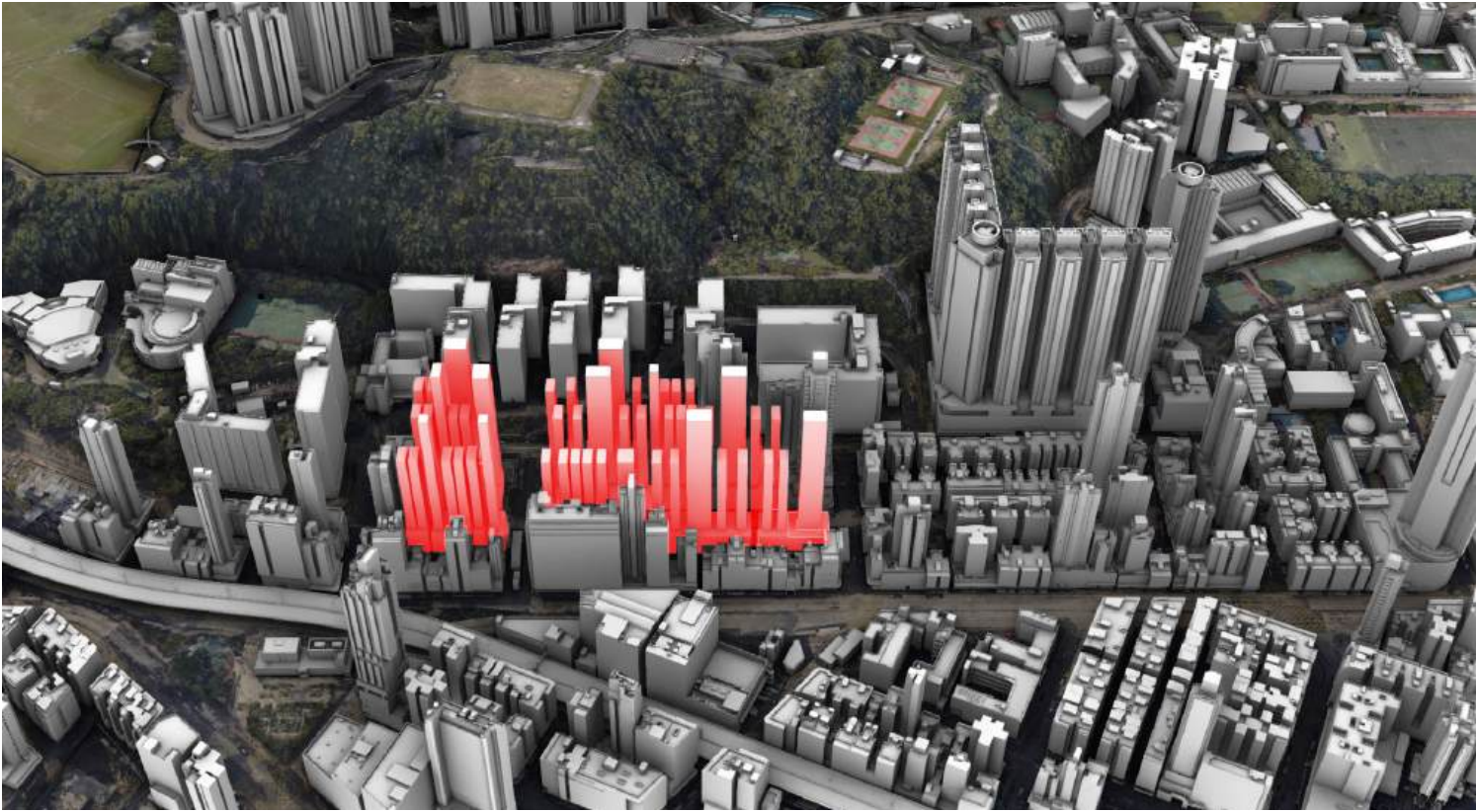
Figure and Ground 1:800



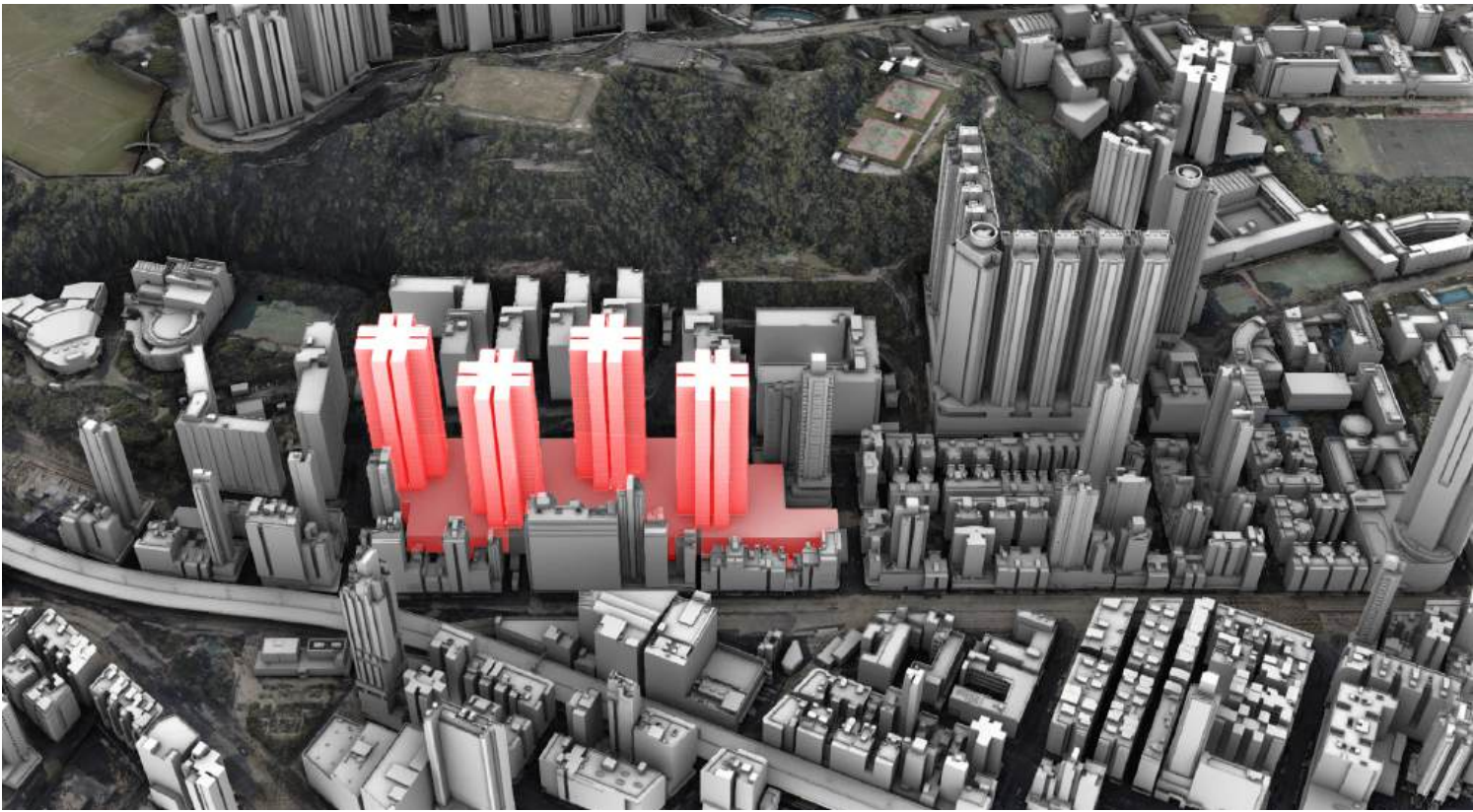
Prototype 1 THE INFRA-



Prototype 2 THE INFILL



new pencil towers

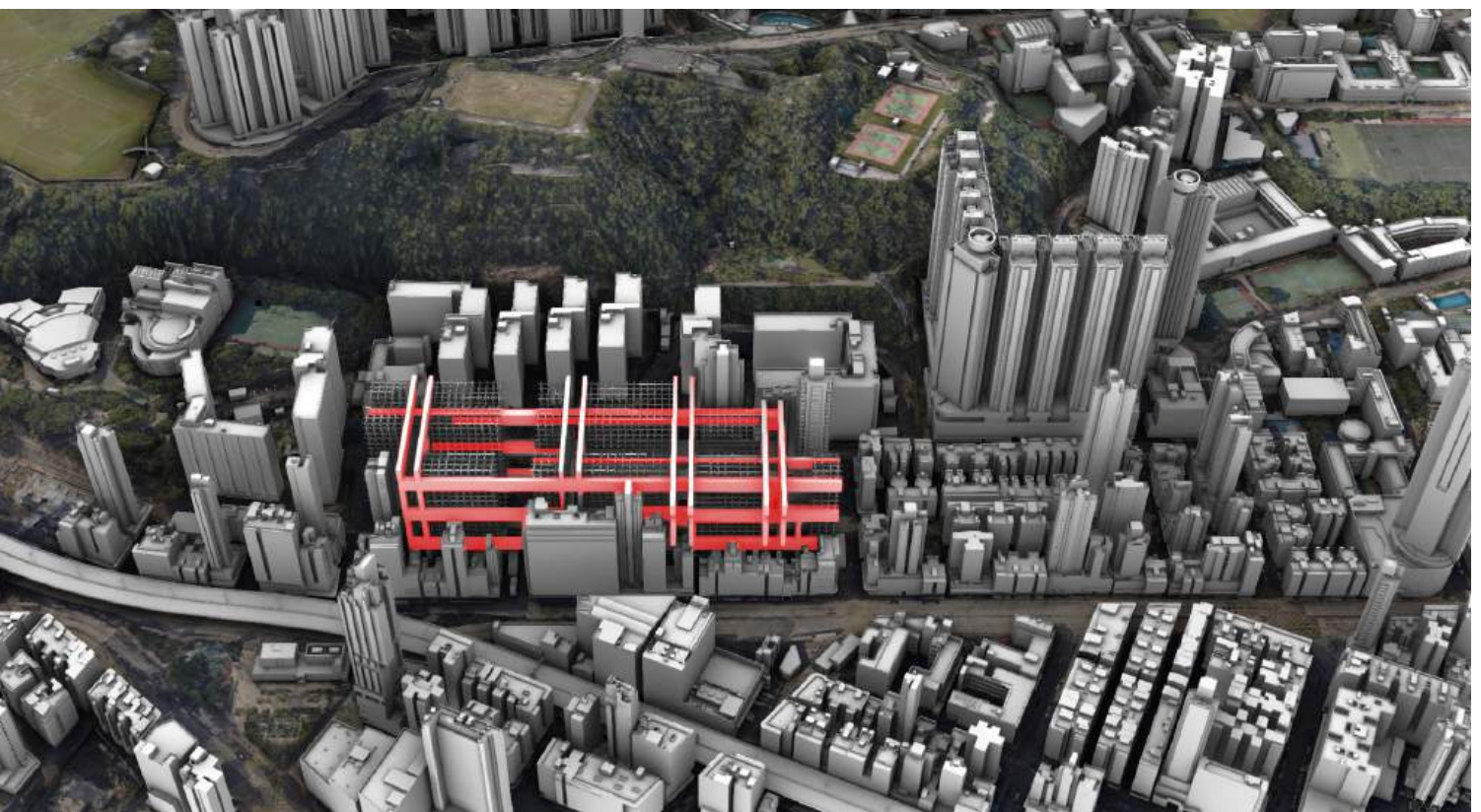


new podium towers

THE INFRA-



durable mega-frame provide sectional voids



durable mega-structure provide interconnectivity

Methods | **Phase 3: A Study of Dimensional Houses**

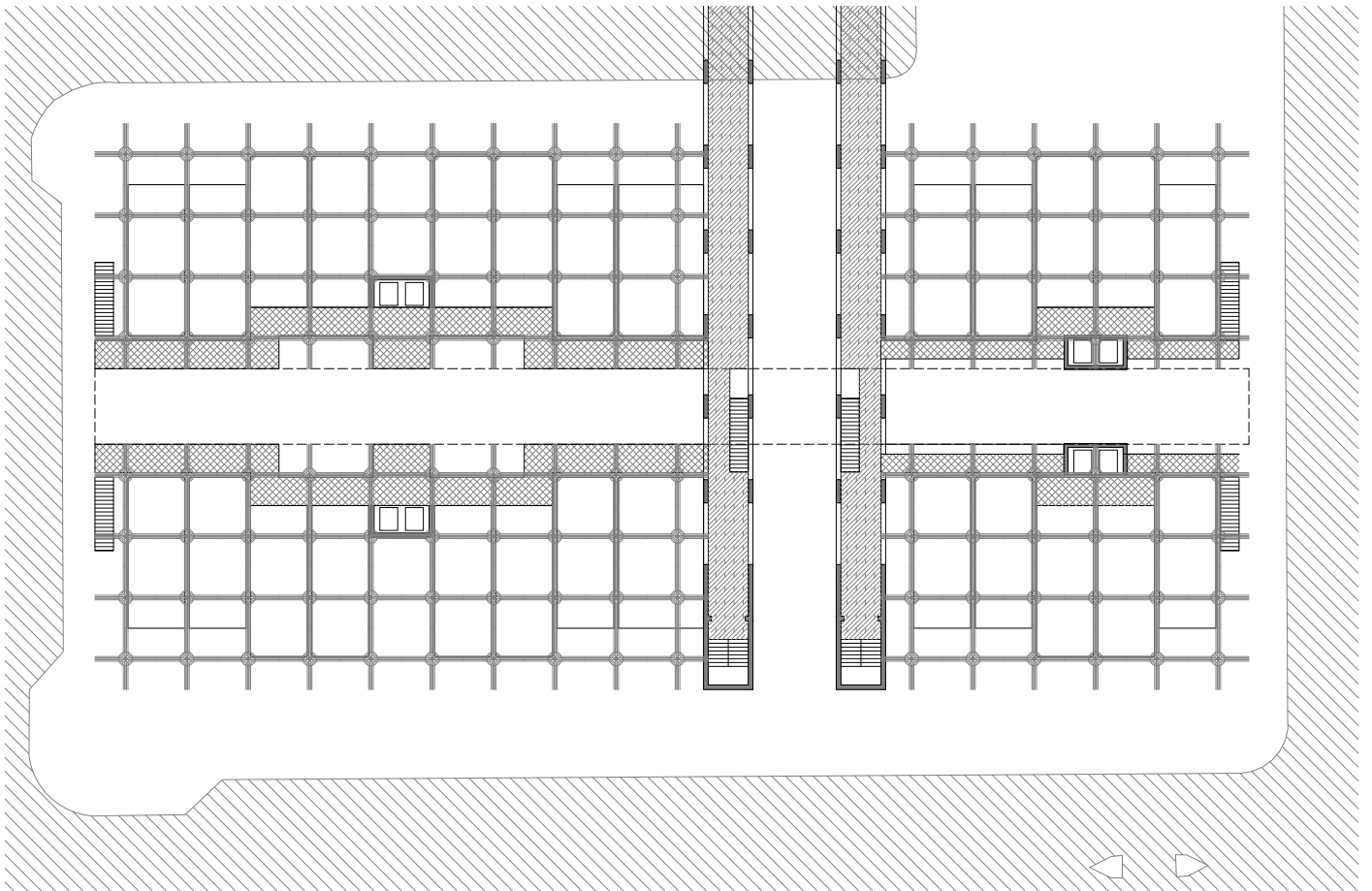
The first prototype proposed a rigid plug-in modules of living. The second prototype introduces an expandable “half-house” concept utilizing permanent structural voids.

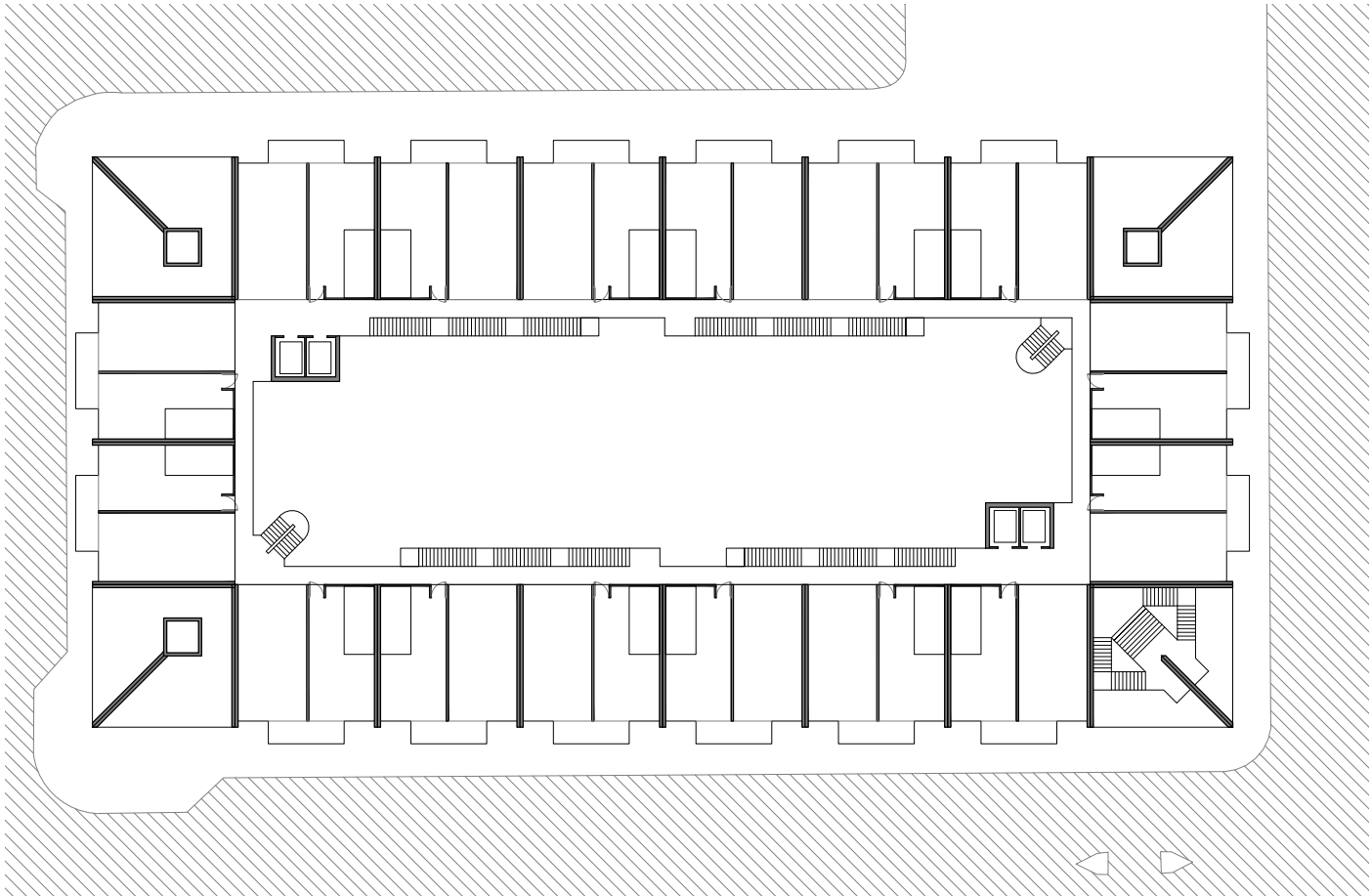
In both case the architectural grid become so important. This phase focuses on testing an “optimal grid” for medium-scale residential housing. The goal is to establish a foundational layout that supports dynamic, incremental growth. By testing various dimensional scenarios, the methodology balances the number of private rooms with necessary communal open space. As family structures change over time, residents can smoothly expand their living areas within this optimized grid, ensuring spatial flexibility without compromising shared community infrastructure.

A catalogs of housing of different volumes is produced for Prototype1.

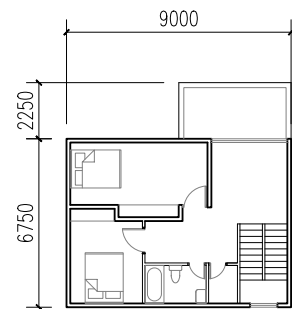
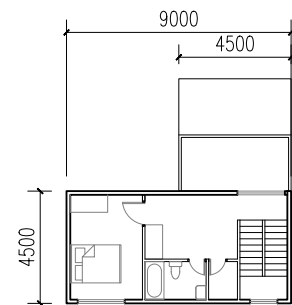
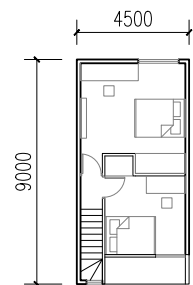
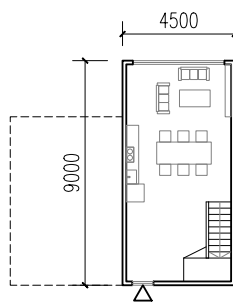
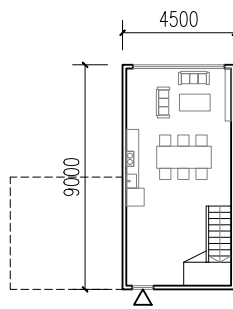
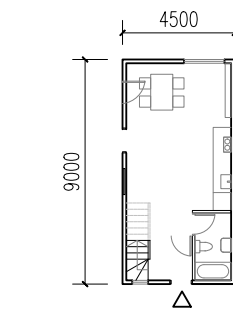
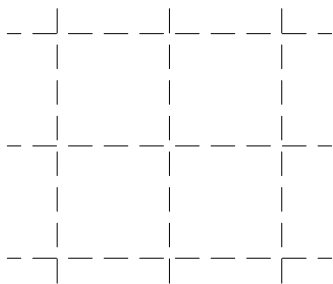
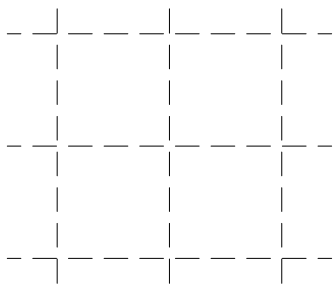
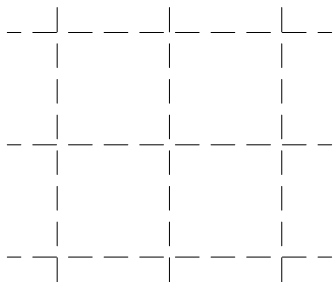
A single-family house is designed within module space for Prototype2.

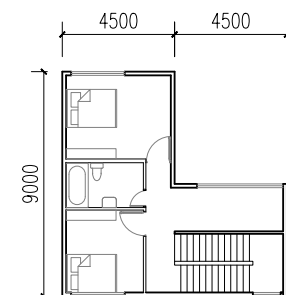
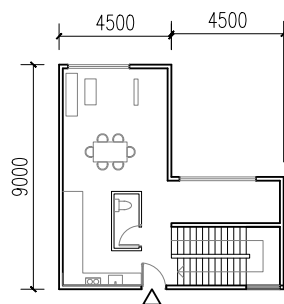
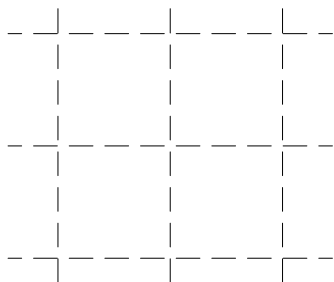
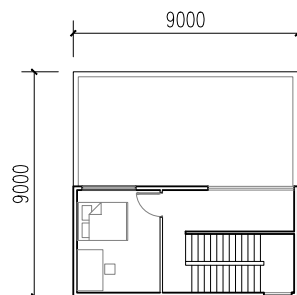
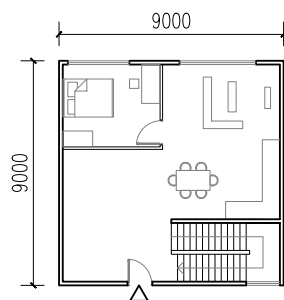
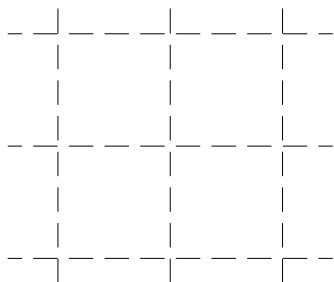
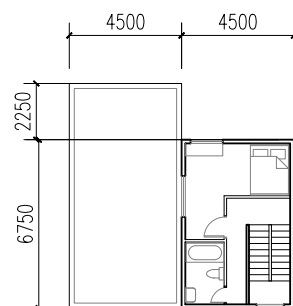
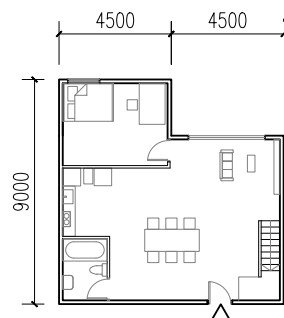
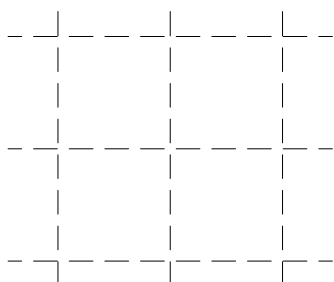
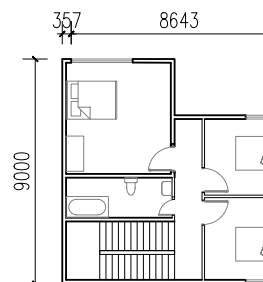
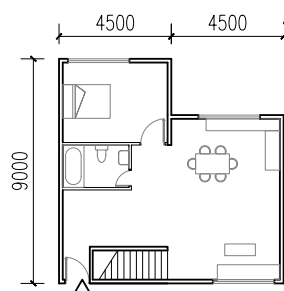
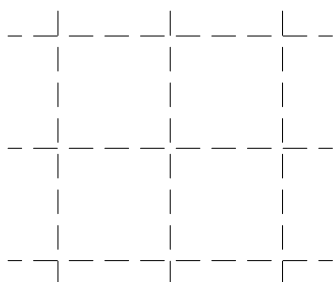
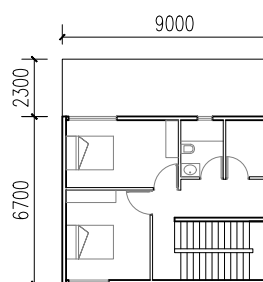
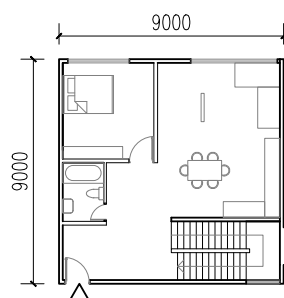
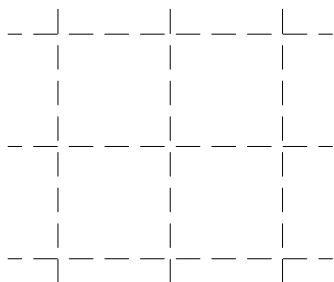


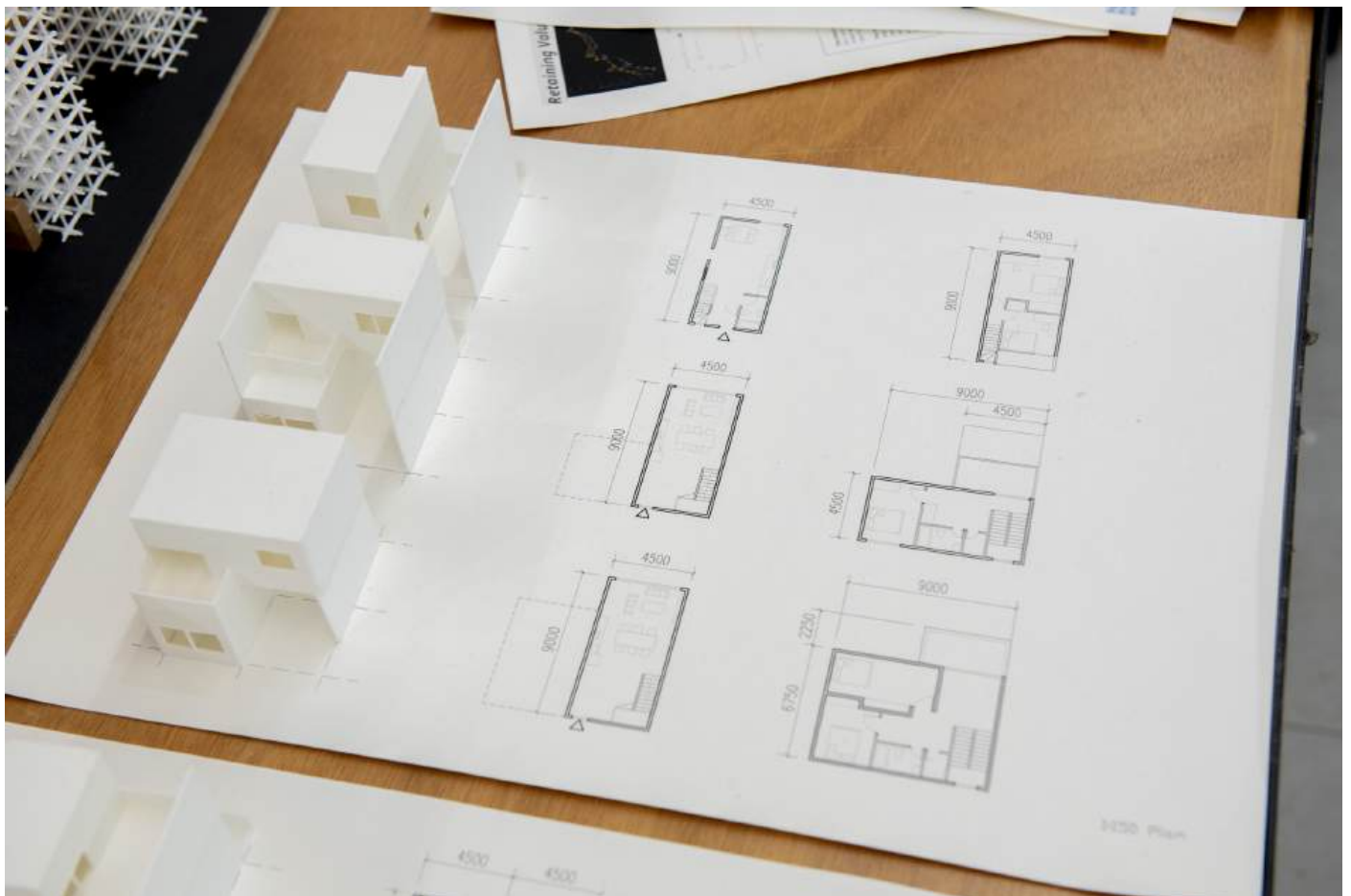
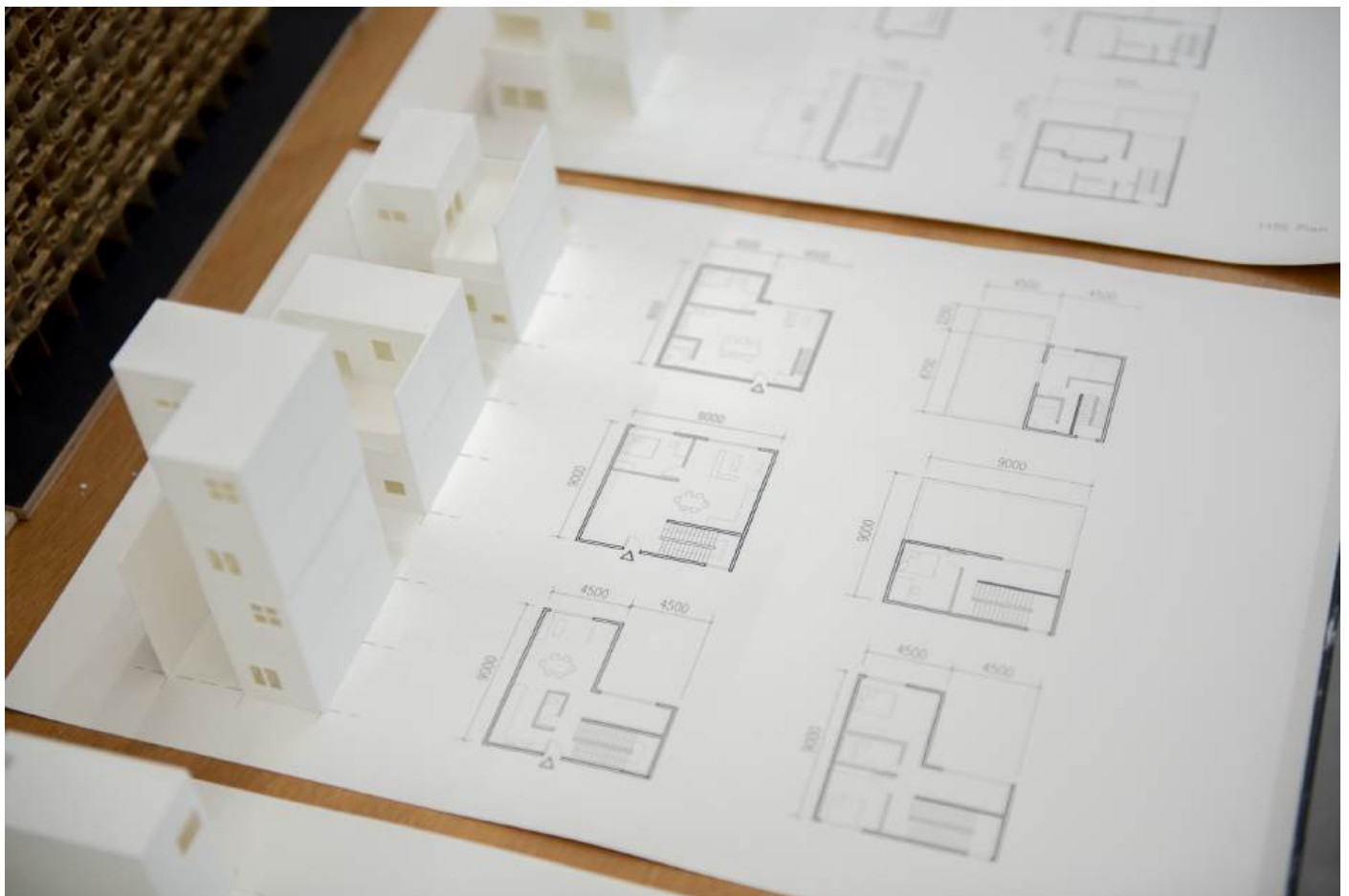


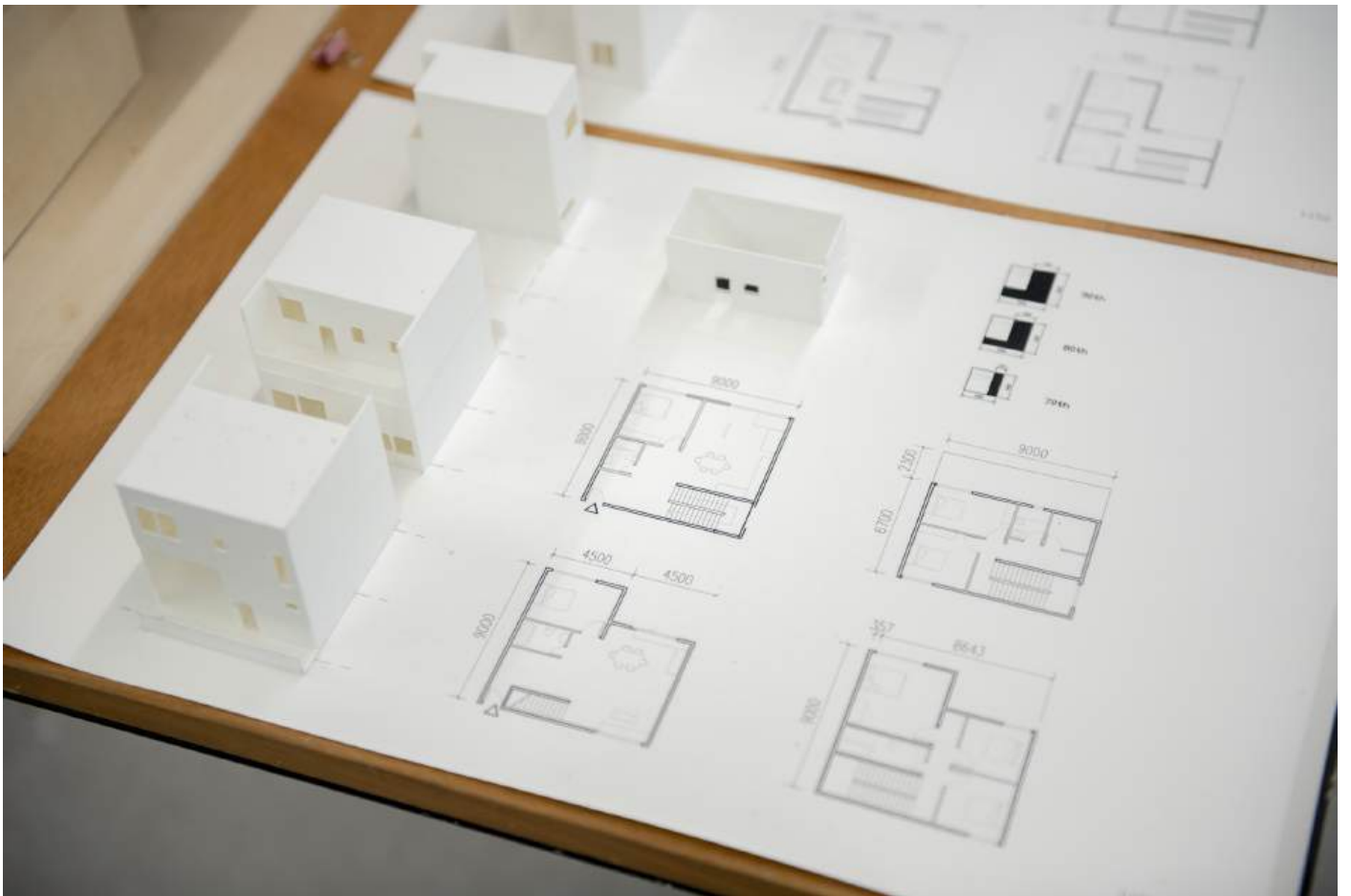


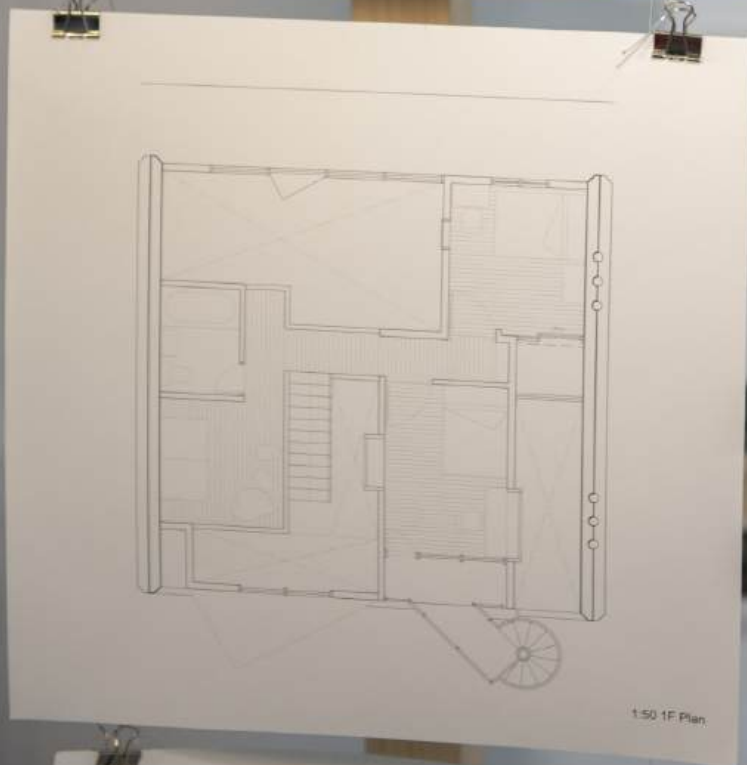
a grid of 4.5m x 4.5m were set up as unit module's
for both prototypes



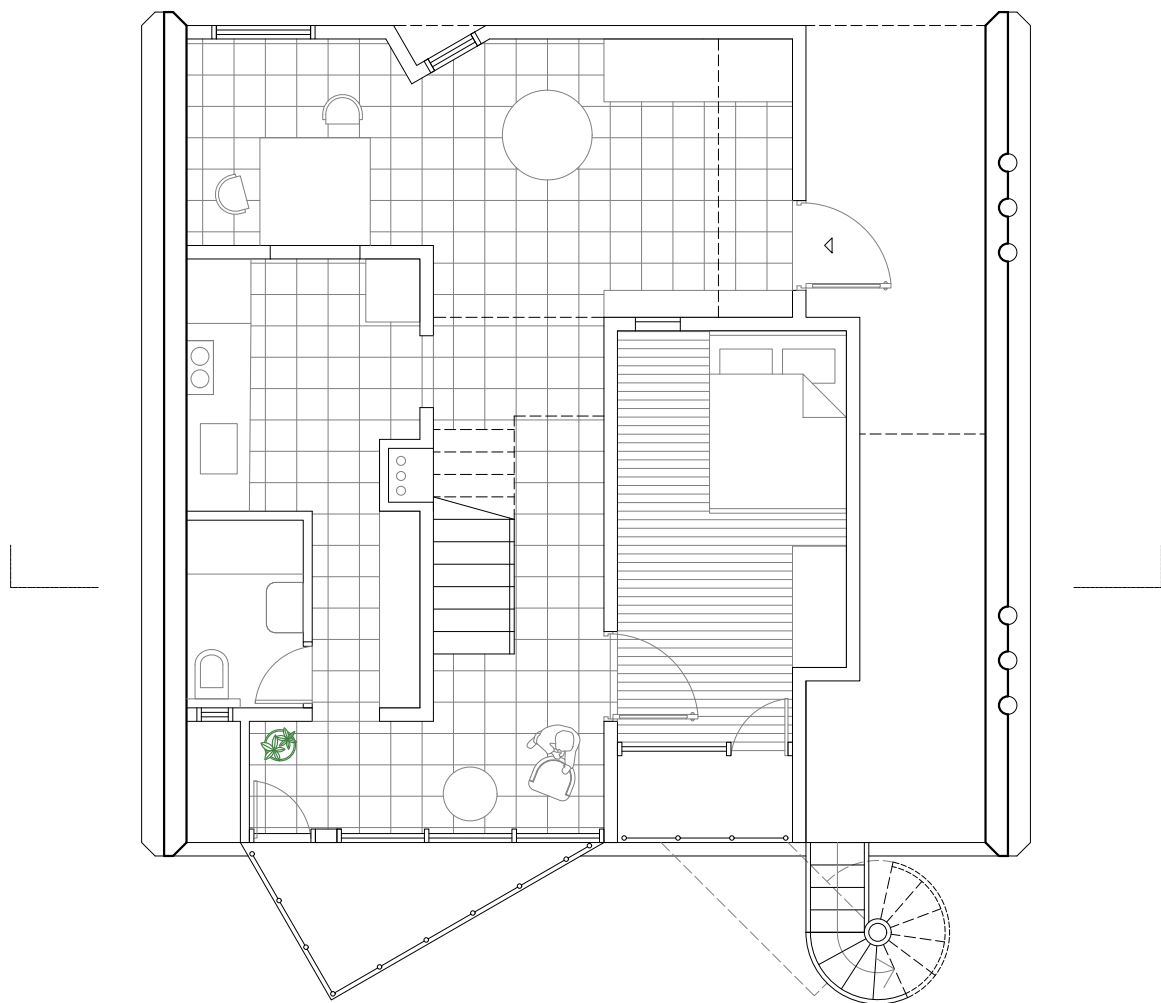






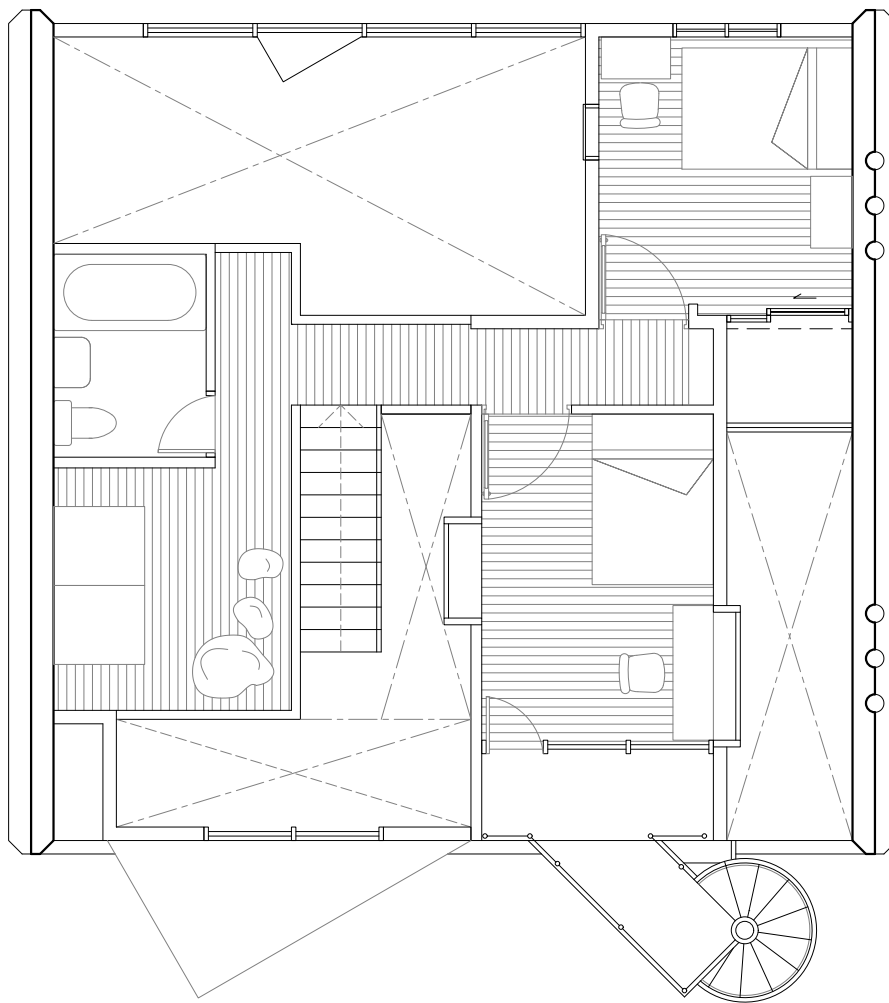






1:50 GF Plan

a 9m x 9m house



1:50 1F Plan







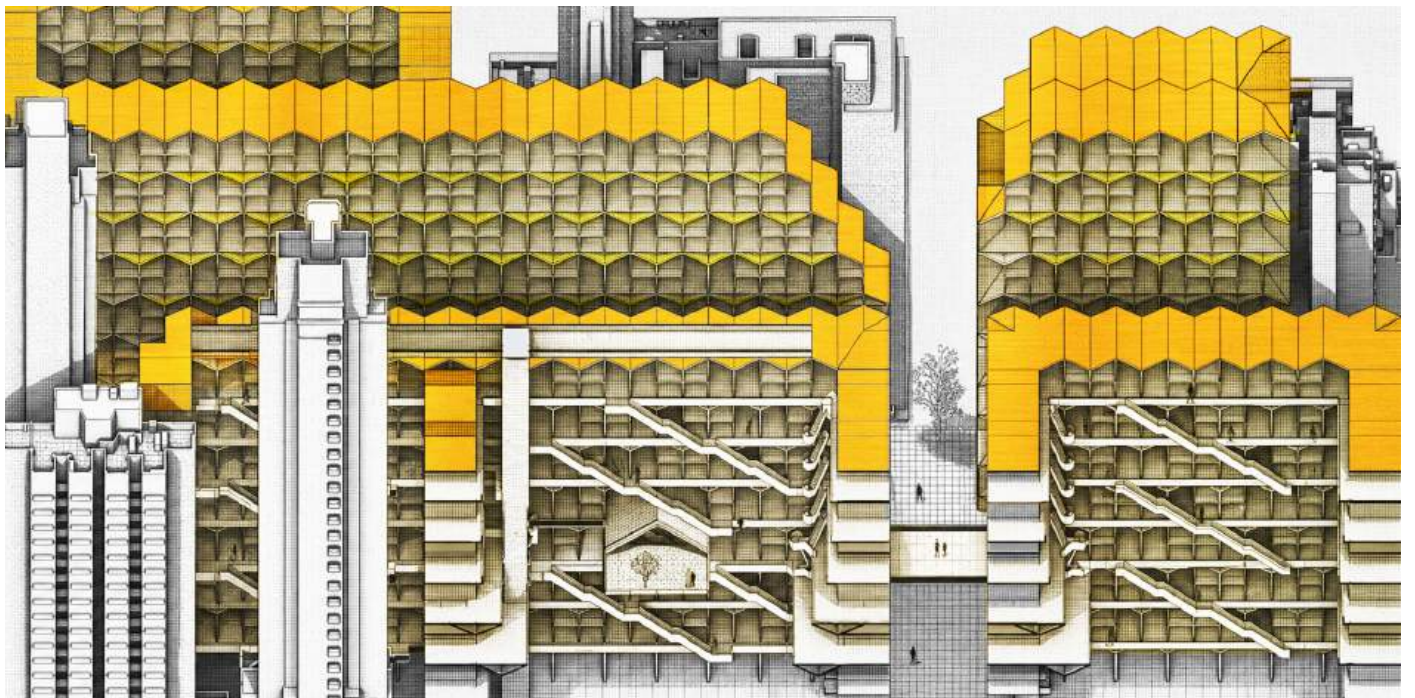


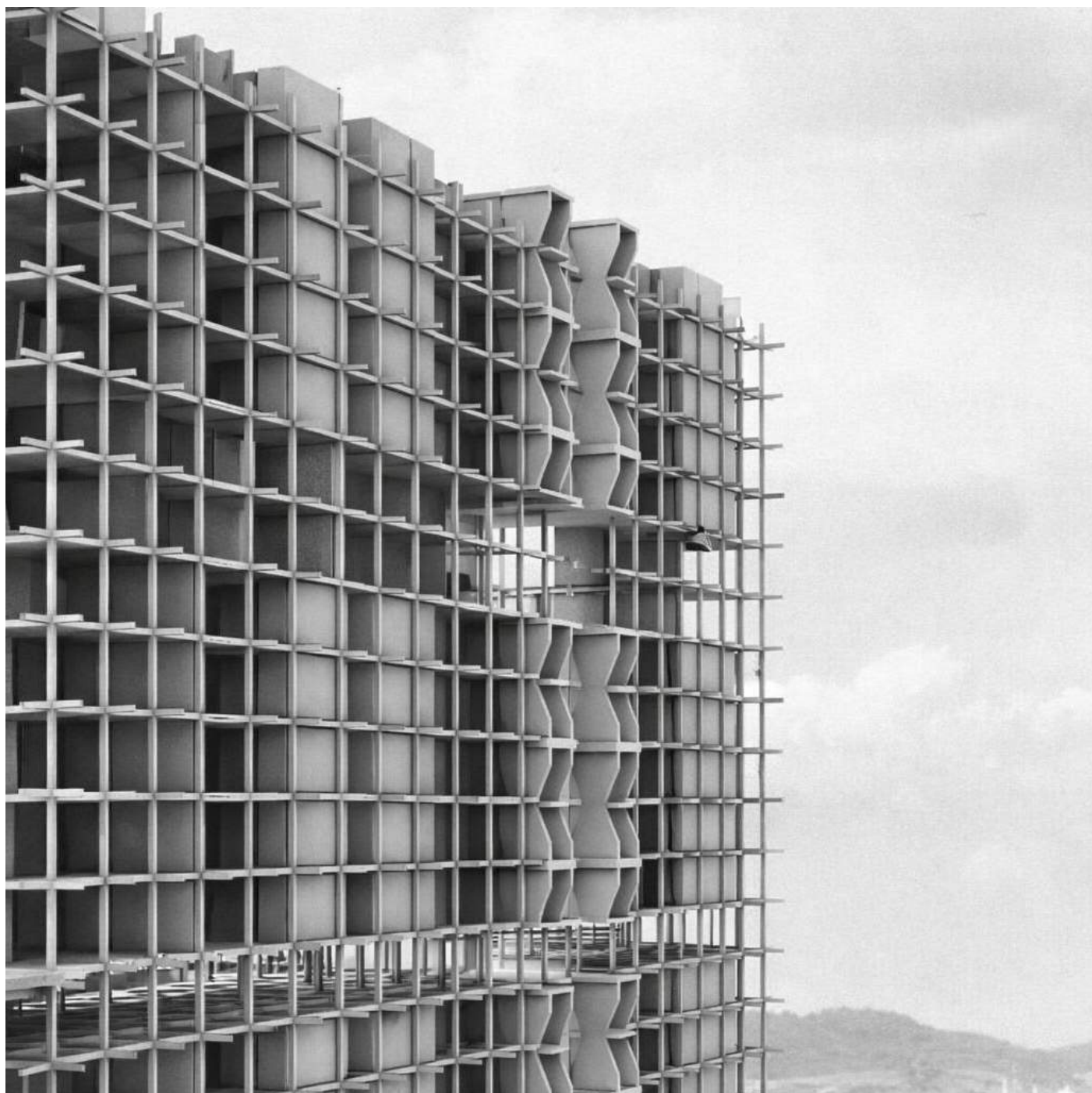




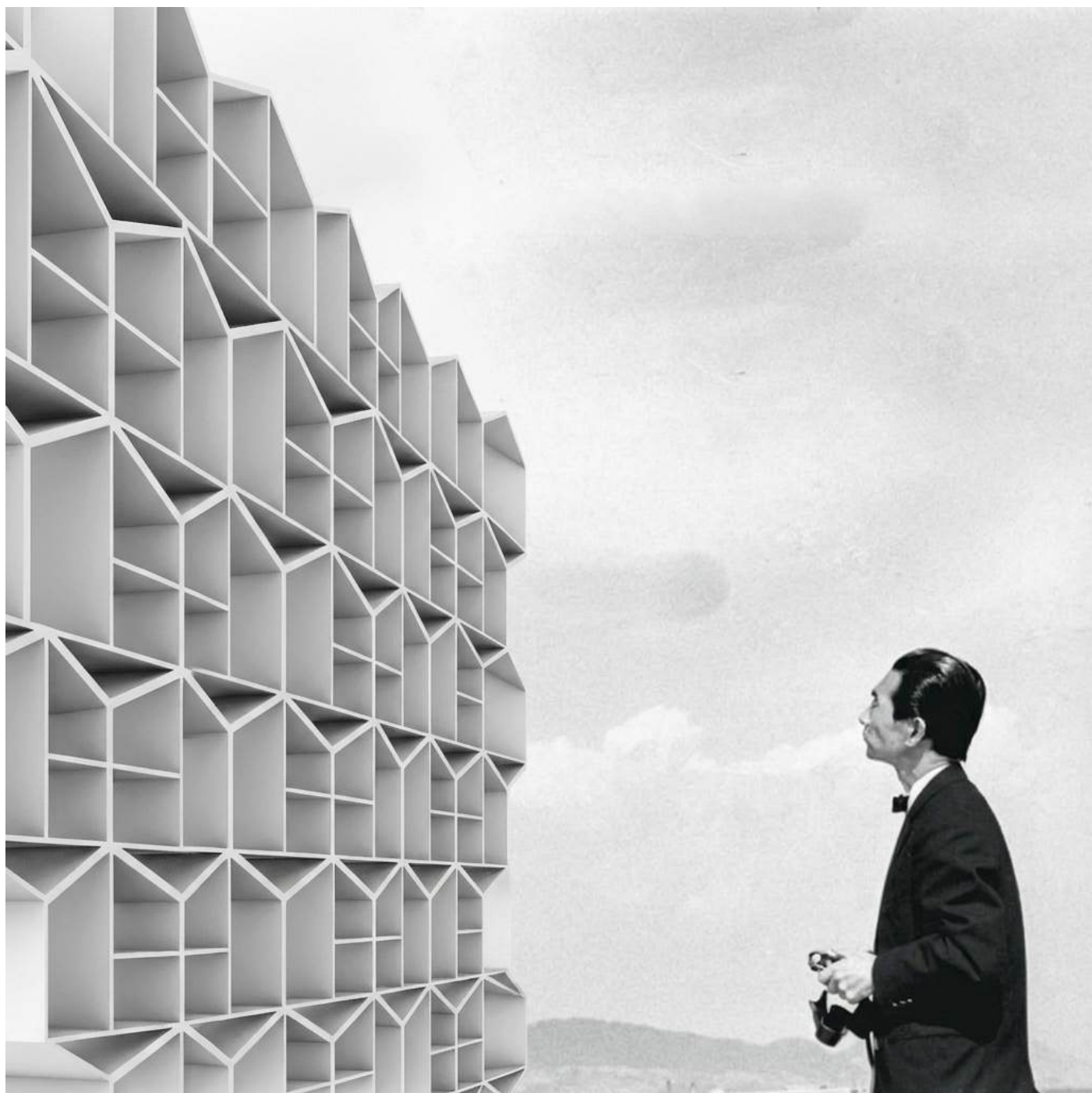
Methods | **Phase 4: Prospective Architecture and Investment-Driven Communal Space**

The spatial organization is designed as a “prospective architecture” that continuously evolves. Rather than remaining static after construction, the building actively adapts to new capital injections from retail investors from the surrounding community. These investors, seeking financial yield or civic and retail opportunities, can pool their governance tokens to fund and construct new communal spaces within the permanent structural grid. This prospective development might include elevated skybridges, cafe, retail activities on the ground floor, productive hydroponic farms, or public tech hubs. By physically separating these high-yield commercial zones from the protected residential sanctuaries, the architecture allows outside investment to constantly upgrade the community's shared infrastructure without ever displacing the middle-class residents.

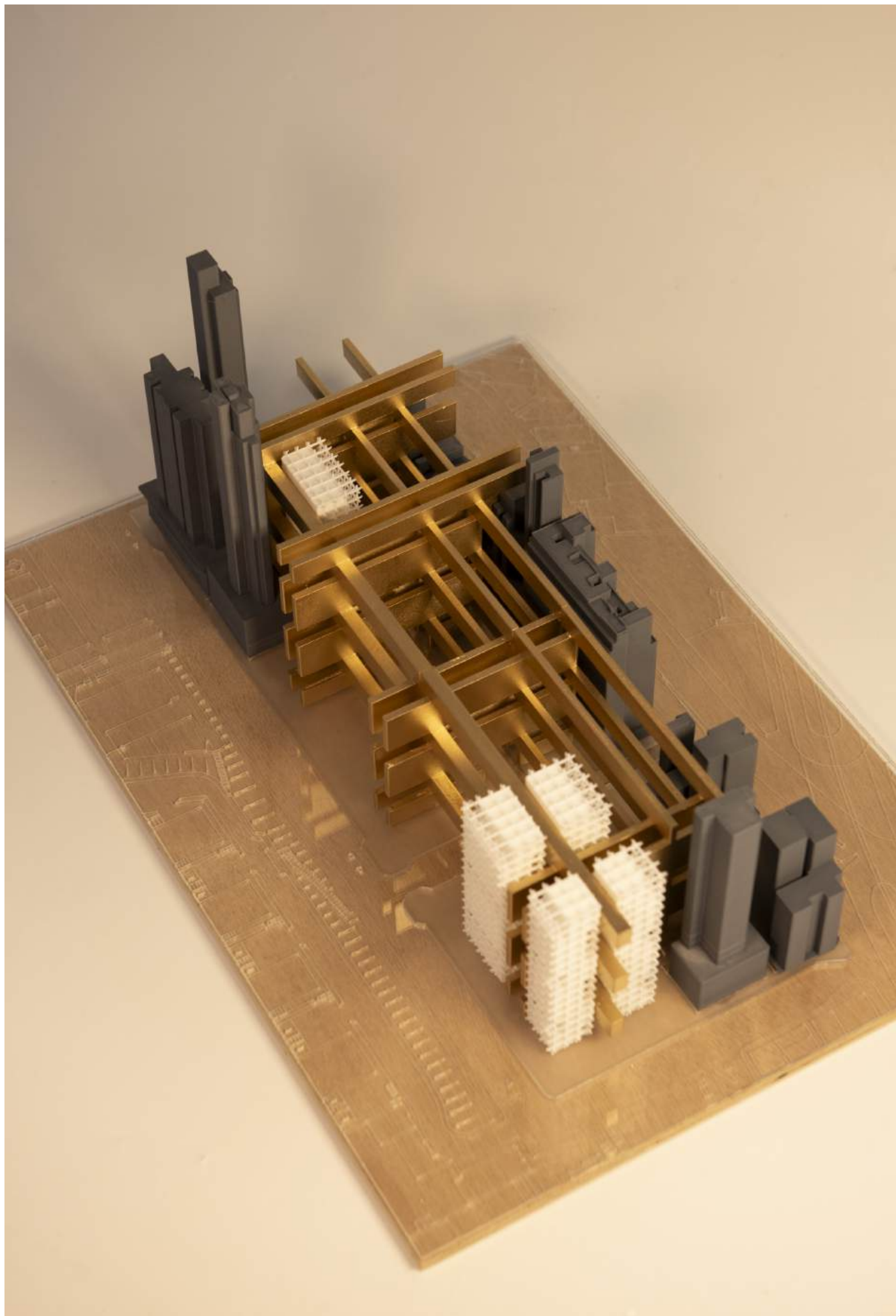


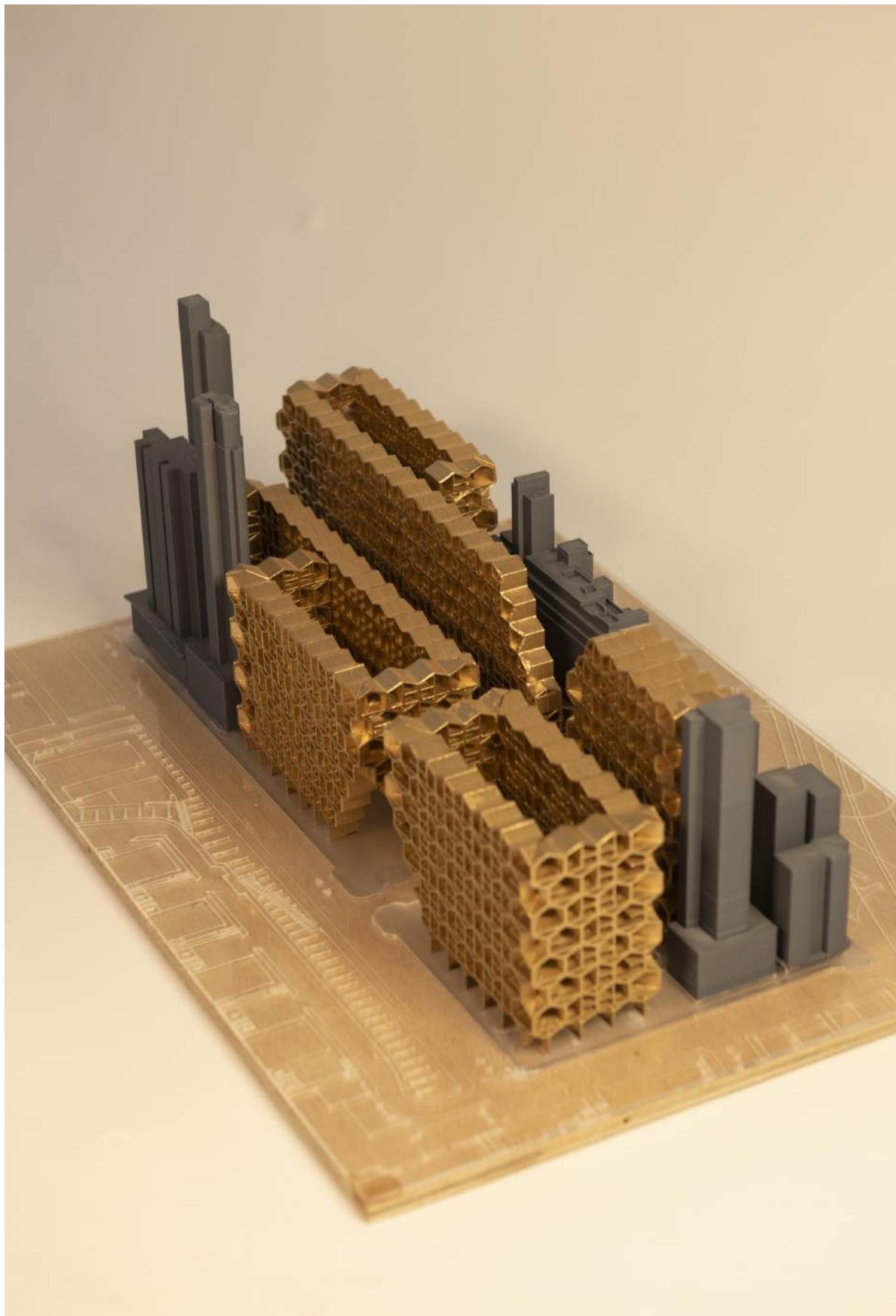


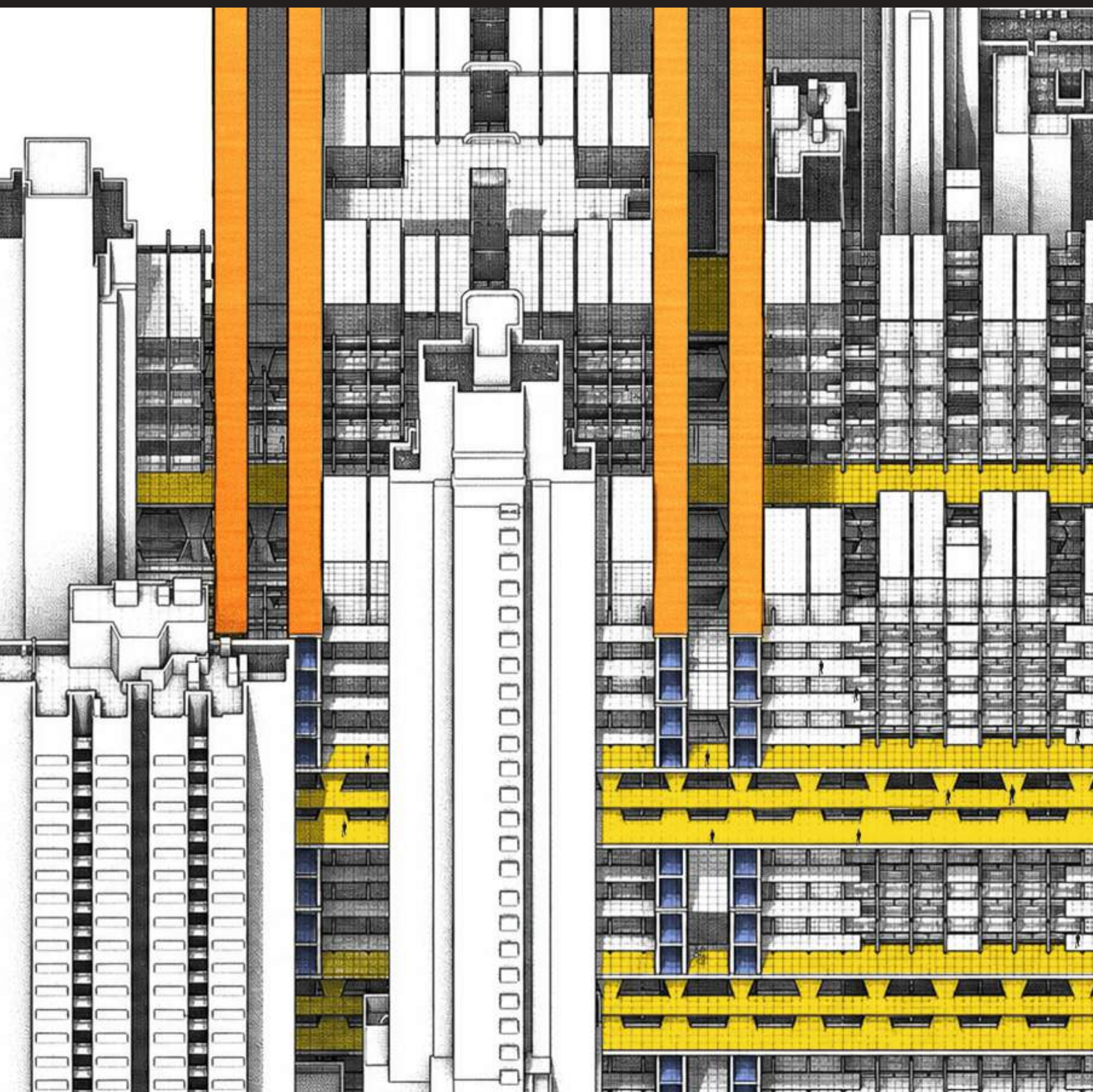
THE INFRA

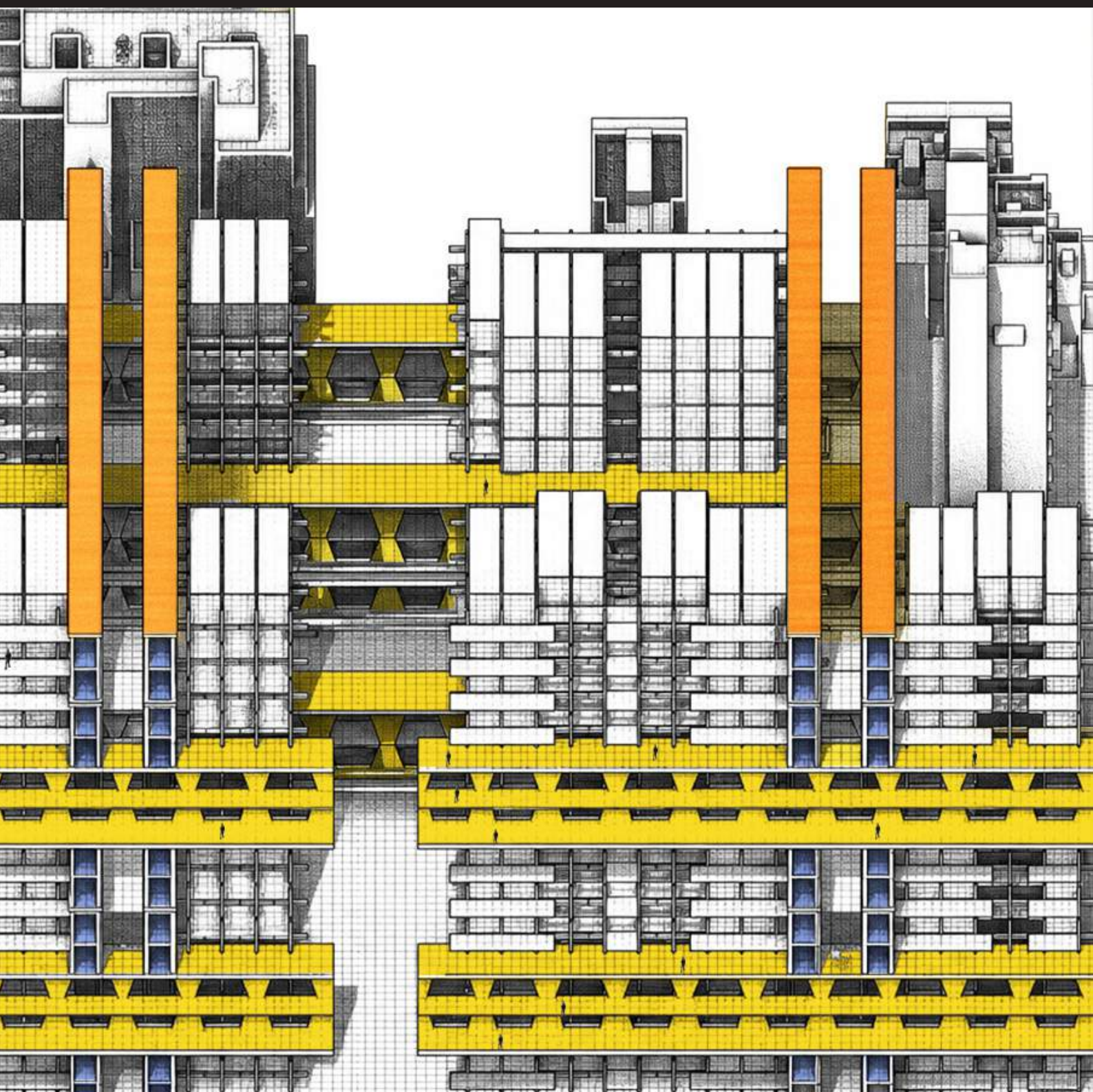


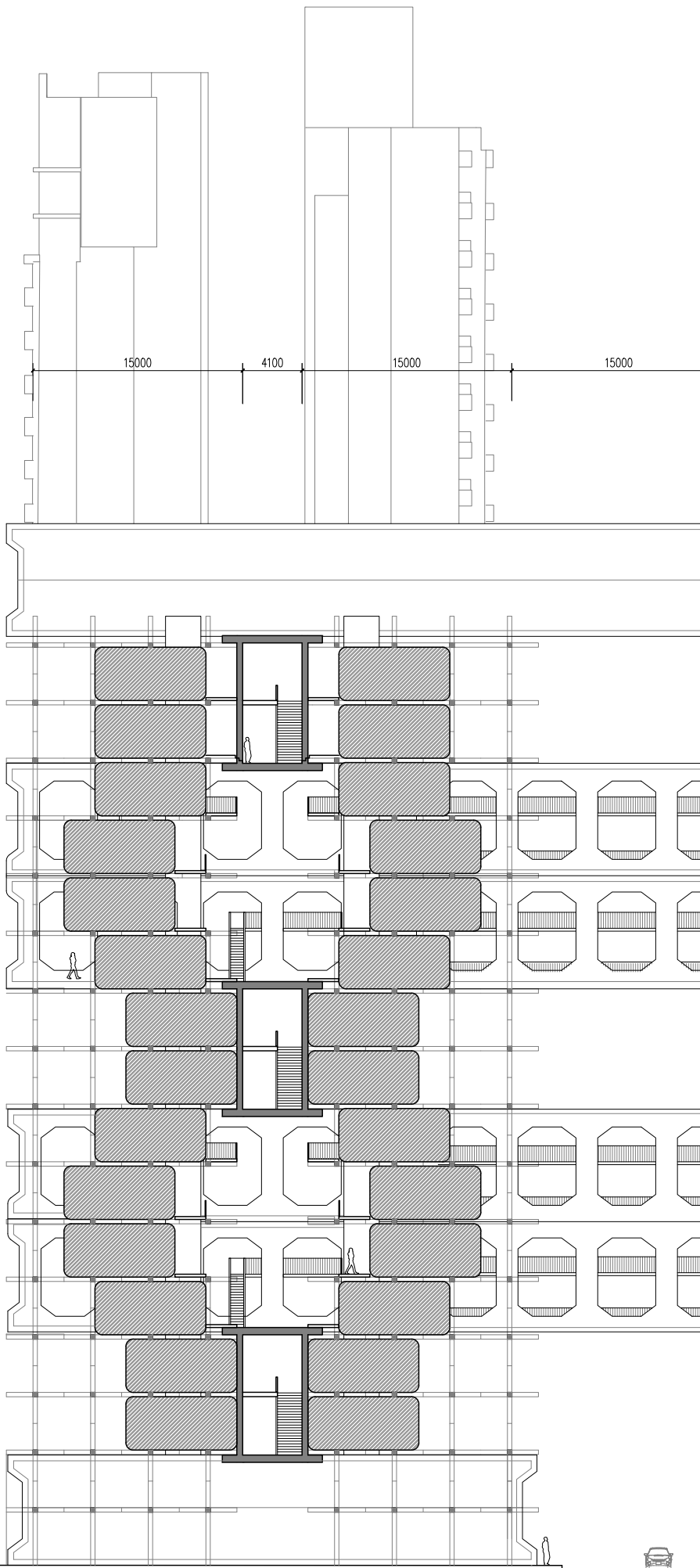
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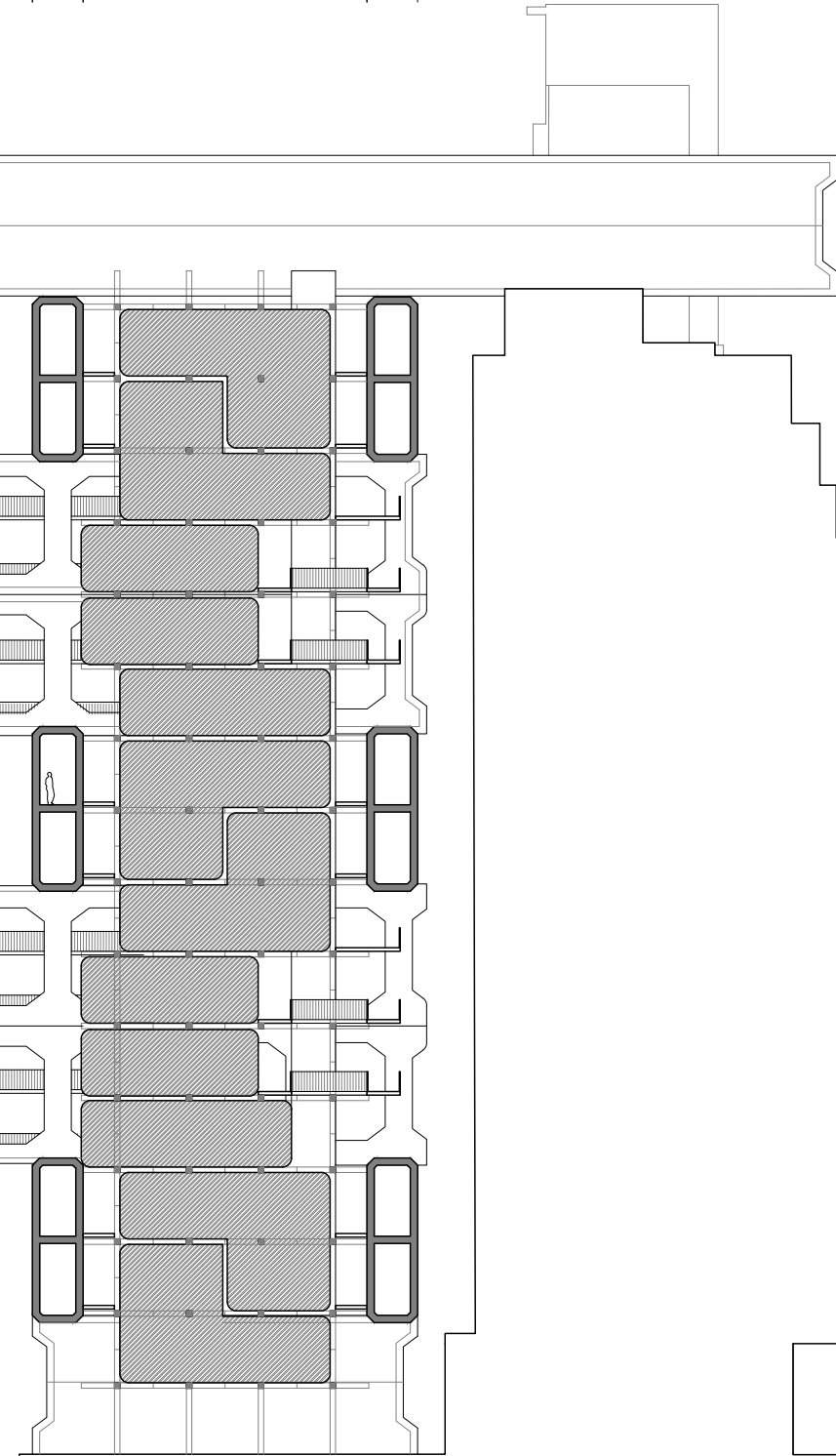




Section THE INFRA

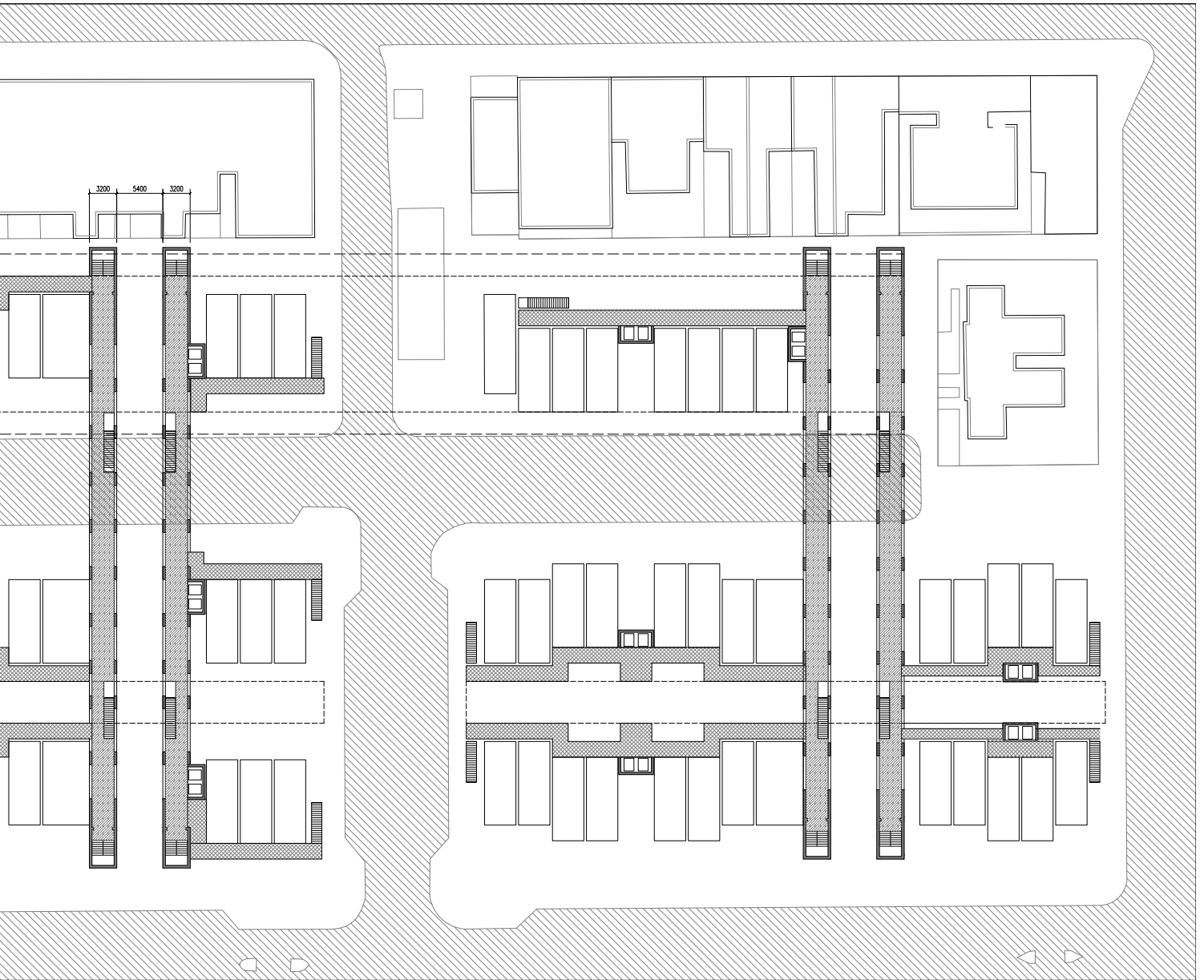
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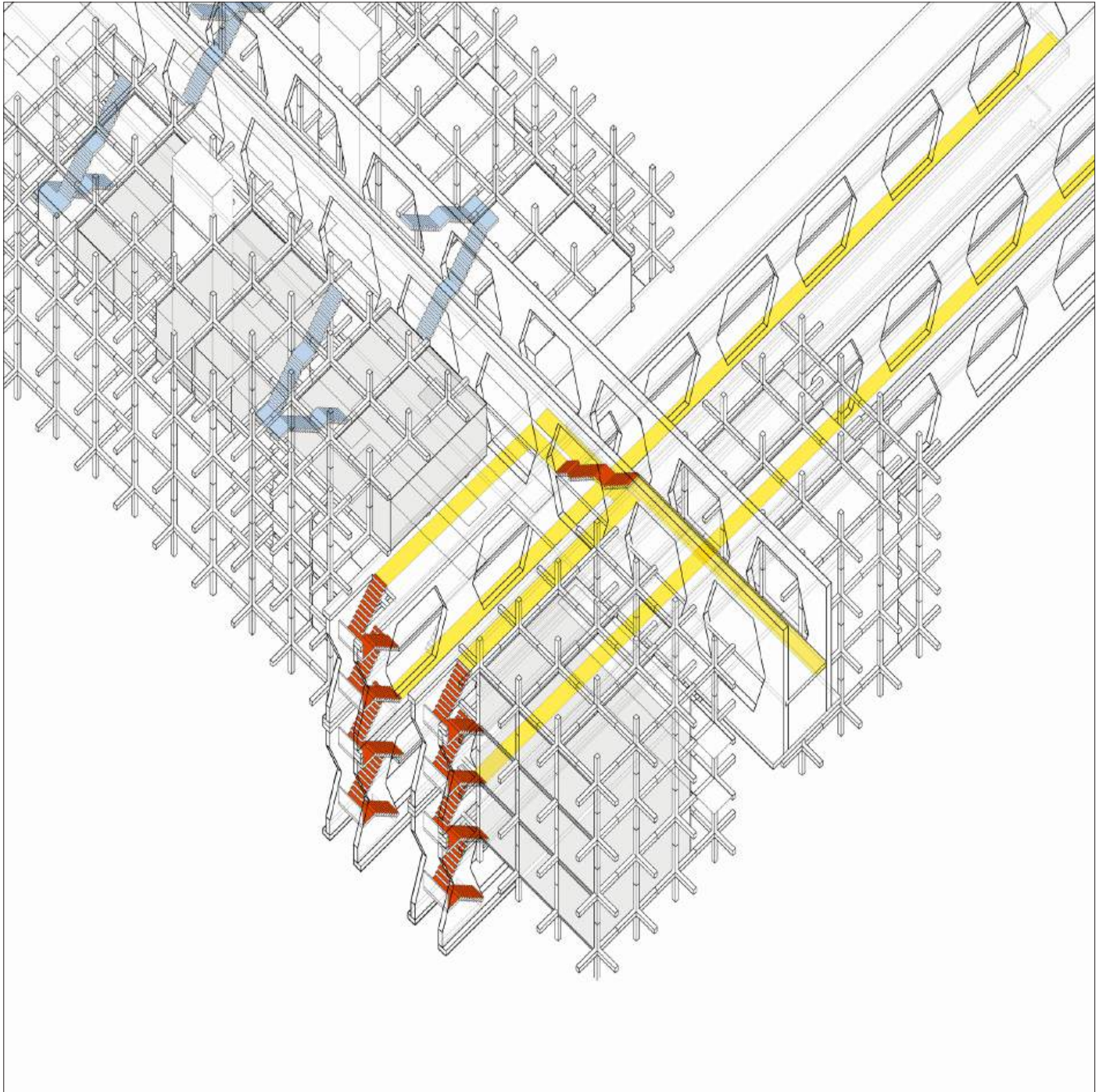
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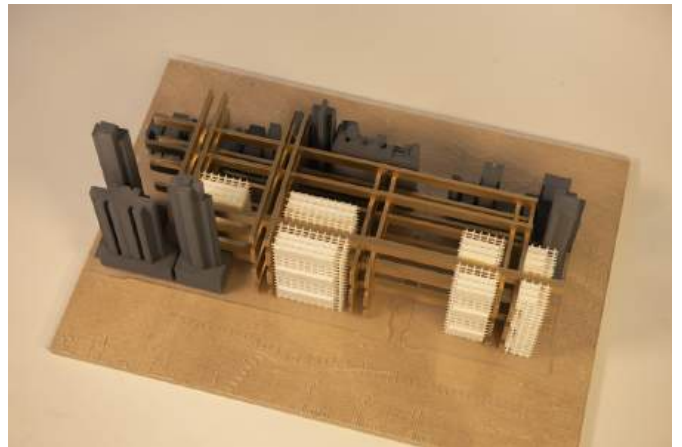
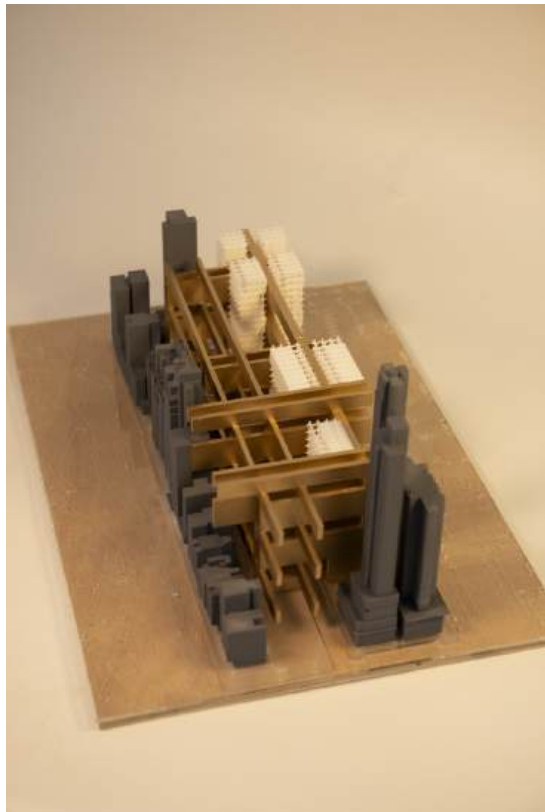


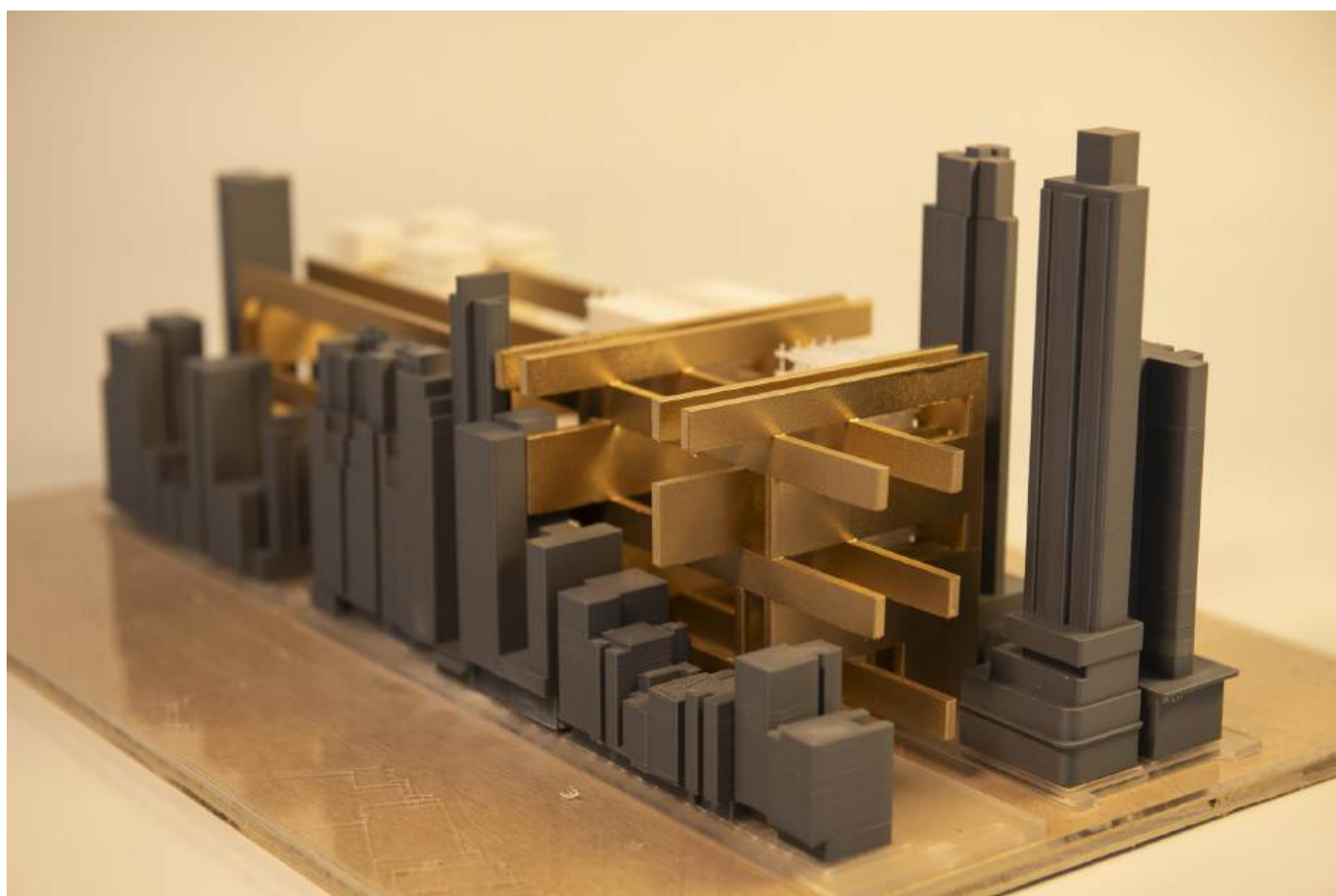


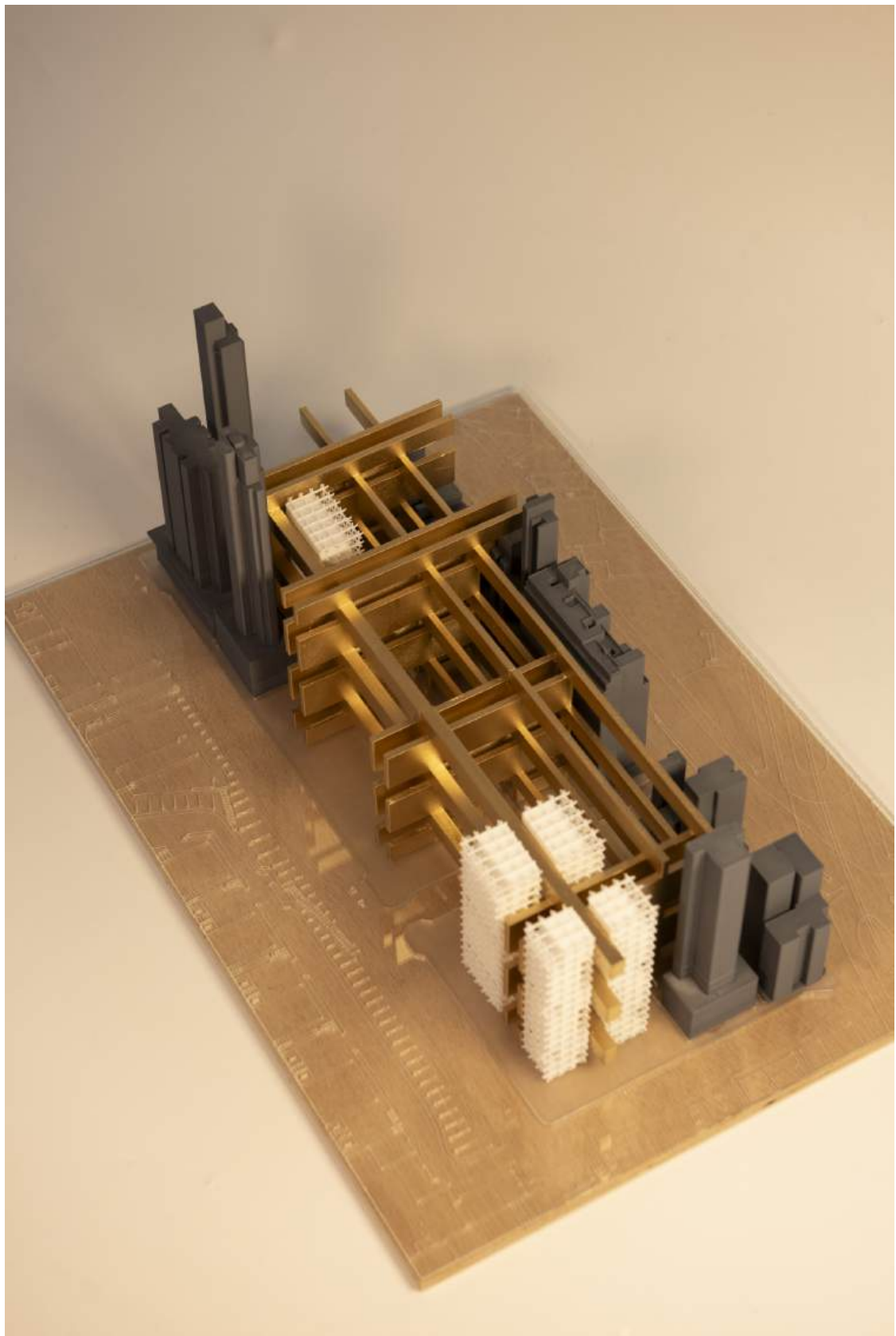
Plan THE INFRA





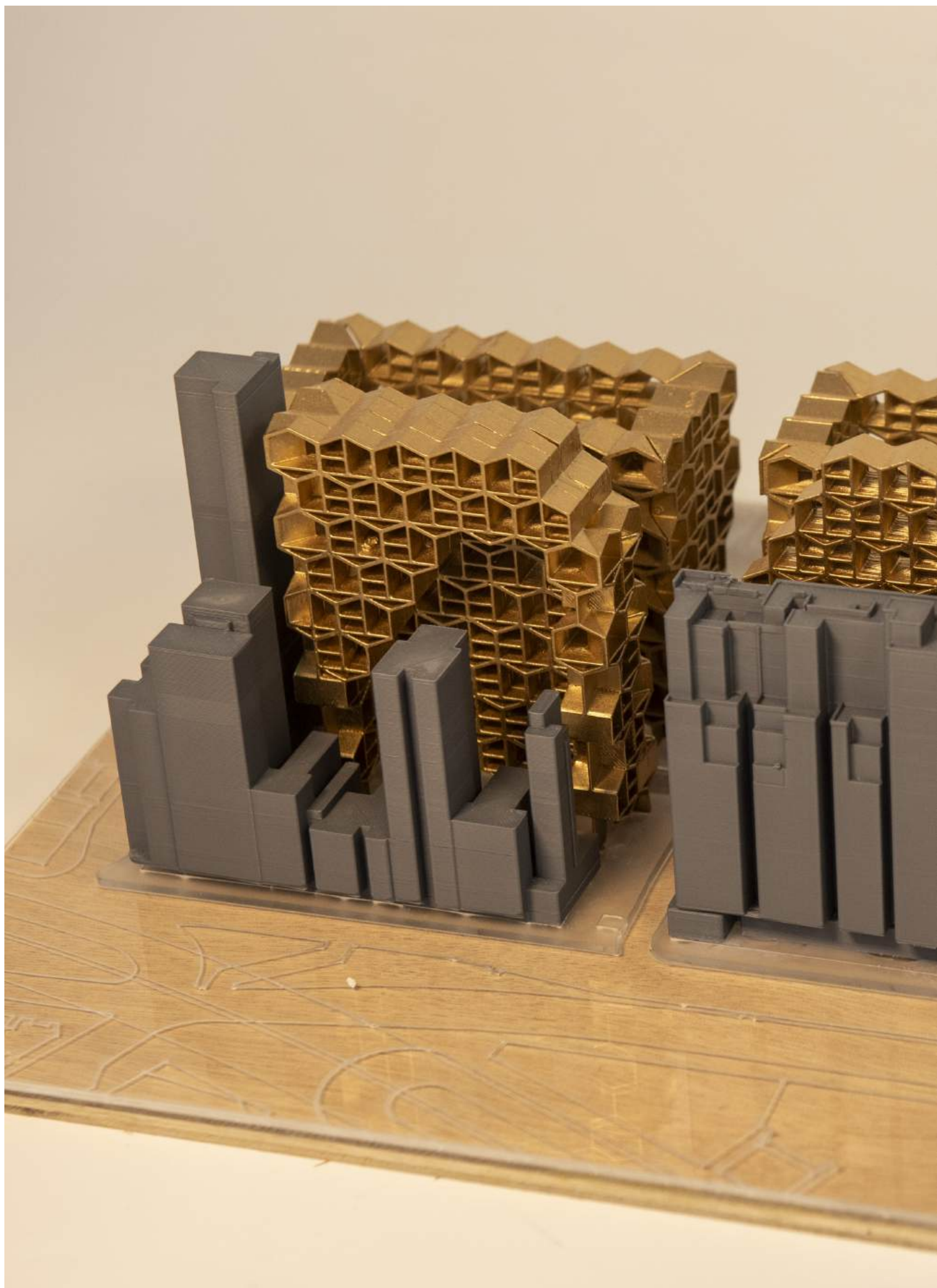




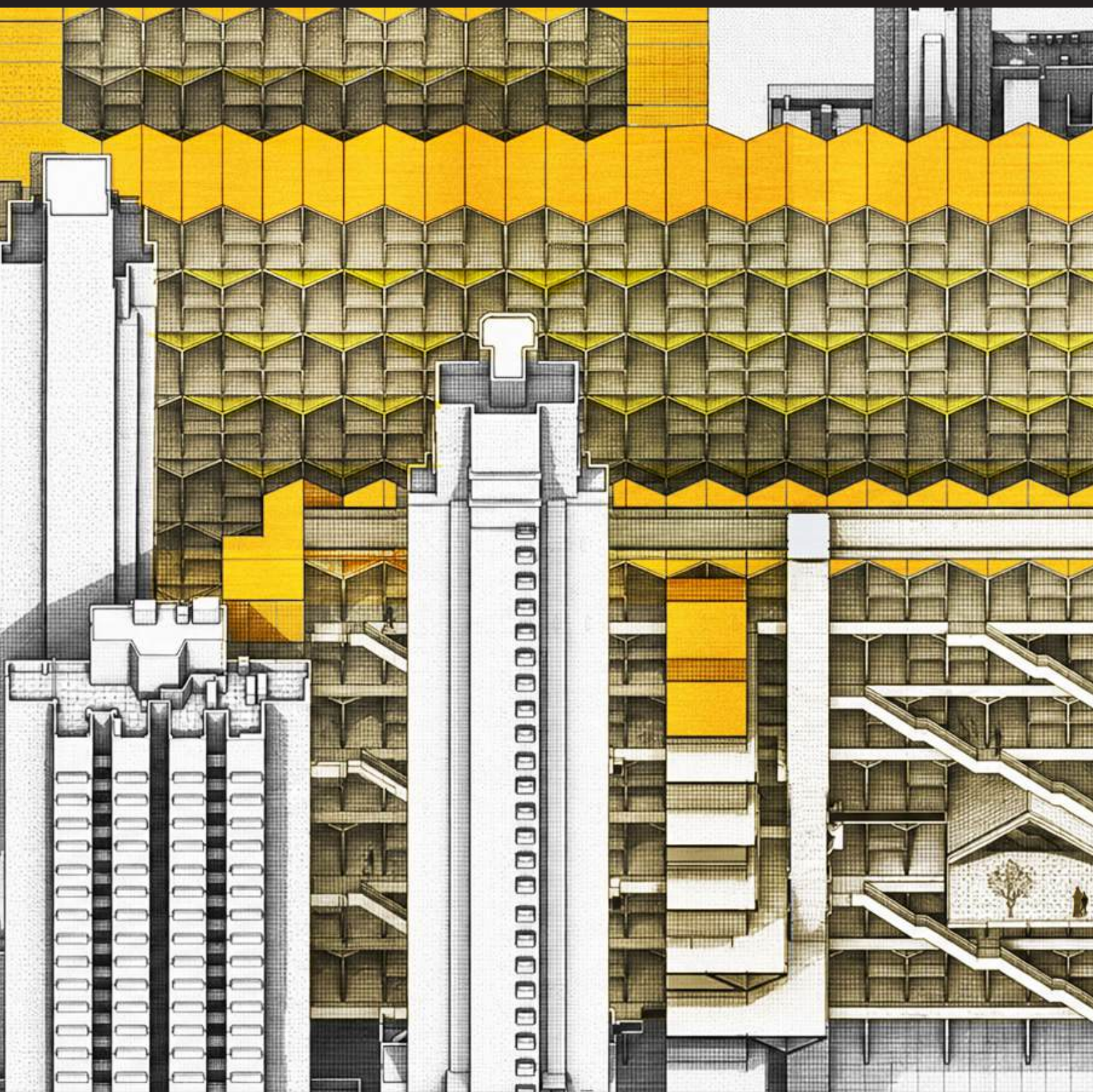




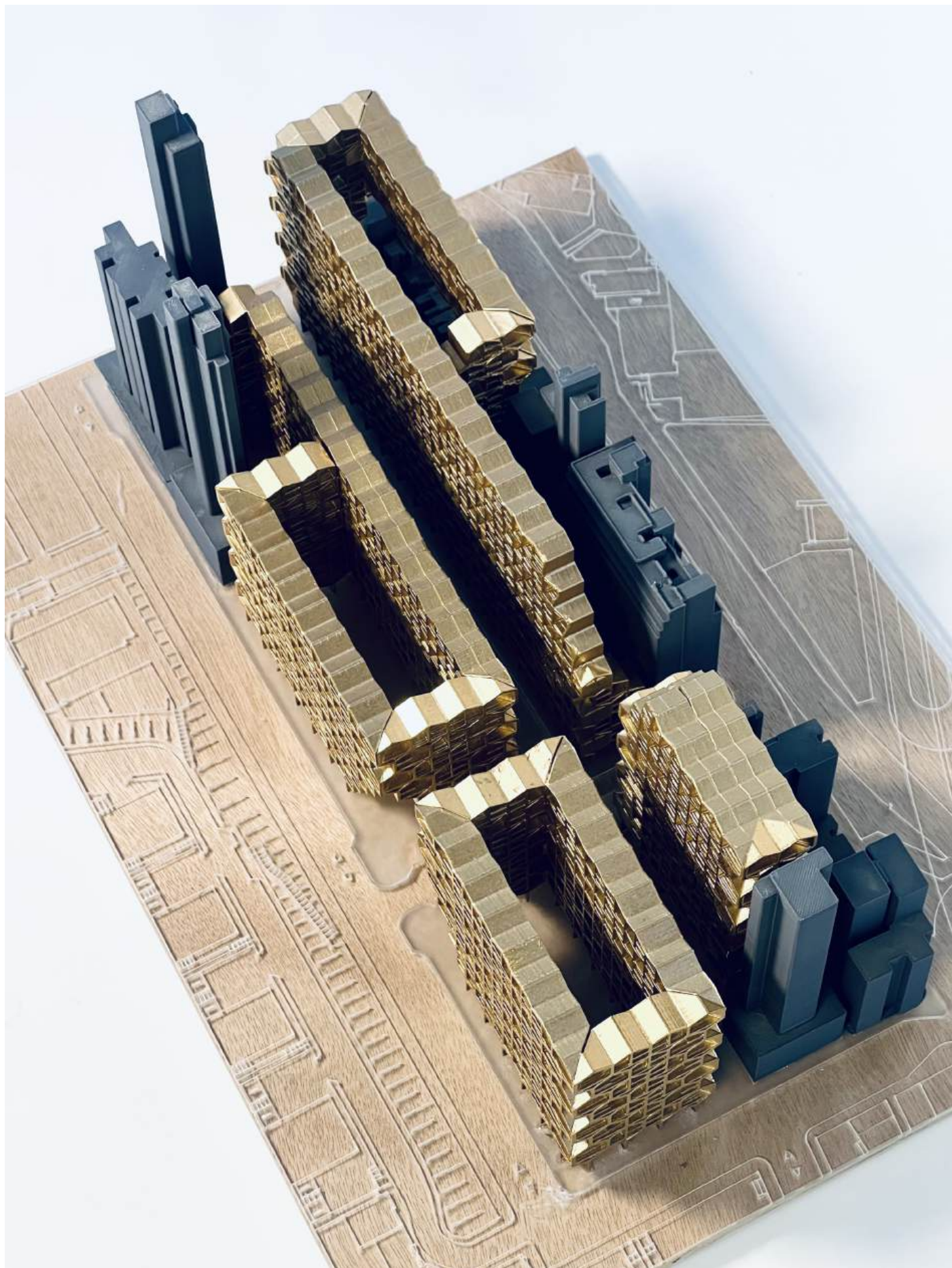


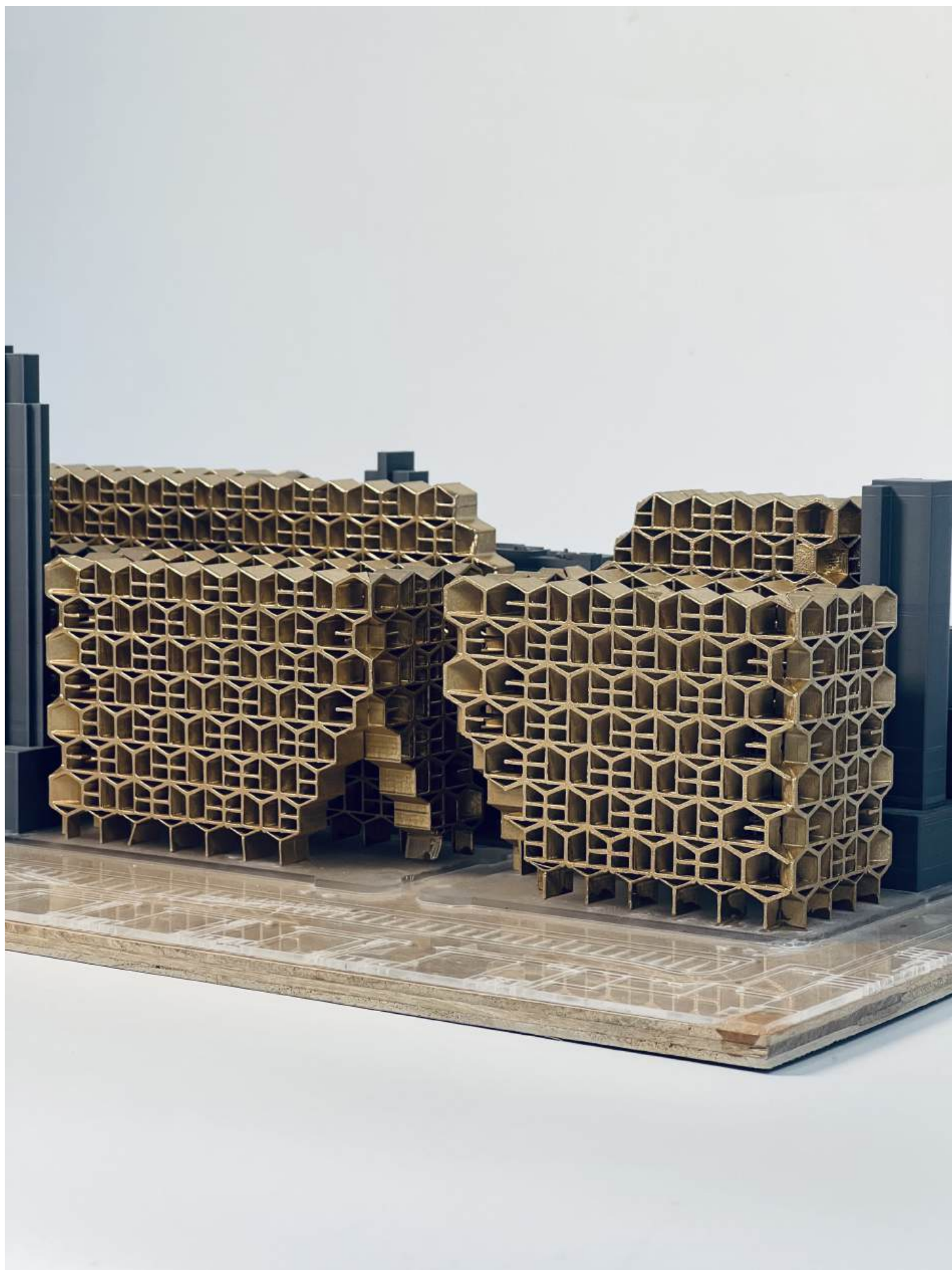


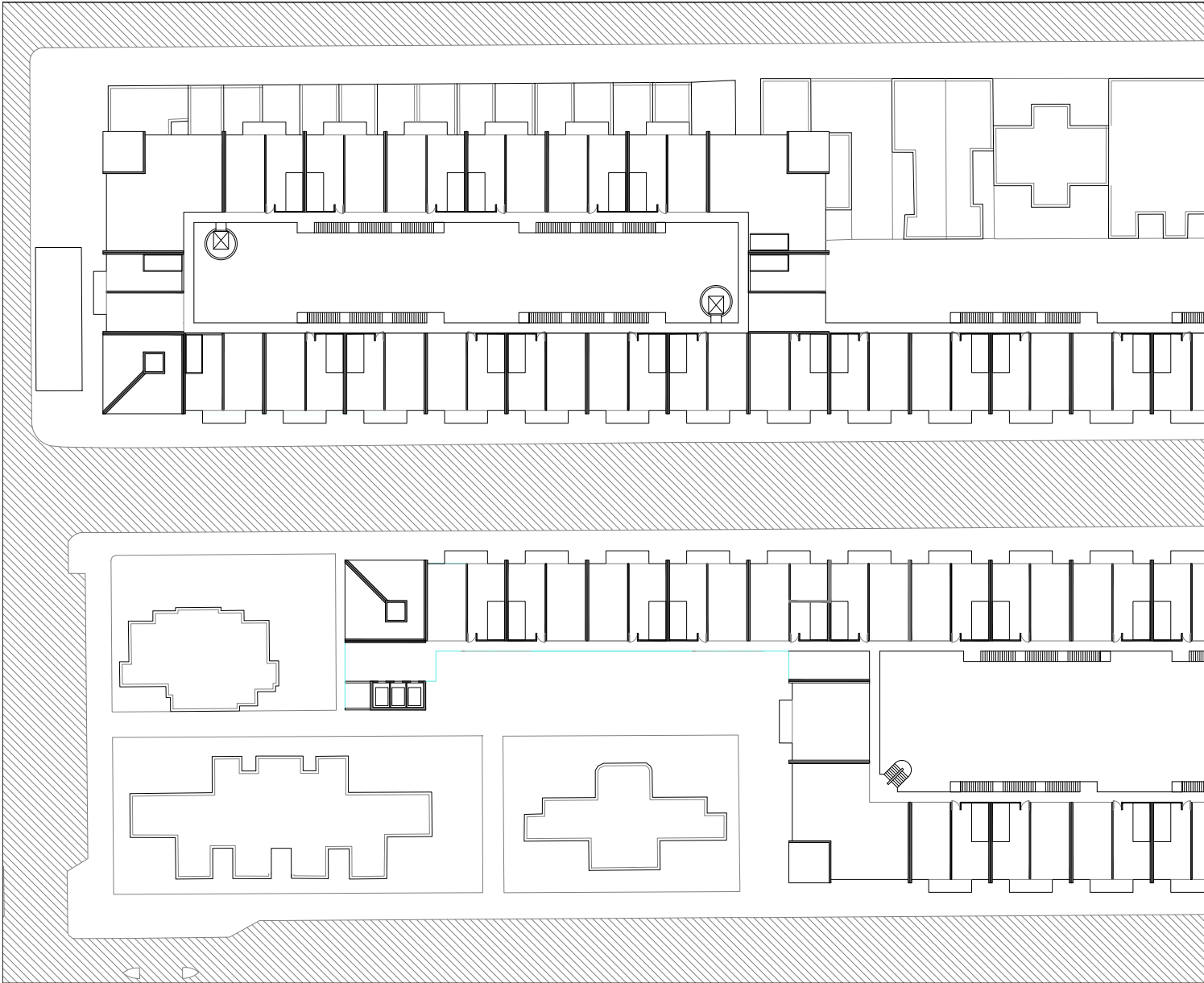




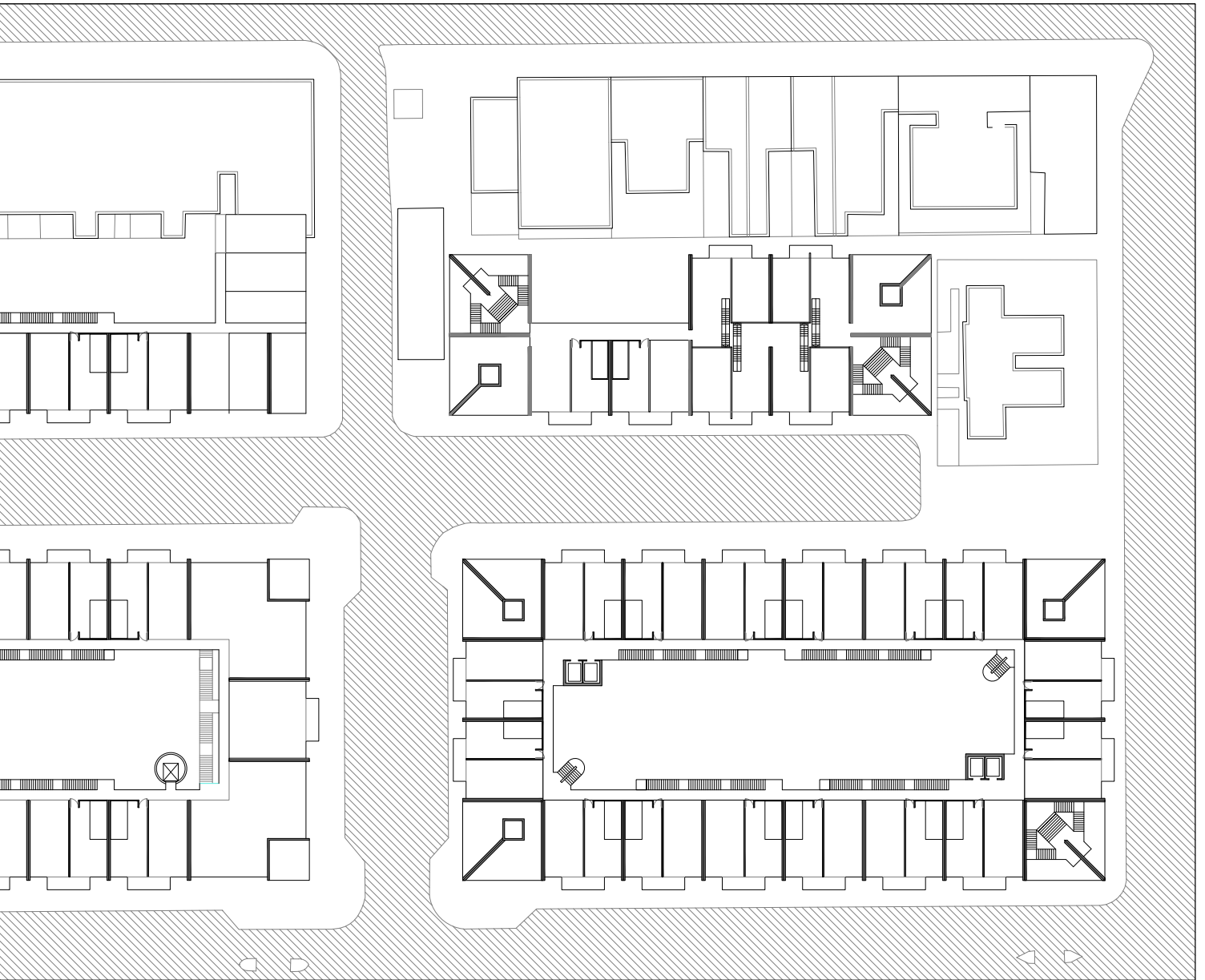






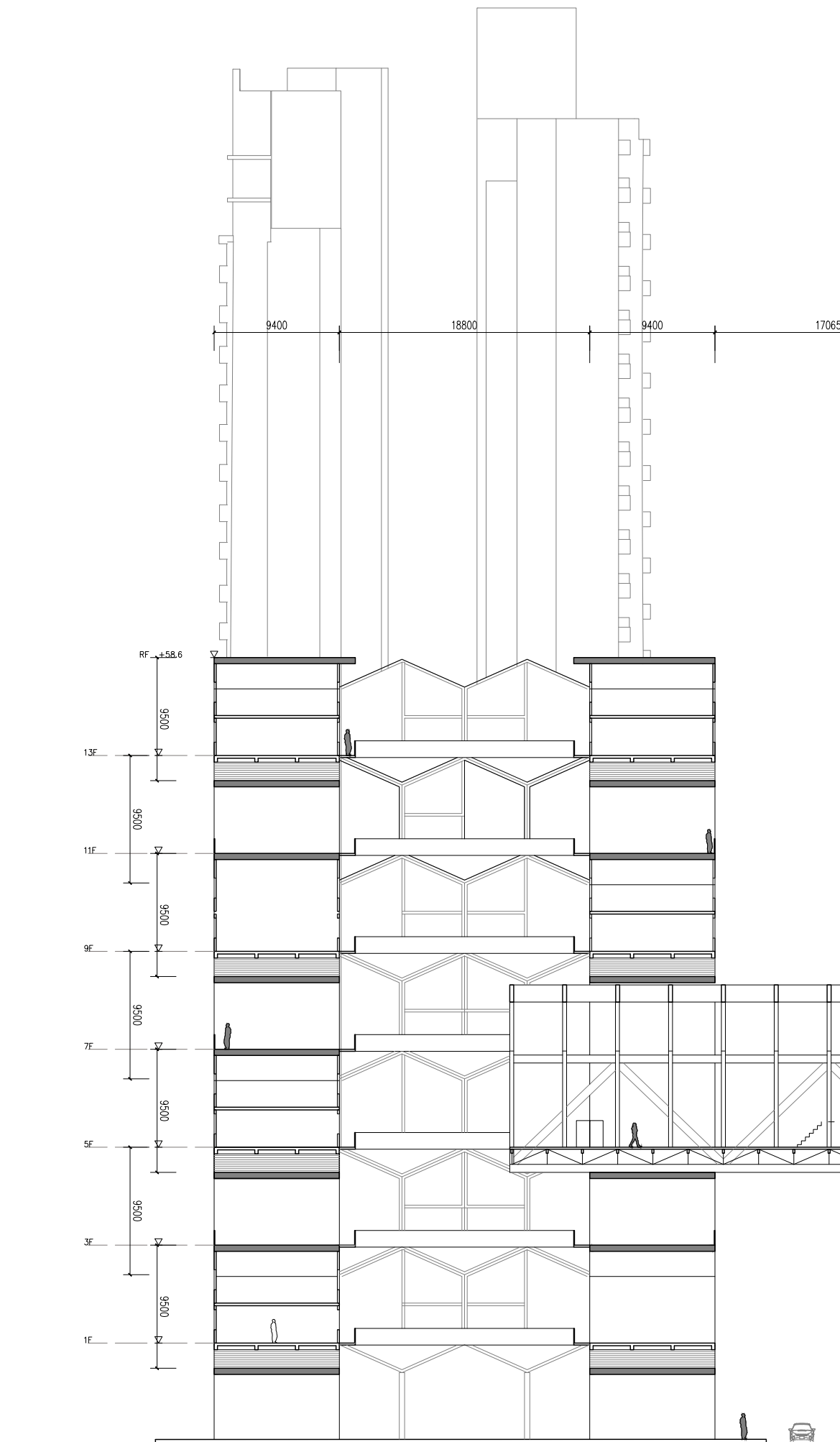


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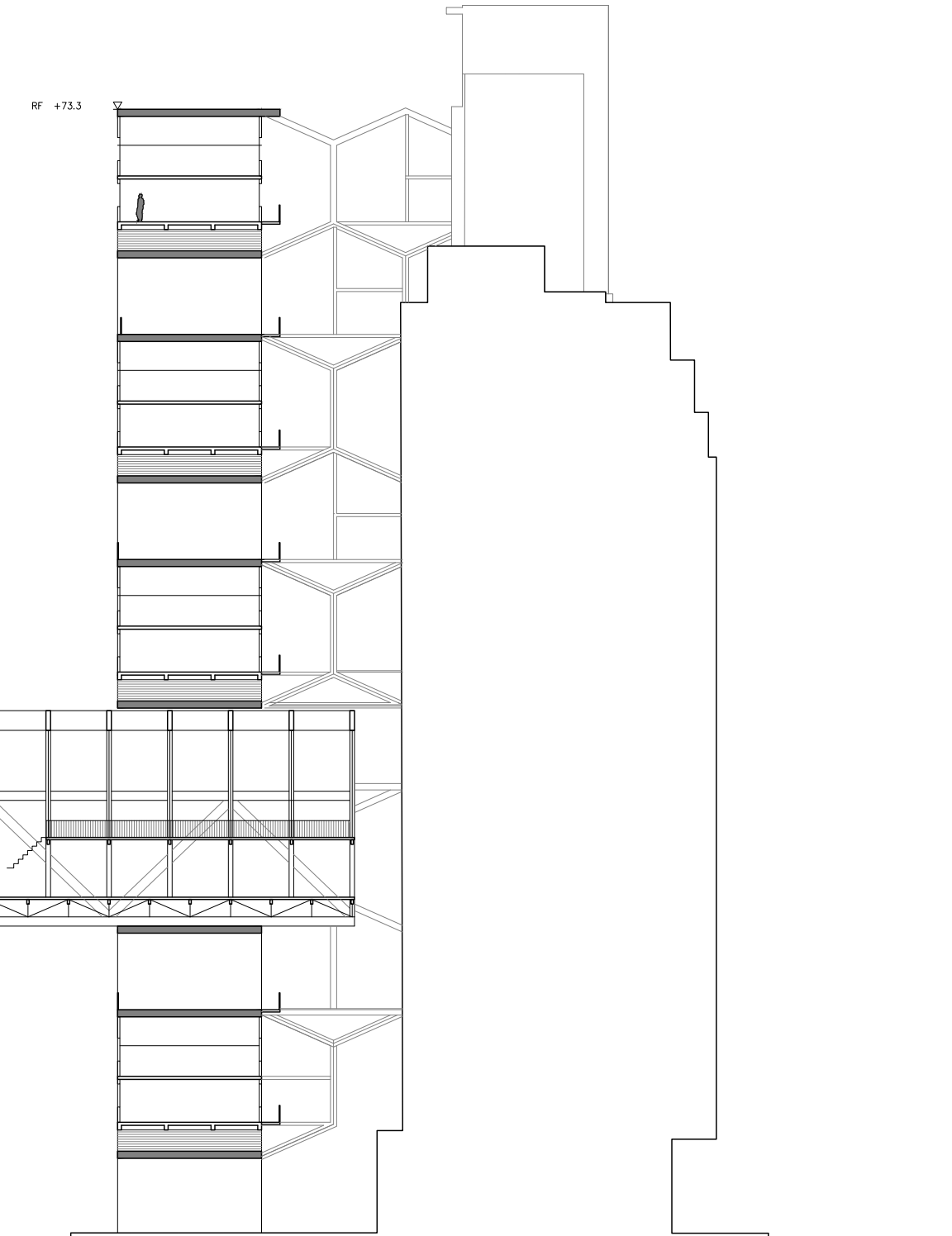


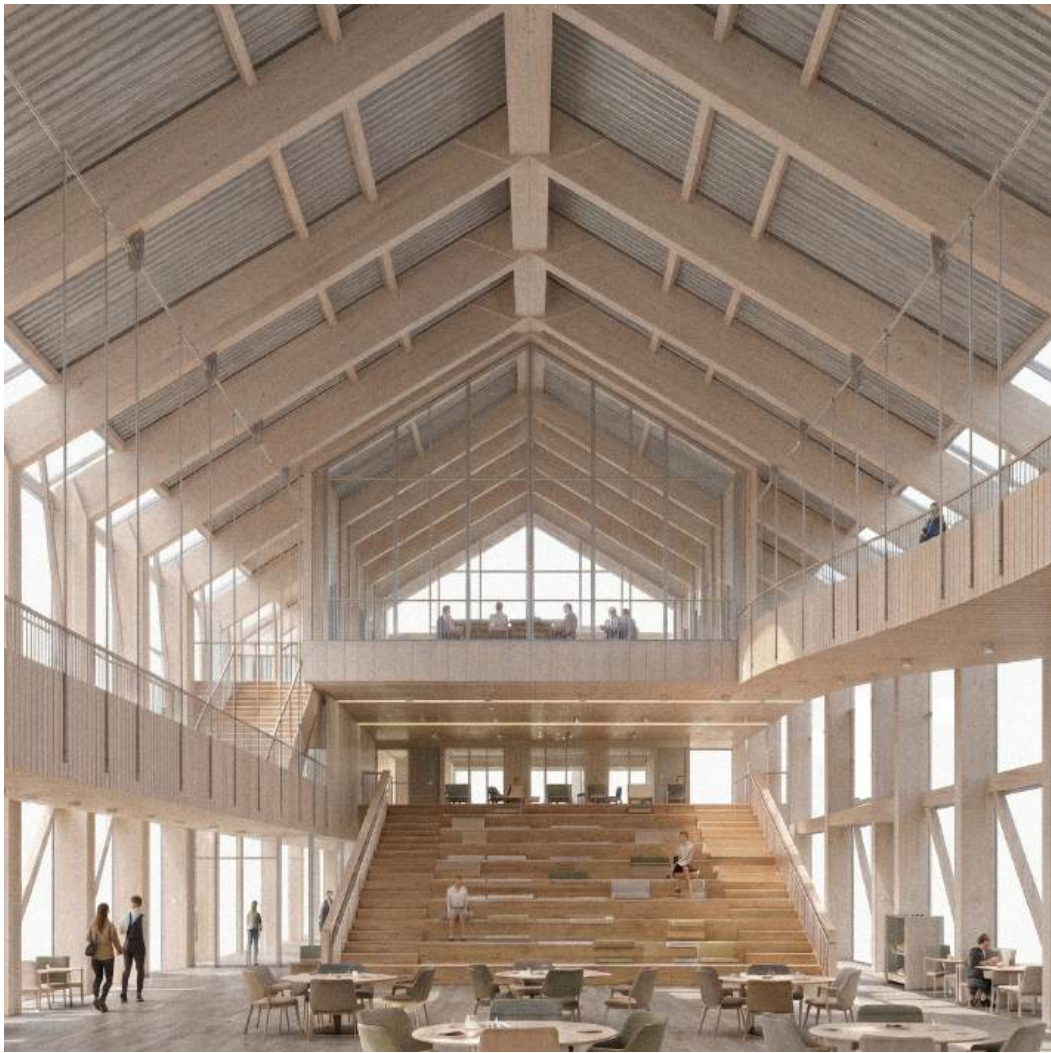
Section THE INFILL

0 5 10m

9400

RF +73.3





Communal Spaces Connecting Building Blocks

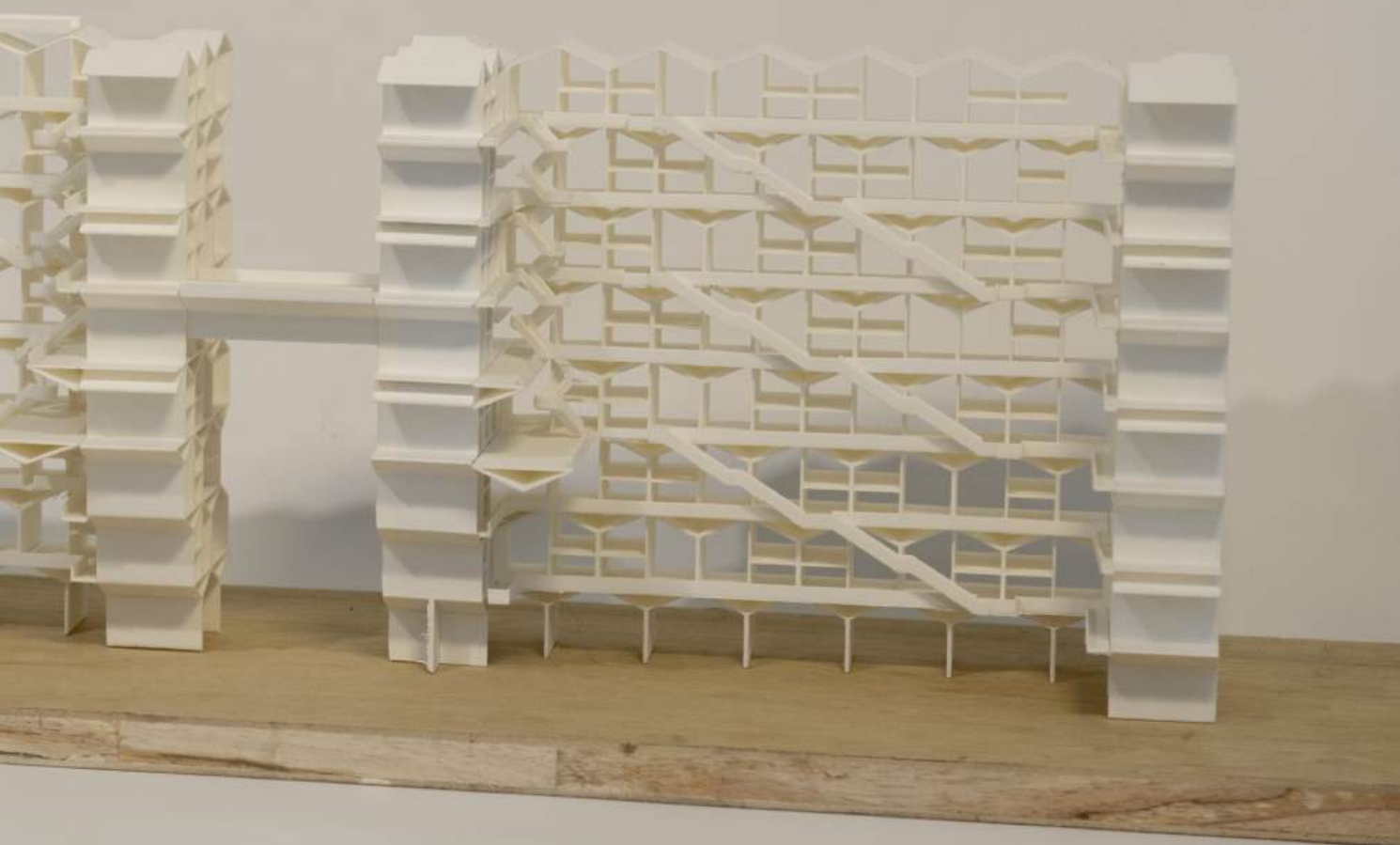


Large Platform and Sky Birdges

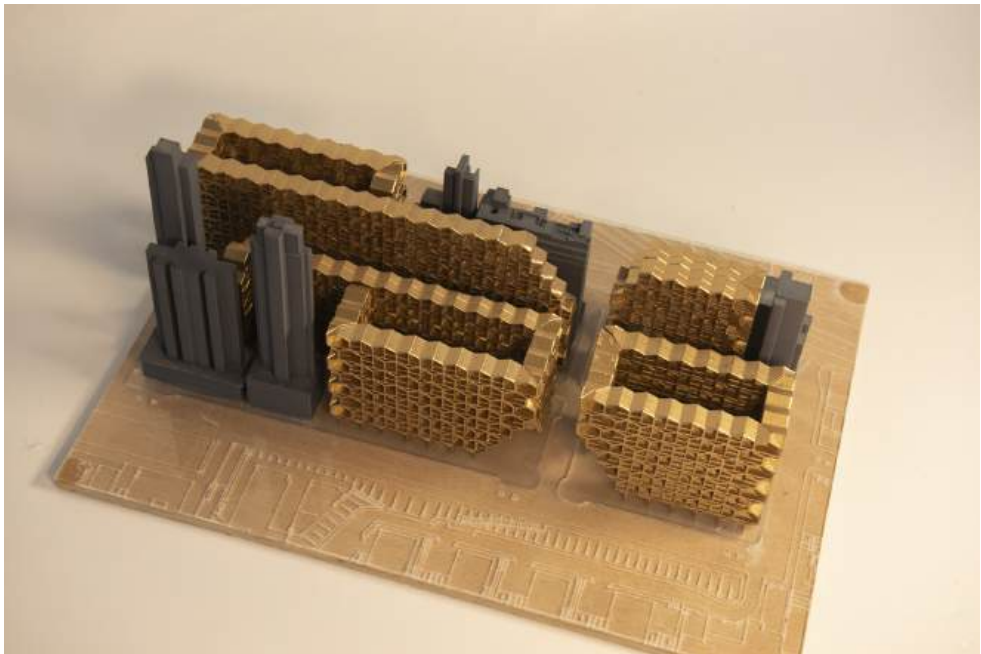


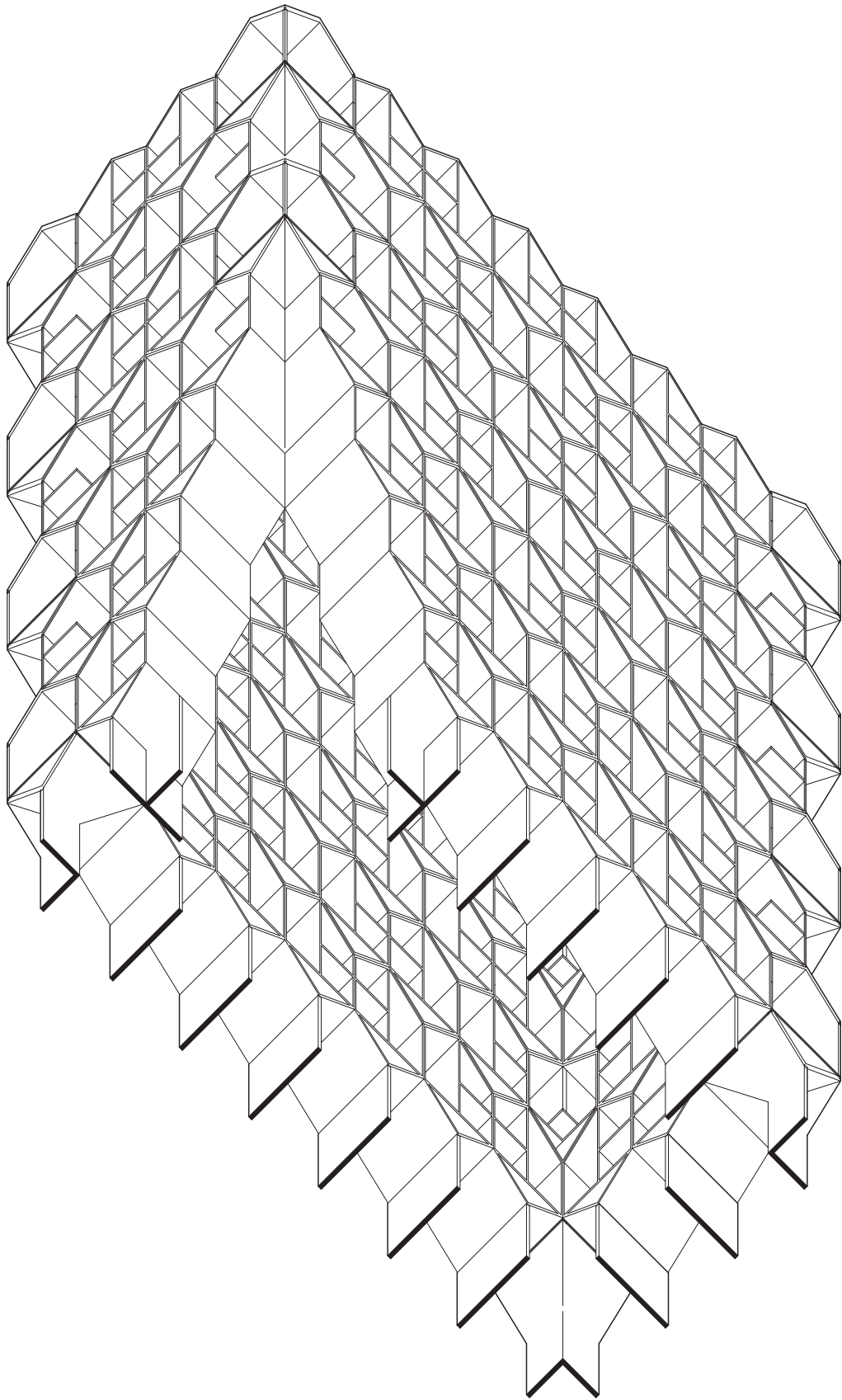
Facade of Autonomy



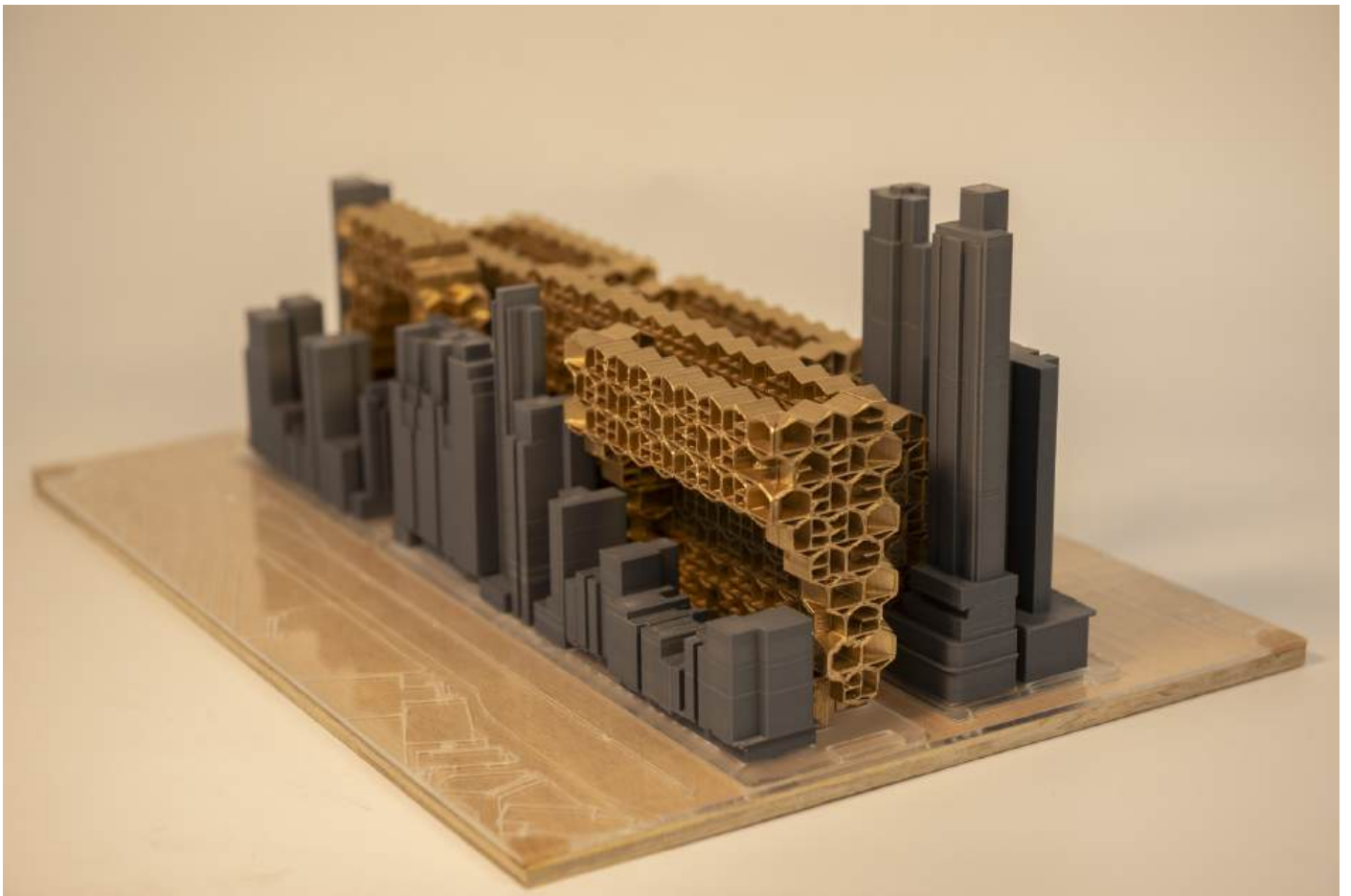


Internal Circulations





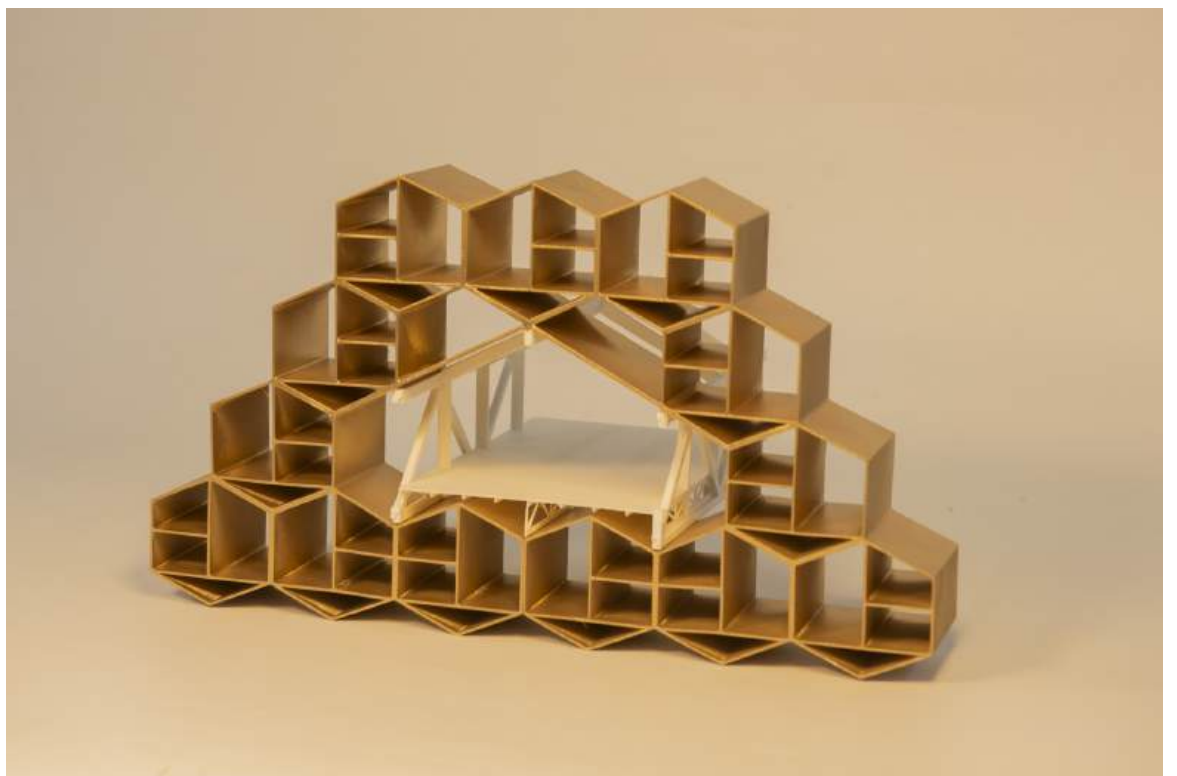
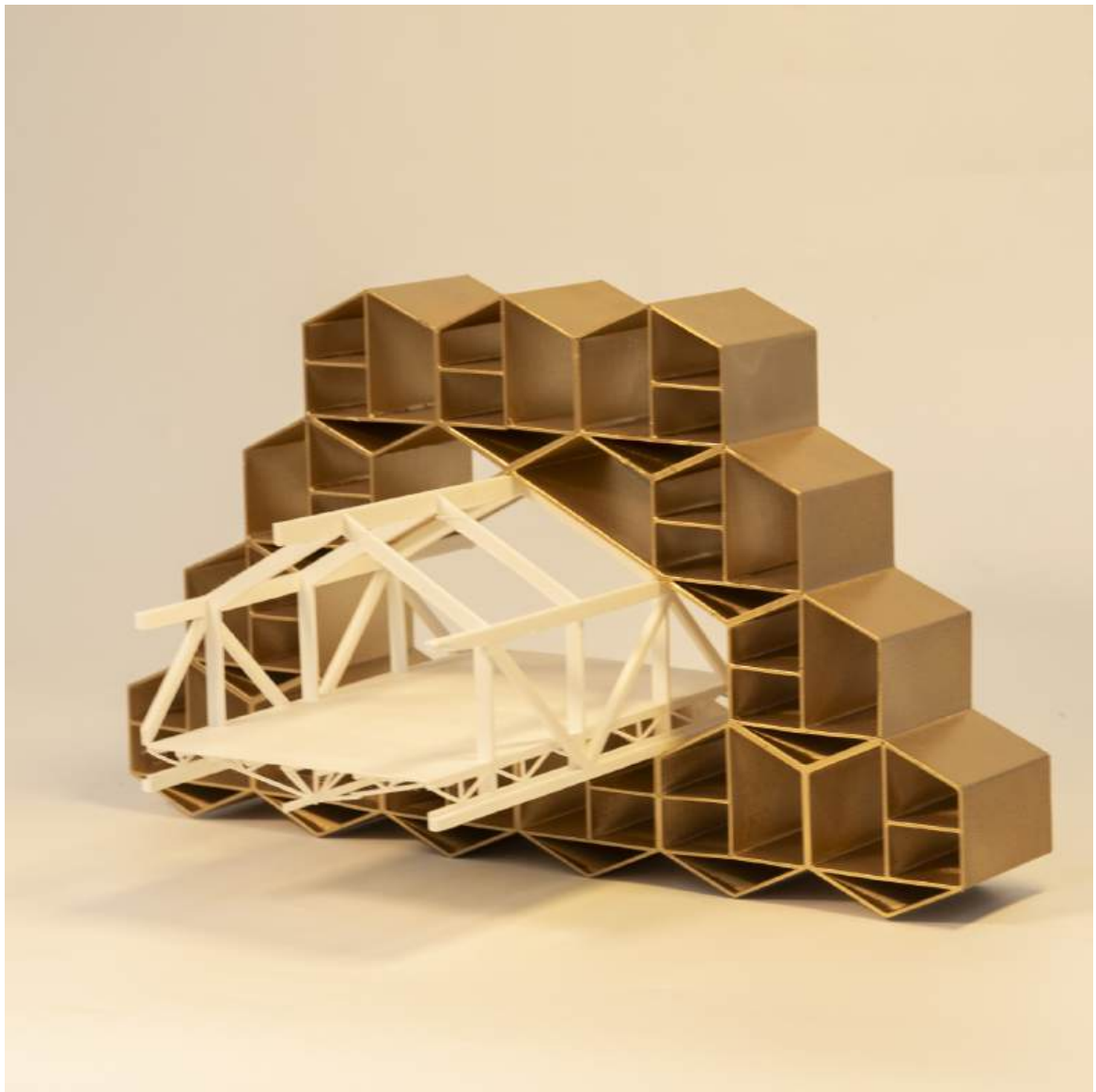
Ground Floor Accessibility

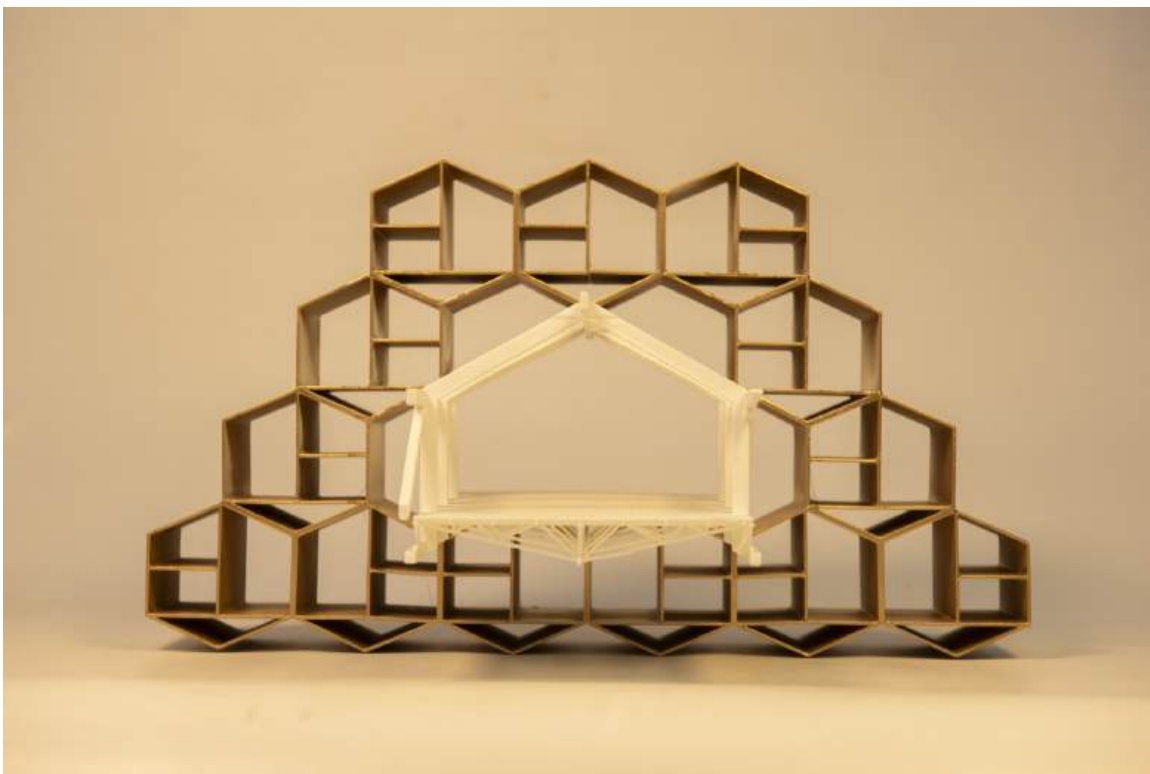


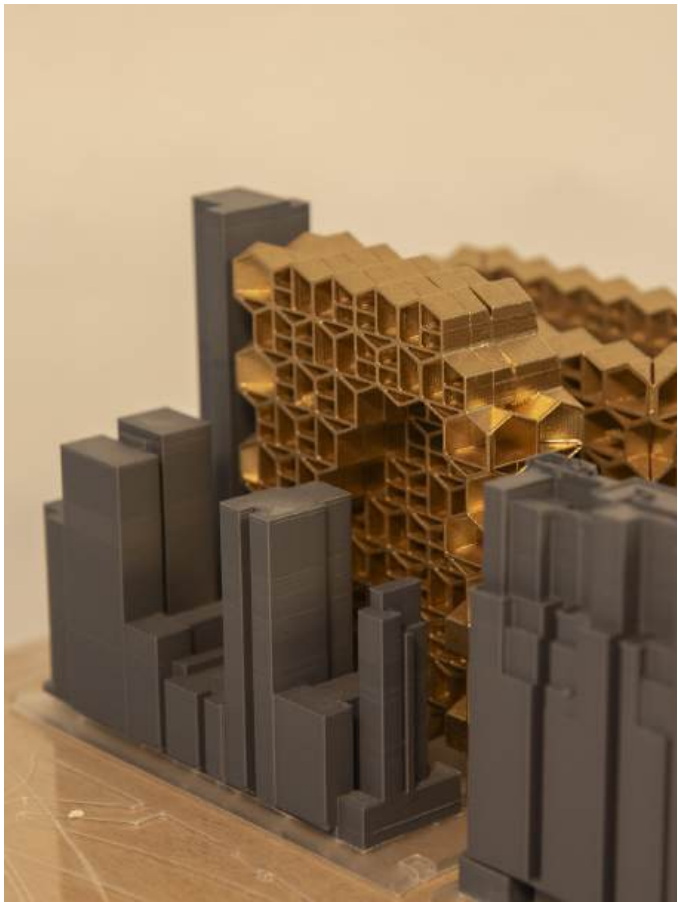
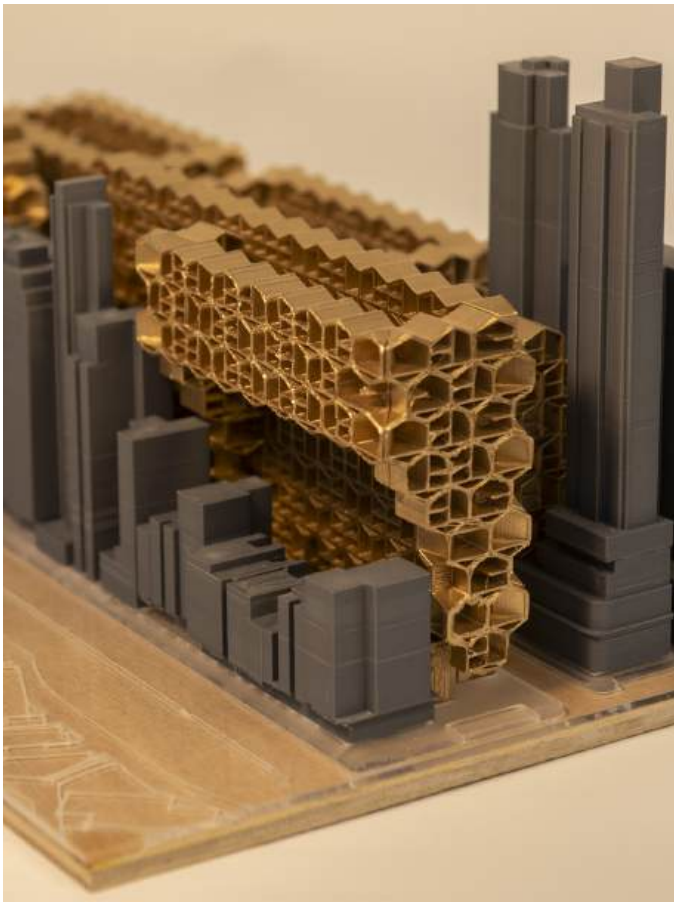
Building Upward

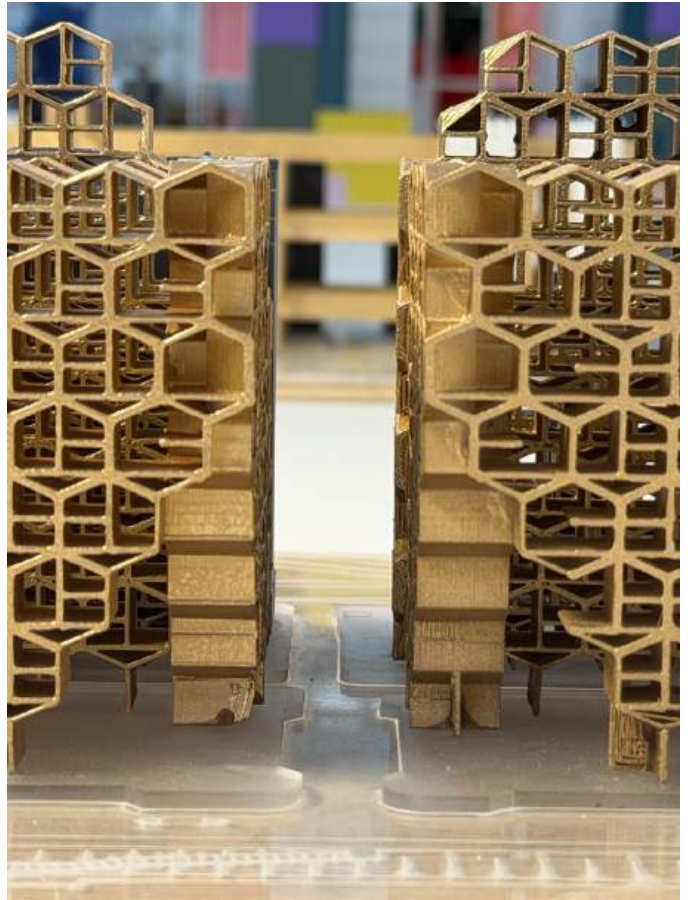
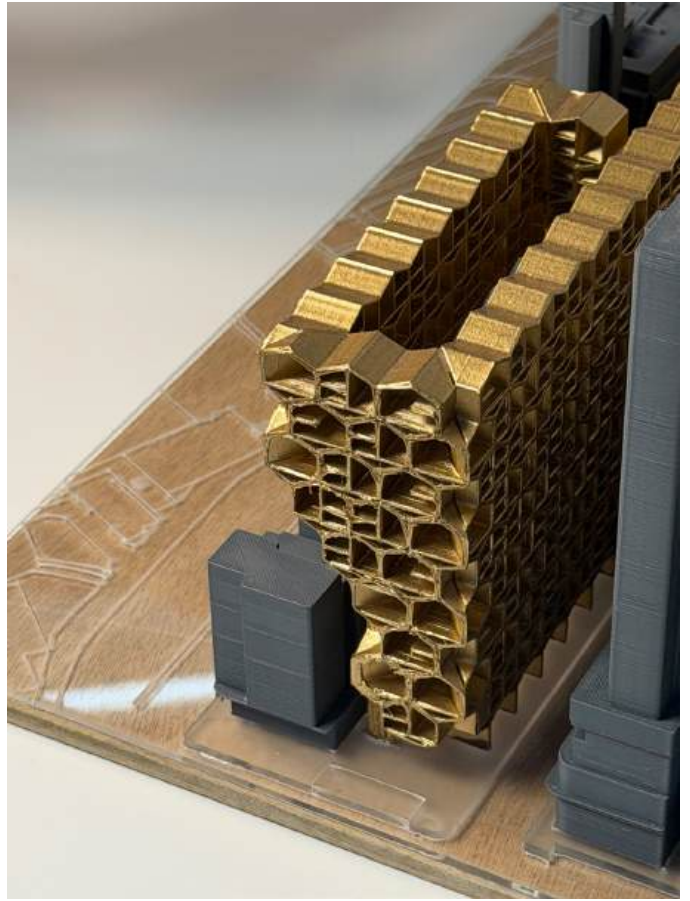
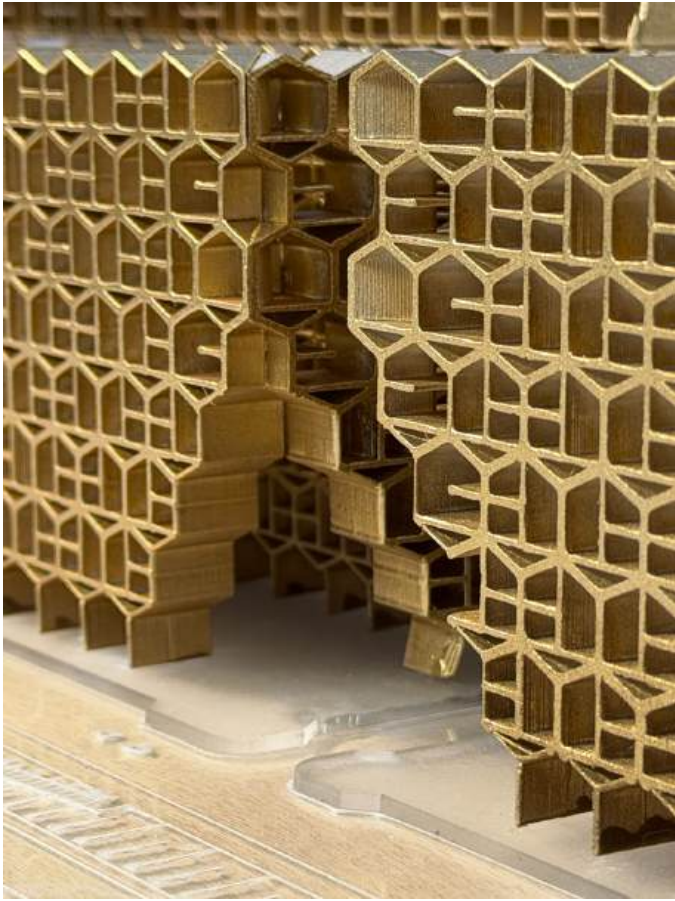


Mega Urban Block





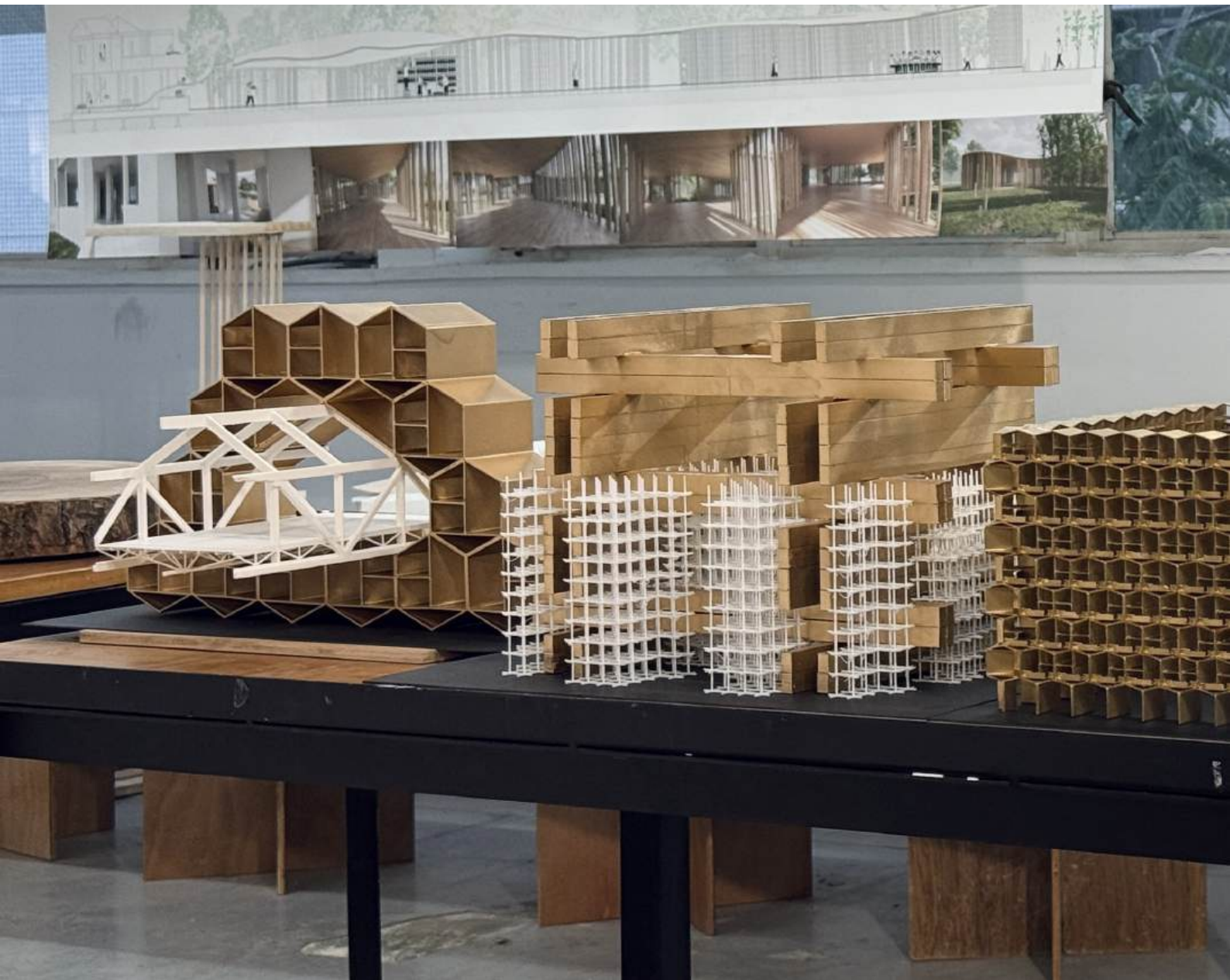




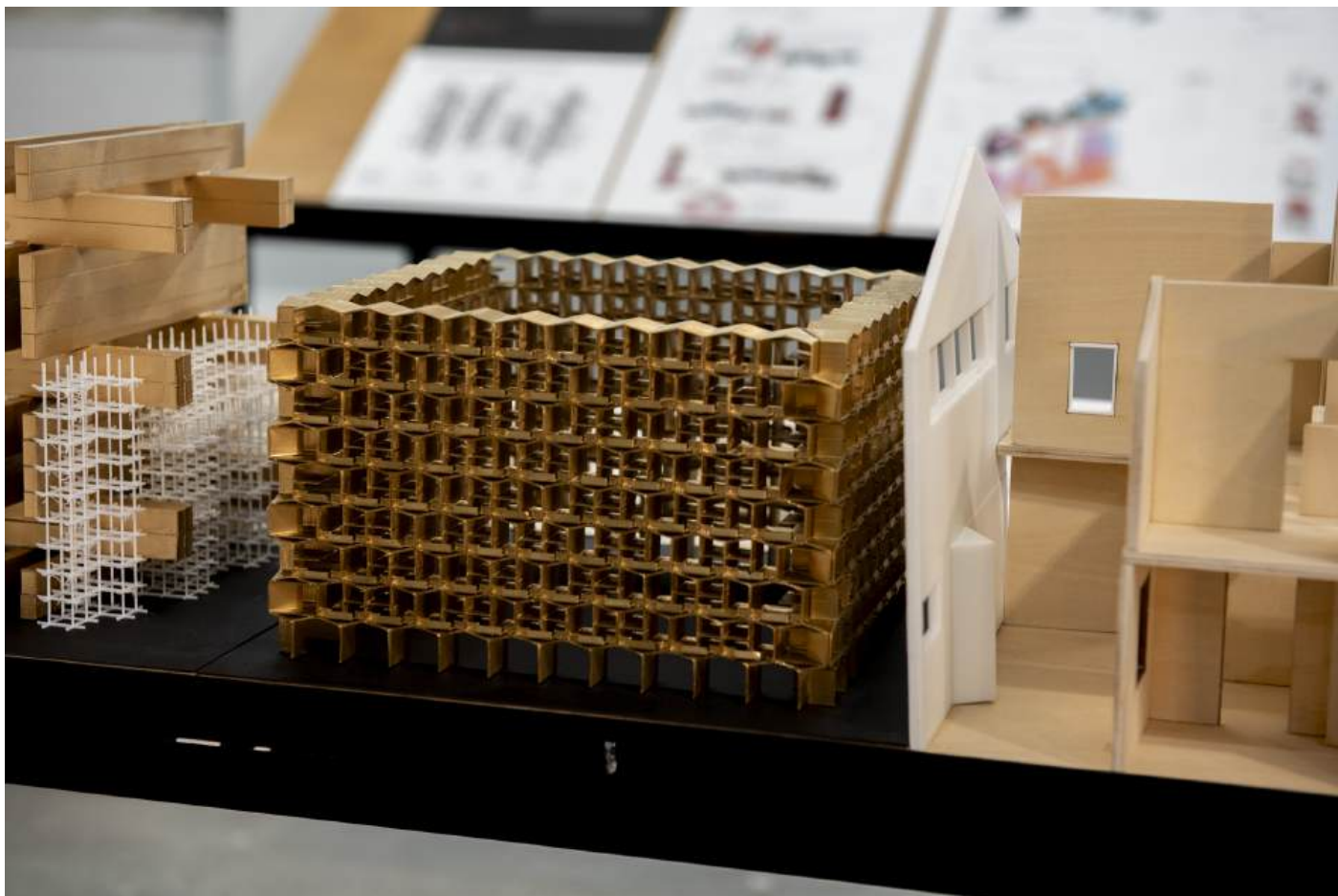
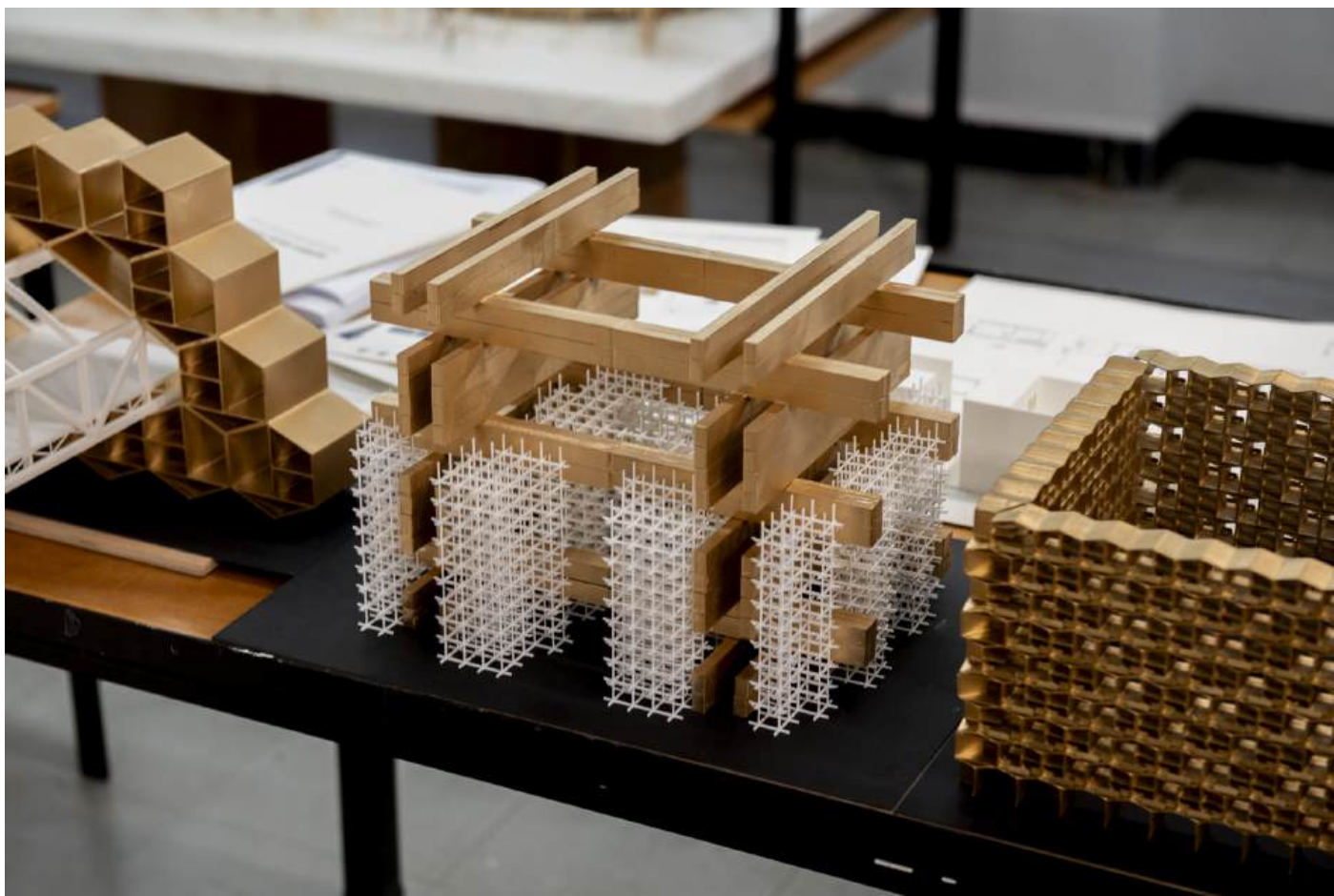
Chapter 6 | **Significance**

The original objective of this thesis was to address the affordability crisis for Hong Kong's sandwich class. Its significance extends far beyond mere cost-reduction. By translating the 100-year lifespan of a building into a liquid financial instrument, Super Liquid Housing offers a scalable blueprint for post-commodity urbanism. The originality of the thesis lies in its embrace of financialization, not as a predatory force, but as a tool for architectural liberation. Instead of fighting the inevitable flow of capital in a city like Hong Kong, the project redirects it. It creates a paradigm where investors find secure yield in the permanence of the structure, and the sandwich class finds sanctuary and spatial autonomy within its voids.

The primary lesson learned throughout this research is that architecture alone cannot solve a housing crisis; it must be coupled with economic redesign. The successes of the project lie in the elegant resolution of PNAP APP-130 regulations within a highly flexible framework. While the model requires a massive initial centralization of capital (The Housing Trust), it ultimately proposes a highly democratic, elastic, and human-centric vision for the future of high-density living.









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Appendix

THESIS MIDTERM REVIEW EVALUATION FORM
TOPIC, METHOD, AND PROCESS

Candidates are to treat this as a mid-term design review and present all materials and information to communicate the architectural design and development progress effectively. In addition, students must demonstrate they have rigorously tested an iterative, "circular" design method that balances design projection and design analysis within the respective contexts, scales, scope and program of their particular hypothesis/question/position.

Students are required to present their methods through the following deliverables:

- **Part I - Topic:** This part of drawings, images, preliminary study models, and other objects/media should be used to demonstrate the originality and significance of their thesis question/hypothesis/position.
- **Part II - Method & Process:** This part of drawings and models should demonstrate rigorous, iterative, and "circular" procedures for testing and analysing their design methods at a minimum of one specific scale;
- Arguments reinforcing how/why the specific scale, scope, context, and program of the test case addresses the hypothesis/question/position that must articulate verbally and visually;
- Evidence of the student's ability to evaluate and discern both the successes and failures of various methodological tests must be articulated verbally and visually; and
- Basic technical competence and expertise as relates to the overall thesis and design methods deployed must be articulated visually and verbally.

Student: To Wai Lam

Charles

ORIGINALITY/ANALYTICAL/CRITICAL ABILITIES

Demonstrates evidence of original thought, strong analytical and critical abilities through models, drawings and specific scales?

Original + unusual insights + research into housing through financial perspective.

DEPTH OF TOPIC

Shows evidence of a thorough grasp of the topic from background examples, reading, and analysis, demonstrated through specific scales, scope, context, and/or program of their design work?

Project has potential to ~~become~~ a test + articulate the frontier between new developmental patterns and the generation of new form.

ORGANISATIONAL/RHETORICAL/PRESENTATION SKILLS

An ability to evaluate their project, demonstrate overall basic technical competence and expertise as expressed verbally and visually?

Very good verbal presentation + excellent graphical diagrams.

Thank you.